WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 5 September 2026 | 30 August 2026 to 5 September 2026 | 150 records pooled from 7 daily issues

150 RECORDS POOLED (150 UNIQUE DOIs)

54% MEASUREMENT SIDE (81 RECORDS)

10 SUBTOPICS ACTIVE

34 POOLED EFFECT ESTIMATES

16 ABSTRACT-FREE (11% OF WEEK)

The week in one paragraph

The biggest regular week this watch has run, and the measurement side took it — but the structural story is indoors. Occupational & indoor went 7 → 19, and what arrived was not exposure epidemiology but *engineering*: ventilation rates, filter media, purifier siting and smoke evacuation.

Scale first, because it conditions everything below. 150 records over seven issues, the largest week outside the founding backfill, on daily totals of 14, 16, 33, 15, **38**, 27 and **7** — a 5.4× range. The 7 on Saturday is not a thin harvest but a **true PubMed zero**: no record entering PubMed on 5 September matched either axis, confirmed independently by the connector against a day that admitted only 544 records in total. Read every week-over-week movement below against that volatility.

The indoor turn is the week's real finding. *Occupational & indoor* at 19 ties the founding backfill week and is more than double any regular week this watch has recorded (previous regular maximum 9). The composition of that cluster is what matters: Jin et al. (31 Aug) optimise air-change rate against purifier placement jointly rather than separately; Wood et al. (31 Aug) and Derwein et al. (4 Sep) both put ventilation rate against filtration and mask blocking in the same model; Jamali et al. (3 Sep) and Lv H et al. (2 Sep) report new filter media (all-cellulose HEPA, polyimide microsphere/nanofibrous). Alongside them sit household solid fuel (Gu et al., Chen Y et al.), surgical smoke (Ma L et al.), welding fume (Yang et al., Wei J et al.), wood dust (Srinivasan et al.) and chalk dust (Mbazima et al., 5 Sep). **This is a control- and mitigation-side literature, not an exposure–response one**, and it is the part of the week most directly usable by an instrumentation programme.

The measurement side is back at its ceiling. *Exposure assessment & modelling* 24 → **47** — the highest this cluster has ever recorded, and level with the all-time single-subtopic weekly high set by *Sensing* in the 2–8 August week — with *Sensing* itself 21 → 34. Together 81 of 150, a measurement-side share of **54%**, tying the highest on record.

The health side thinned correspondingly, and the effect estimates thinned with it. *Respiratory & allergic* (9 → 5) and *Burden, policy & mitigation* (13 → 7) both hit their lowest weekly counts recorded. **34 pooled effect estimates against 150 records** is the weakest estimate density of any week this watch has run — one estimate per 4.4 records, against one per 2.7 last week. **Operationally the week was clean**: 150 records, **150 unique DOIs**, so unlike last week there was no double-publication to repair.

Subtopic trend and inflection

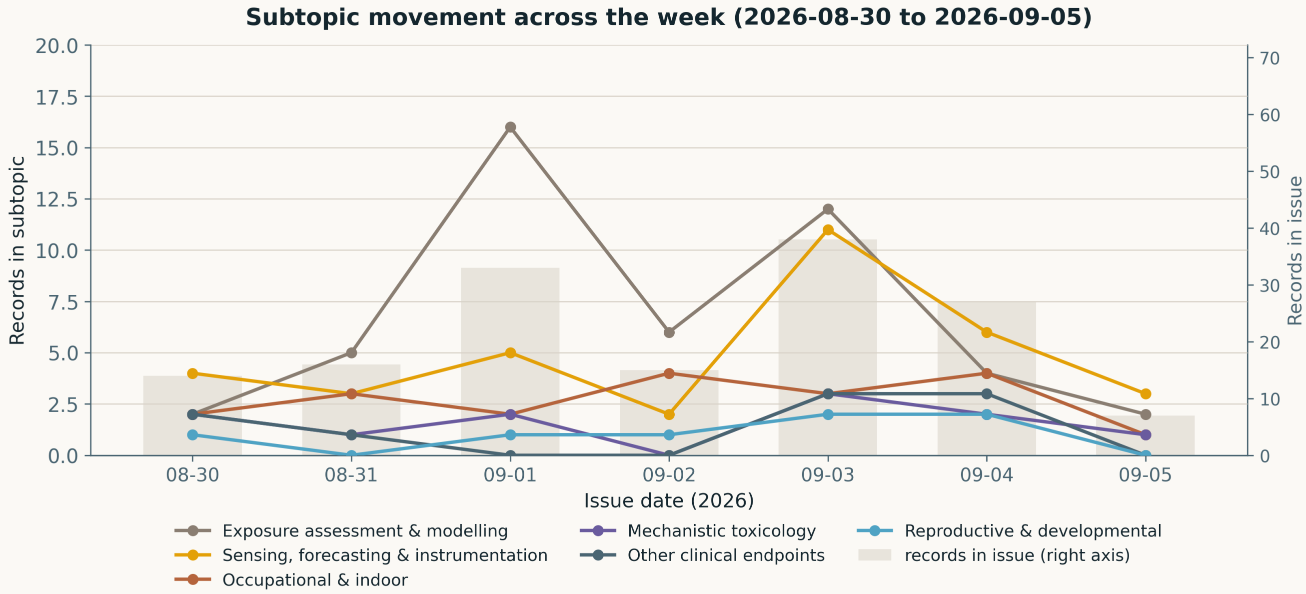


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **Read the denominators first**: daily totals were 14, 16, 33, 15, **38**, 27 and **7**. The 3 September peak of 38 and the 5 September floor of 7 are both harvest-side artefacts rather than field signals — the first a heavy Crossref-by-ISSN deposit day, the second a weekend on which PubMed admitted no PM record at all. The Saturday point should not be read as a decline.

Expanding. *Occupational & indoor* **7** → **19** is the week's largest movement in absolute and relative terms, and its 19 ties the founding backfill week against a regular-week maximum of 9. *Exposure assessment & modelling* **24** → **47** sets a cluster record; *Sensing, forecasting & instrumentation* **21** → **34** recovers the ground it lost in the two preceding weeks. *Reproductive & developmental* (**3** → **7**) is at its highest regular-week level, and *Cardiovascular & metabolic* (**5** → **7**) continues the slow recovery the 29 August rollup flagged.

Contracting. *Burden, policy & mitigation* **13** → **7** and *Respiratory & allergic* **9** → **5** are both the lowest weekly readings this watch has recorded for those clusters. *Neuro / mental health* fell **7** → **4**; that is low but not a record (the 2–8 August week returned 2). *Mechanistic toxicology* was effectively flat at **12** → **11**, holding a three-week band of 11–12.

The inflection worth naming. For four weeks the measurement-side share has run 41–54% with no trend, and the 29 August rollup said explicitly that no claim about the field should rest on it. That still holds. What is *not* noise is the indoor cluster: a 7 → 19 jump concentrated in ventilation, filtration and source-capture engineering, arriving in the same week that *Burden, policy & mitigation* hit its floor. The mitigation literature did not shrink — it moved from the policy register into the engineering register. **Worth a second week of watching before it is called a trend**, because a single MDPI or Elsevier volume release can manufacture a cluster of this size, and the 27 August precedent in the last rollup is exactly that failure.

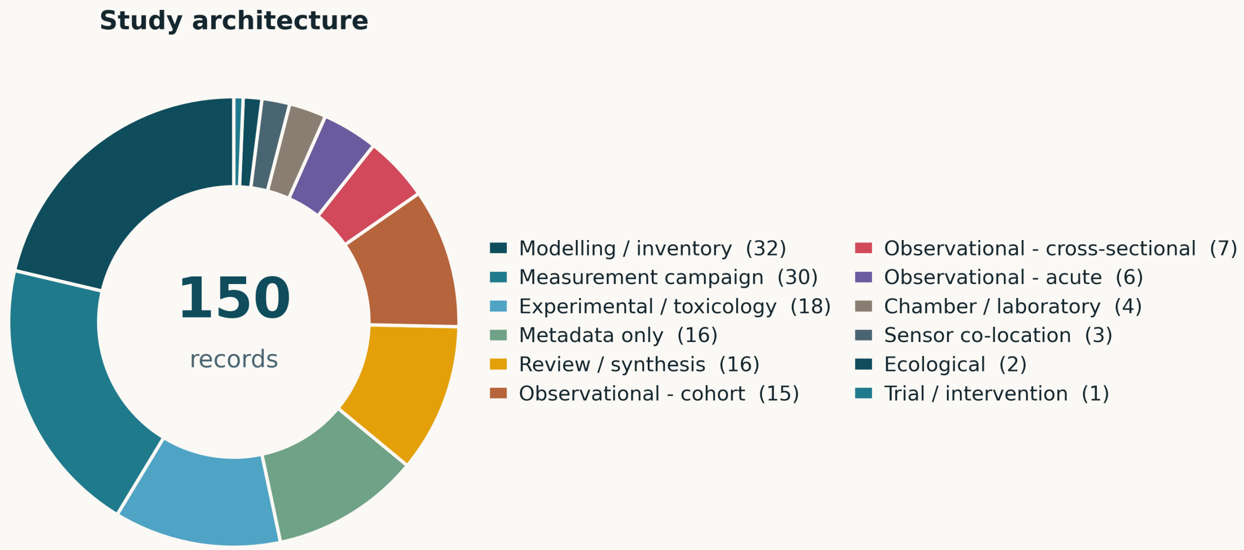


Fig. Study architecture over the pooled week. *Modelling / inventory* (32) leads *Measurement campaign* (30) — the first week this watch has recorded modelling ahead of measurement — with *Experimental / toxicology* at 18 and *Metadata only* at 16. Primary observational epidemiology (cohort, cross-sectional, acute) totals 28, fewer than modelling alone. Only **one** interventional record appears in the entire 150.

Pooled analytics

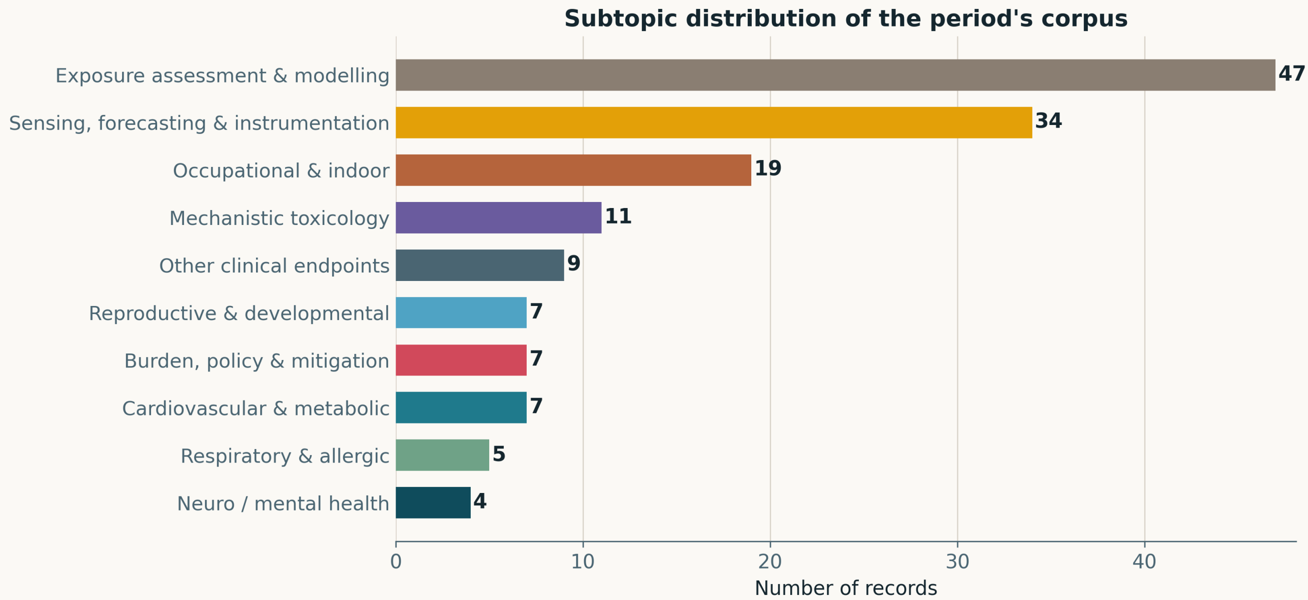


Fig. Pooled subtopic distribution, 150 records. All ten clusters were active, for the sixth consecutive week. The two measurement clusters alone (47 + 34 = 81) outweigh the eight health and policy clusters combined (69).

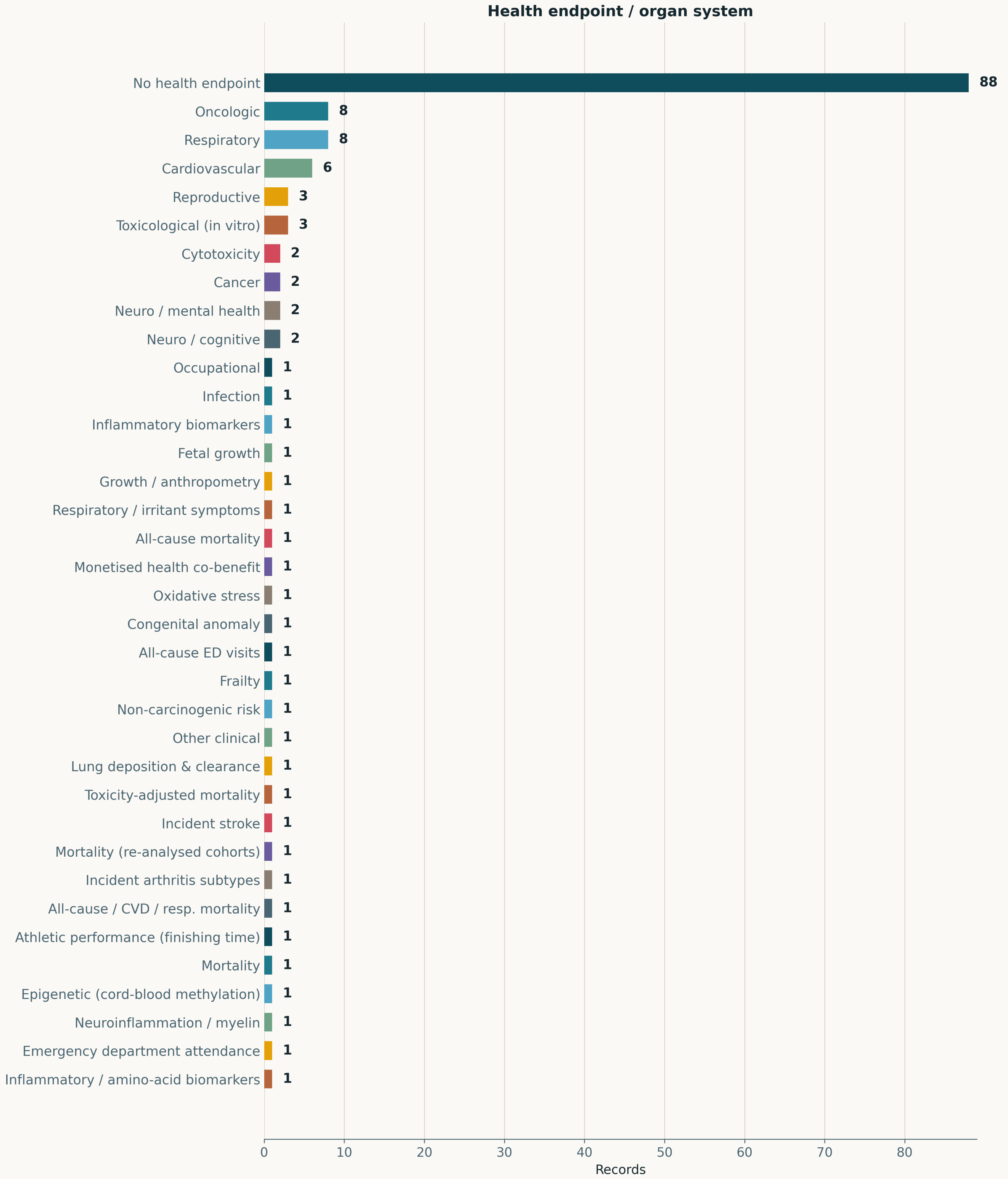


Fig. Health endpoint by organ system. The three largest bars carry *no* human endpoint — instrument (16), aerosol chemistry (15) and exposure (14) — and with monitoring and modelling added, **57 of 150 records have no health endpoint at all**. Respiratory (8), carcinogenic risk (7) and cardiovascular (6) lead the endpoints that exist. Note that *Cancer* and *Oncologic* remain split in `ENDPOINT_CANON`, so oncological work is understated by this panel; logged 3 September, still unfixed.

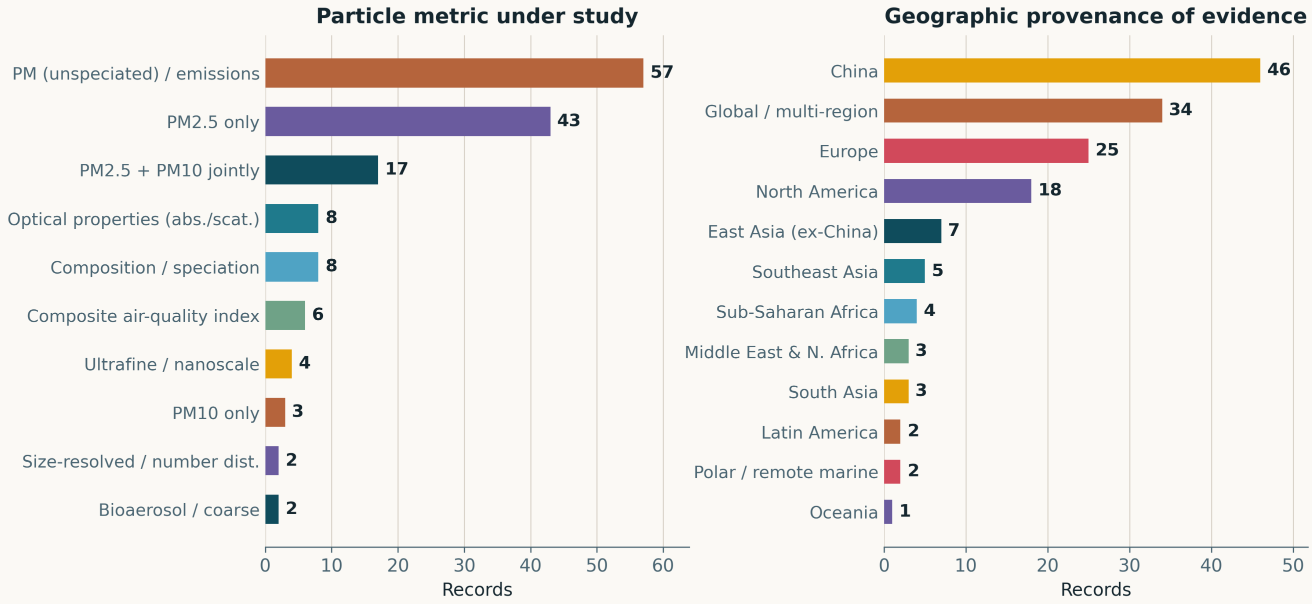
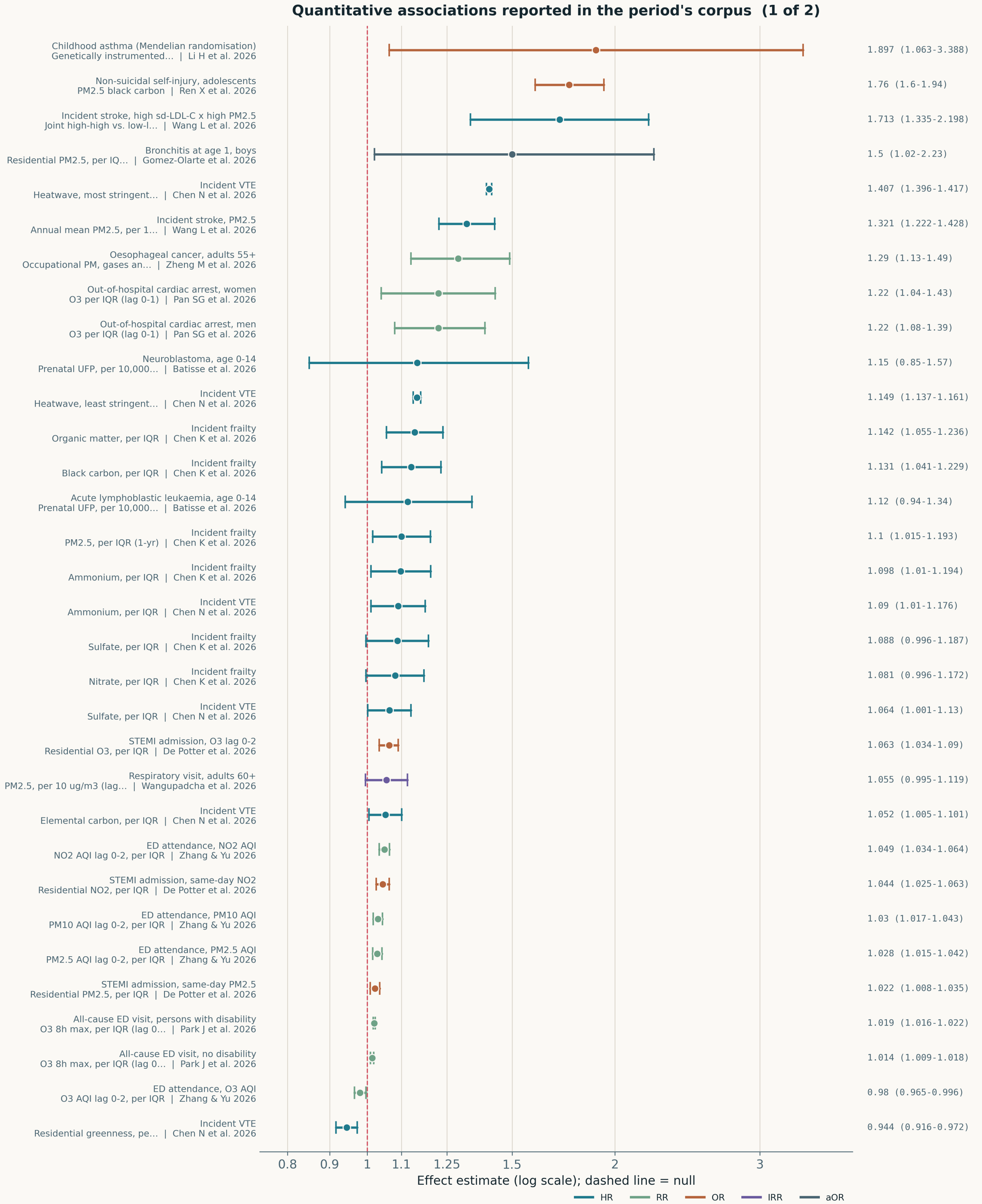


Fig. Left — particle metric. Unspeciated PM and emission factors (57) lead $PM_{2.5}$-only (43), with $PM_{2.5}$+PM_{10} jointly at 17. Ultrafine remains marginal at 4 despite two of the week's strongest records being ultrafine-resolved. Right — geographic provenance. **China supplies 46 of 150 (31%)**. The panel understates concentration in the anglophone west because the raw geography field is still unnormalised — *North America* (11), *United States* (7) and *Canada* (1) are distinct labels, as are *Europe* (17), *United Kingdom* (6), *Germany* (3), *Belgium* (2) and others. That defect is carried from daily-issue authoring and is logged again below.


Quantitative associations reported in the period's corpus (1 of 2)
Childhood asthma (Mendelian randomisation)
Genetically instrumented... | Li H et al. 2026
1.897 (1.063-3.388)
Non-suicidal self-injury, adolescents
PM2.5 black carbon | Ren X et al. 2026
1.76 (1.6-1.94)
Incident stroke, high sd-LDL-C x high PM2.5
Joint high-high vs. low-l... | Wang L et al. 2026
1.713 (1.335-2.198)
Bronchitis at age 1, boys
Residential PM2.5, per IQ... | Gomez-Olarte et al. 2026
1.5 (1.02-2.23)
Incident VTE
Heatwave, most stringent... | Chen N et al. 2026
1.407 (1.396-1.417)
Incident stroke, PM2.5
Annual mean PM2.5, per 1... | Wang L et al. 2026
1.321 (1.222-1.428)
Oesophageal cancer, adults 55+
Occupational PM, gases an... | Zheng M et al. 2026
1.29 (1.13-1.49)
Out-of-hospital cardiac arrest, women
O3 per IQR (lag 0-1) | Pan SG et al. 2026
1.22 (1.04-1.43)
Out-of-hospital cardiac arrest, men
O3 per IQR (lag 0-1) | Pan SG et al. 2026
1.22 (1.08-1.39)
Neuroblastoma, age 0-14
Prenatal UFP, per 10,000... | Batisse et al. 2026
1.15 (0.85-1.57)
Incident VTE
Heatwave, least stringent... | Chen N et al. 2026
1.149 (1.137-1.161)
Incident frailty
Organic matter, per IQR | Chen K et al. 2026
1.142 (1.055-1.236)
Incident frailty
Black carbon, per IQR | Chen K et al. 2026
1.131 (1.041-1.229)
Acute lymphoblastic leukaemia, age 0-14
Prenatal UFP, per 10,000... | Batisse et al. 2026
1.12 (0.94-1.34)
Incident frailty
PM2.5, per IQR (1-yr) | Chen K et al. 2026
1.1 (1.015-1.193)
Incident frailty
Ammonium, per IQR | Chen K et al. 2026
1.098 (1.01-1.194)
Incident VTE
Ammonium, per IQR | Chen N et al. 2026
1.09 (1.01-1.176)
Incident frailty
Sulfate, per IQR | Chen K et al. 2026
1.088 (0.996-1.187)
Incident frailty
Nitrate, per IQR | Chen K et al. 2026
1.081 (0.996-1.172)
Incident VTE
Sulfate, per IQR | Chen N et al. 2026
1.064 (1.001-1.13)
STEMI admission, O3 lag 0-2
Residential O3, per IQR | De Potter et al. 2026
1.063 (1.034-1.09)
Respiratory visit, adults 60+
PM2.5, per 10 ug/m3 (lag... | Wangupadcha et al. 2026
1.055 (0.995-1.119)
Incident VTE
Elemental carbon, per IQR | Chen N et al. 2026
1.052 (1.005-1.101)
ED attendance, NO2 AQI
NO2 AQI lag 0-2, per IQR | Zhang & Yu 2026
1.049 (1.034-1.064)
STEMI admission, same-day NO2
Residential NO2, per IQR | De Potter et al. 2026
1.044 (1.025-1.063)
ED attendance, PM10 AQI
PM10 AQI lag 0-2, per IQR | Zhang & Yu 2026
1.03 (1.017-1.043)
ED attendance, PM2.5 AQI
PM2.5 AQI lag 0-2, per IQR | Zhang & Yu 2026
1.028 (1.015-1.042)
STEMI admission, same-day PM2.5
Residential PM2.5, per IQR | De Potter et al. 2026
1.022 (1.008-1.035)
All-cause ED visit, persons with disability
O3 8h max, per IQR (lag 0... | Park J et al. 2026
1.019 (1.016-1.022)
All-cause ED visit, no disability
O3 8h max, per IQR (lag 0... | Park J et al. 2026
1.014 (1.009-1.018)
ED attendance, O3 AQI
O3 AQI lag 0-2, per IQR | Zhang & Yu 2026
0.98 (0.965-0.996)
Incident VTE
Residential greenness, pe... | Chen N et al. 2026
0.944 (0.916-0.972)
0.8
0.9
1
1.1
1.25
1.5
2
3
Effect estimate (log scale); dashed line = null
HR
RR
OR
IRR
aOR

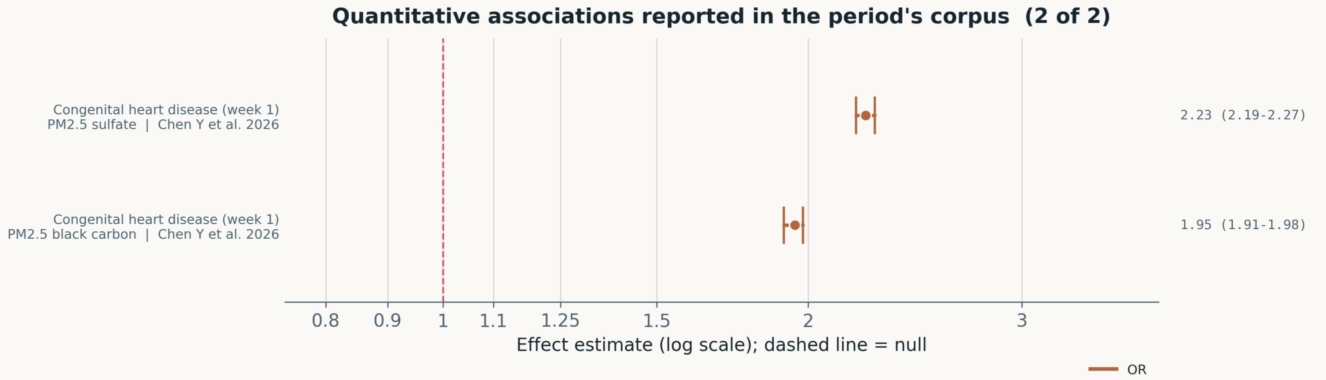


Fig. All 34 pooled effect estimates on a common logarithmic axis. **These are not mutually comparable and are not meta-analysed**: contrasts include per-IQR, per-1 μg/m^3, per-quartile and AQI-unit changes, on HR (16), RR (9), OR (7), aOR (1) and IRR (1). **Heterogeneity is structural, not statistical**, so no I^2 is reported — pooling these would be a category error. Two features are readable anyway. First, **precision tracks sample size to the point of implausibility**: Chen Y et al.'s congenital heart disease ORs (2.23, 2.19–2.27) and Chen N et al.'s tightest VTE estimates (1.407, 1.396–1.417) carry intervals so narrow that they imply either enormous registries or under-propagated uncertainty, and the abstracts do not let a reader tell which. Second, **only two of the 34 cross the null** (Batisse et al.'s paediatric leukaemia and neuroblastoma HRs) — a distribution that reflects what abstracts choose to report at least as much as what the studies found.

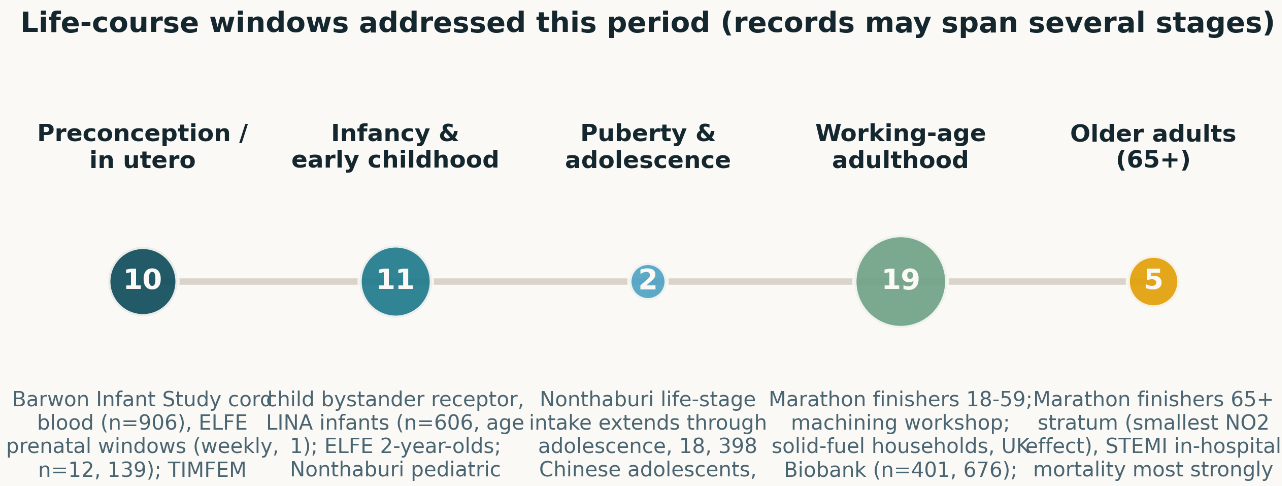


Fig. Life-course windows across the pooled week. Working-age adulthood dominates, consistent with the occupational and indoor surge described above.

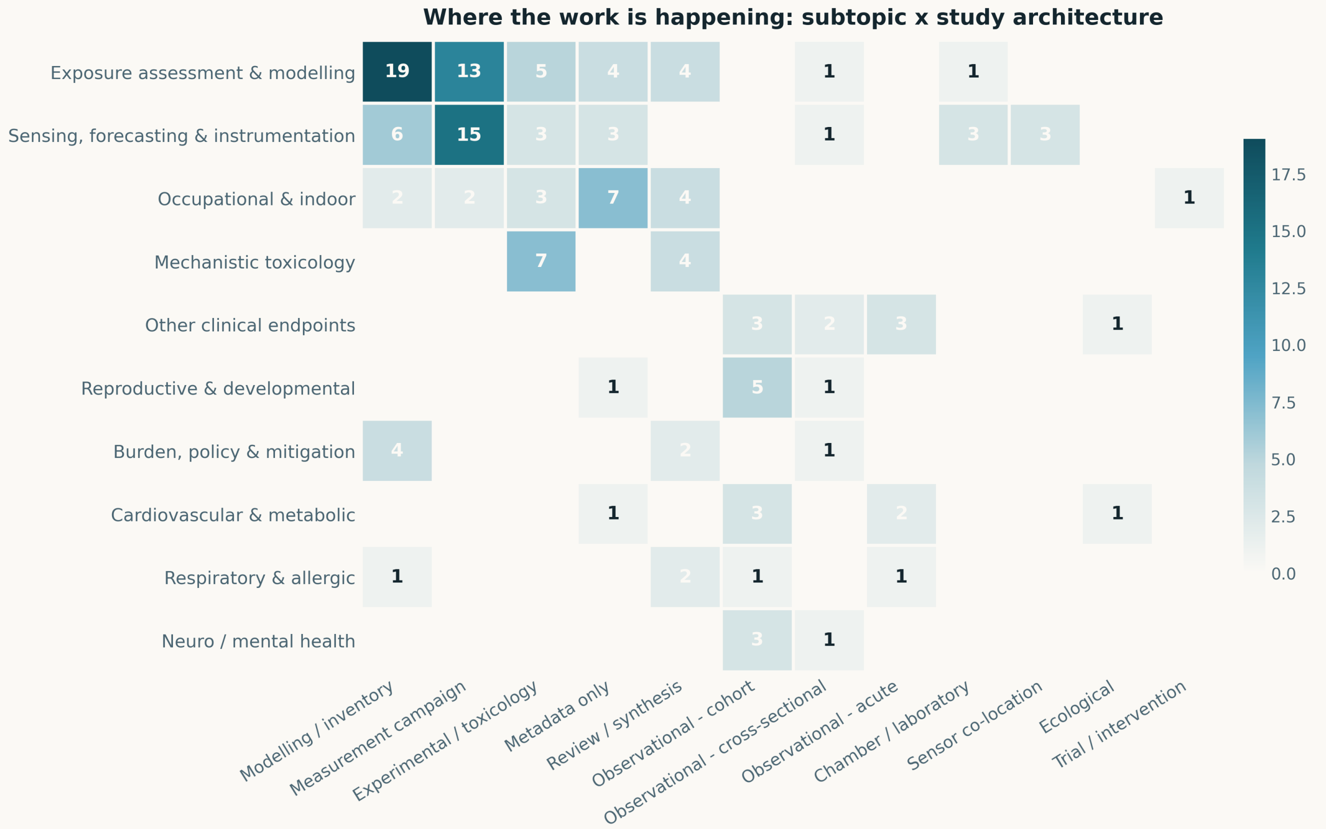


Fig. Subtopic × architecture, pooled. The standing structural finding survives another week at $n = 150$: the *Sensing* and *Exposure* rows carry almost no health-endpoint linkage, and *Mechanistic toxicology* remains almost entirely experimental. Instrumentation and epidemiology are still being written by different people in different journals.

Venues, trial watch, and what the pipeline did

Venues. *Environ Sci Technol* at **23 of 150** is the most concentrated single-venue week this watch has recorded, and it is a harvesting artefact as much as an editorial one — ES&T is on the Crossref by-ISSN list and publishes in daily batches, so it enters this corpus at full rate while journals reached only through PubMed enter at PubMed's cadence. *Atmos Chem Phys* (8), *J Environ Sci* (*China*) (8) and *Atmosphere* (8) follow, then *Environ Pollut* (7). **Research Square supplied 5 preprints**, all labelled as such in the register.

Recurring authors. Surname clustering is dominated by high-frequency Chinese surnames — Wang (7), Chen (6), Li (4), Liu (4) — which is a naming artefact, not a research-group signal, and this watch has no author-disambiguation field to resolve it. **No genuine repeat-group cluster is claimed for this week.** Adding a first-author affiliation field remains the open item that would make this section meaningful; it has been on the list since 3 September.

Trial watch — one new interventional registration in the period. NCT07805824 (first posted 4 September) is a cluster-randomised trial of a workplace heat and air-pollution mitigation package across **20 restaurants and 200 oven workers** in Najaf, Iraq, with a delayed-intervention comparator over 24 weeks. Its primary endpoint is an objective gingival melanin index; its **sole secondary endpoint is change in personal $PM_{2.5}$ exposure, measured both gravimetrically and in real time**. `analyze_endpoints` was run: 1 primary, 1 secondary, 0 other. **Registry-hygiene flags are recorded on the record and should be read before citing it**: it is labelled PHASE4 with no drug; `start_date` equals `primary_completion_date` equals 2026-01-01 while status is COMPLETED; and the public `detailed_description` still carries the submitter's own pre-submission checklist. The endpoint design is usable; the timeline fields are not. Two further records were *updated* rather than registered in the period — **NCT04153539**, the Fudan busy-road versus traffic-free-park crossover in 69 healthy young adults, and **NCT05016271** — and **NCT07797413** (Savannah port-adjacent nutrition feasibility trial, $n = 20$) was carried in the August monthly. `trials.json` now holds **26 trials across 11 windows**.

A weekend zero in the registry is structural. ClinicalTrials.gov posted **no updates registry-wide** on 5 September, nor on 29 or 30 August, against 973–1,001 on each intervening weekday. Every Saturday and Sunday daily issue will therefore report a trial zero for calendar reasons alone. This was verified this week across six dates and should stop the weekend zeros being mistaken for an absence of PM trial activity — a misreading the daily issues were previously set up to invite.

What the harvesting legs did. PubMed carried five of seven days and returned a verified zero on the sixth-largest[1] — on the *smallest* day. Crossref-by-ISSN was the week's workhorse and the reason ES&T dominates the venue table. **OpenAlex returned HTTP 429 on every day of the week**, unchanged since 28 July, and is now best described as a dead leg rather than a degraded one. **arXiv returned HTTP 200 every day but its newest submission has been frozen at 2026-08-20 all week**, so it contributed nothing to any issue in this period — worth a look at the query before next week. **Consensus returned a nil for the seventh consecutive run**, but the 4 September check confirmed why: its hits are the right literature with the wrong dates, and one of them was already in this corpus. Post-filtering Consensus on Crossref `created` inside the window remains the recommended fix and is still not implemented.

Open gaps

- **Ultrafine remains 4 of 150** while two of the week's strongest records — Sawan et al. on airport-attributable UFP and Shi S et al. on ship-plume number modes — are ultrafine-resolved. The metric that carries the source signal is still the metric almost nobody measures.
- **One interventional record in 150.** The evidence base moving into the engineering register this week is almost entirely observational or bench-scale; the ventilation and filtration papers optimise against models and chambers, not against health outcomes in occupied buildings.
- **No health endpoint in 57 of 150 records**, and the sensing/exposure rows of the heatmap remain empty of endpoint linkage for the sixth week. This is now the longest-standing structural finding in the series and it has not moved.
- **Estimate density fell to one per 4.4 records.** Whether that is a real thinning of quantitative epidemiology or an artefact of a measurement-heavy week is not decidable from one week; the four-week series is 3.3, 2.7, 2.9, 4.4.
- **Pipeline defects carried forward**: unnormalised geography labels (which understate western concentration in f3), the `Cancer`/`Oncologic` split in `ENDPOINT_CANON`, no first-author affiliation field, and a 6-element `LIFECOURSE` array in the 2026-08-03 corpus where the schema requires 5 — outside this window, but it will corrupt any rollup that spans it.

[1] Ordering by daily count: 38, 33, 27, 16, 15, 14, 7 — Saturday 5 September is the smallest.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation 34 records

Carried from the daily issues — full summaries (34 of 34 records in this subtopic)

Alderotti et al. 2026 Sci Total Environ

Raman spectroscopy as a non-destructive biomonitoring readout

Mature *Quercus ilex* sampled along a high/intermediate/low NO_2 and PM_{10} gradient in Tuscany. Leaf Raman spectra show coordinated modulation of pigment (chlorophyll, carotenoid) and inducible flavonoid bands, cross-validated against destructive biochemistry and chlorophyll fluorescence, and consistent with *acclimation* rather than photoinhibitory damage. A composite Raman index tracked site-level NO_2 most closely, which the authors read as nitrogen driving the response. The attraction is a rapid, non-destructive, scalable field readout; the limitation is that a three-site gradient cannot separate NO_2 from PM_{10} from any co-varying urban factor, so the index is at best a composite urban-exposure indicator and not a pollutant-specific one.

doi: 10.1016/j.scitotenv.2026.182253

Alraddadi et al. 2026 Atmosphere

Explainable one-hour-ahead AQI forecasting across eight Saudi cities

Attention-based bidirectional recurrent networks against a persistence baseline in eight climatically distinct cities. The deep models **matched persistence on RMSE and** R^2 (cross-city mean $R^2 \approx 0.90$) and beat it on MAE by roughly a fifth, with the margin largest in the most variable cities. Event-based evaluation flagged **84–91% of high-AQI exceedances**. Surface meteorology added little on average and its value was city- and architecture-specific. SHAP and LIME put particulate matter as the dominant driver everywhere. The persistence comparison is the part worth copying: at one-hour horizon hourly AQI is so autocorrelated that an unbenchmarked R^2 of 0.90 means nothing, and most papers in this genre do not report the baseline.

doi: 10.3390/atmos17090858

Blaga et al. 2026 [backfill] PLOS Clim

City-scale calibration of a 44-node $PM_{2.5}$ network for regulatory-compliant assessment

44 Clarity Node-S sensors across Bucharest against National Environmental Protection Agency reference stations; nine co-located spatial clusters, one used to fit and eight held out. Manufacturer pre-calibration **systematically underestimated** $PM_{2.5}$, worst in summer when secondary aerosol formation dominates. A two-step seasonal model on temperature and relative humidity alone moved Pearson r from **0.06 to 0.65** (dry season) and from $-$**0.28 to 0.89** (humid season). Applied network-wide, calibrated sensors average $>$**60 days/yr** above the EU 25 $\mu g\,m^{-3}$ daily limit against 35 permitted; pre-calibration, only 8 of 44 (18%) exceeded. Weakness: calibration degrades in hot dry low-concentration conditions and inter-sensor variability in calibration performance remains large, so a single network-wide model is not defensible. The 1:8 train/validate split is the design detail worth copying.

doi: 10.1371/journal.pclm.0000780

Cai J et al. 2026 Environ Sci Technol

Subway air microbiome is homogeneous at network scale and heterogeneous at platform scale

Stratified cluster sampling of **43 underground stations across 13 lines** of the Shanghai subway, collecting total suspended particles and $PM_{2.5}$ *simultaneously* at entrances, halls and platforms, then integrating taxonomy, transcriptional potential, $PM_{2.5}$ chemistry, microclimate and network topology. Bacteria dominate both size fractions and most transcriptional signal; eukaryotes are strongly enriched in TSP, **a clean demonstration of particle-size filtering of the biological load**. Dominant taxa (*Staphylococcus*, *Bacillus*) show weak abundance–transcription coupling, so abundance is a poor proxy for activity. Diversity and composition are essentially uniform across the network — consistent with passenger mixing — while finer-scale structure appears within stations. For exposure work this means a single station is a reasonable network sentinel for community composition and a poor one for microenvironment. Weakness: one city, one season not stated, no human outcome.

doi: 10.1021/acs.est.6c00174

Chaparro et al. 2026 Sci Total Environ

Biological and technological sensors for airborne particles at heritage sites

Ships on metadata only. The title indicates parallel biomonitoring and instrumental sensing of airborne fine particles for preventive conservation of cultural heritage, but no abstract was deposited and the record is not in PubMed, Europe PMC or Semantic Scholar on this run, so the sensor models, the biomonitor species and any agreement statistics between the two are unavailable. Counted and listed; no numbers claimed. On the retry list — cross-validation of a biological against an instrumental particle sensor is directly relevant to low-cost network validation and rarely published.

doi: 10.1016/j.scitotenv.2026.182285

Chen C et al. 2026 J Hazard Mater

Amine-containing particles from fireworks by single-particle mass spectrometry

2023 Spring Festival field campaign with single-particle aerosol mass spectrometry plus source samples. Amine-containing particles rose **96.2%** during the firework period, peak size shifted $\mathbf{0.42 \rightarrow 0.44}\,\mu$**m**, and internal mixing with sulfate increased; fireworks raised metal and elemental-organic-carbon classes while depressing biomass-burning and secondary classes. **N-methylaniline** is proposed as a tracer separating fireworks from firecrackers. A multi-model comparison identifies modified aerosol acidity, not liquid water content, as the dominant control on gas–particle partitioning of amines, with ALWC showing a non-linear threshold. One festival, one city; the tracer claim needs a second site before it is usable.

doi: 10.1016/j.jhazmat.2026.143446

Chen et al. 2026k Environ Sci Technol

Lake-breeze effects on urban air quality resolved by a city-wide low-cost sensor network

One year (2022) of the Microsoft Eclipse LCS network plus meteorological stations over Chicago, composited into **42 lake-breeze events** and 51 weak events confined within a few kilometres of the shore. Weekday events reduce O_3 by up to ∼6 ppb along the northeast shoreline — attributed to NO titration in a shallow boundary layer, inflow of cleaner lake air, and suppressed photochemistry at lower temperature — while *raising* $PM_{2.5}$ by up to $\sim 3\,\mu g\,m^{-3}$ via accumulation of local emissions and semivolatile partitioning. The limitation that matters: event-composited means over an LCS network inherit that network's calibration bias, and no reference-monitor validation of the $PM_{2.5}$ increment is reported, so the $3\,\mu g\,m^{-3}$ figure is a network-relative anomaly rather than a traceable concentration. Directly relevant precedent for using a dense LCS deployment as the primary instrument in a mesoscale study.

doi: 10.1021/acs.est.6c09290

Cui C et al. 2026 Environ Sci Technol

Eighteen years of CCN reconstructed by XGBoost with SHAP attribution

An observation-driven XGBoost framework reconstructs CCN number concentration at a North China Plain mountain site, 2007–2025. Particle number size distribution carries **55.7–66.3%** of predictive importance, with the controlling size range shifting from 100–200 nm at 0.2–0.4% supersaturation to **50–100 nm at 1.0%**. No monotonic CCN trend, but summer bulk activation ratio and hygroscopicity parameter both **declined from 2009 to 2025**. Coupling PMF with the ML model attributes that to falling SO_2 influence and rising contributions from new particle formation and growth — a legible consequence of desulfurisation policy. Reconstruction over an 18-year record inherits every instrument change in that record, and SHAP importance is not a causal decomposition.

doi: 10.1021/acs.est.6c04501

Delkash 2026 Environ Pollut

Barometric pressure variability in community air monitoring near a landfill

Ships on metadata only — the Crossref deposit has no abstract and Semantic Scholar returns the record with a null abstract field. The title describes a receptor-scale multi-method analysis of how barometric pressure variability imprints on community monitoring data around an elevated-temperature landfill, across two mitigation interventions. Counted and listed; no numbers claimed. This is the single most operationally relevant record of the day for network work — pressure-driven emission modulation is a known confounder in fenceline and community monitoring and is almost never quantified — so it goes to the top of the retry list.

doi: 10.1016/j.envpol.2026.129084

Galioto et al. 2026 Atmosphere

Physics-informed ML for street-canyon NO_2 in Palermo

Six years of hourly data (2020–2025) on one of Europe's densest traffic corridors, resolved to **∼11 m**: COPERT-Italy emission factors over the OpenStreetMap network, dispersed with the SIRANE street-network model, then fused with measurements in an ML stage and mapped in GIS. Observation–model scatter over **44,752 hours** centres on the 1:1 line. The genuinely useful move is inverting the chain to recover corridor traffic volume from observed NO_2 — a template for using a fixed monitor as a traffic sensor, which transfers to PM only where the non-exhaust fraction is separately handled. Analyte is gas-phase; single corridor; and a physics-informed hybrid validated against the same stations that trained it needs an out-of-corridor test before the inversion can be trusted.

doi: 10.3390/atmos17090865

Giang et al. 2026 J Radiol Prot

Fifteen consumer radon monitors, 98 days, one occupied house

45 units (15 models × 3) co-located with an AlphaGuard PQ2000 and two alpha-track detectors in the lowest lived-in space of an occupied Ontario home, with $PM_{2.5}$, CO_2, temperature and RH logged alongside; reference mean **99 Bq/m^3** (range 62–141). Cumulative performance: **8 of 15 models** had two or more units within ±25% relative percent error, the best reaching 3–5% after 13 weeks, while others ran from **−47% to +220%**. Intra-model CV was mostly under 15%; real-time RMSE for the best units was 5–6 Bq/m^3 with Spearman ρ=0.70–0.77 on two-day averages. Different analyte, and the QA design is directly transferable: this is the between-model spread estimate that consumer PM networks routinely lack. One home, one season, one concentration regime.

doi: 10.1088/1361-6498/aea16a

Guth et al. 2026 Atmosphere

Time-resolved emission factors for 19 prescribed burns, measured with low-cost sensors

Emission factors for CO, $PM_{2.5}$, EC, OC and speciated organics across **19 prescribed fires** in Colorado and southeastern Georgia, using multiple low-cost monitors per burn to get temporal resolution that filter-based EF campaigns cannot reach. CO EFs are driven by combustion phase — **+3260%** at low modified combustion efficiency, i.e. smouldering. Broadcast and pile burns are compared directly, which is rare. The methodological point for a distributed network is that the LCS bias that disqualifies a sensor for regulatory-equivalent concentration is tolerable here because the measurand is a ratio within a plume; the risk is that any RH- or size-dependent bias in the PM channel propagates straight into the $PM_{2.5}$/CO EF ratio, and the abstract does not report the sensor correction applied.

doi: 10.3390/atmos17090867

He J et al. 2026 ACS Sens

Room-temperature ppb NO_2 chemiresistor by molecular packing control

Alkyl chains of varying length on cyano-substituted distyrylbenzene tune molecular conformation and film microcrystallinity, exposing more surface-accessible functional groups and raising NO_2 sensitivity; the optimised device reaches a **30 ppb** detection limit at room temperature, and response is *enhanced* by humidity through NO_2–H_2O synergy. Demonstrated in a flexible wireless vehicle-exhaust monitor. Carried here on the sensing axis with the analyte stated plainly: this measures a gas, not a particle, and contributes nothing to PM mass or size. It is included because room-temperature operation and positive rather than negative humidity dependence are both unusual for the co-located gas channels that sit in multi-pollutant low-cost nodes. No long-term drift or cross-sensitivity data.

doi: 10.1021/acssensors.6c00977

Hong et al. 2026 Indoor Air

Predicting fungal and bacterial bioaerosol concentrations across building types

9,377 indoor air samples across 27 multiuse facility types in South Korea, modelled with a generalised additive mixed model to capture non-linear temperature and humidity effects, then a random forest on the resulting sensitivity classes. Fungi respond mainly to relative humidity (16 facility types), bacteria mainly to temperature (7); peak microbial levels occur at **55–60% RH**, supporting 60% as the practical dampness threshold. Stratifying by meteorological sensitivity before prediction raised F1 from 0.53 to **0.65 (bacteria) and 0.75 (mould)**. Culture-based counts, so the non-viable fraction is invisible, and the facility-type classes are administrative rather than physical. Directly relevant to anyone adding T/RH channels to an indoor node and expecting them to carry bioaerosol information.

doi: 10.1155/ina/6724151

Ichikawa et al. 2026 Atmos Chem Phys

Drone sampling agrees with a flux tower to within 30%, and that is the useful number

Multi-year warm-season BVOC measurements at multiple heights on a 30 m flux tower in a *Quercus serrata*-dominated suburban Tokyo forest, with supplemental drone sampling and comparison against MEGAN. Isoprene is **over 90% of measured BVOC composition** at peak summer, monoterpenes stay low with weak vertical gradients, and daily average isoprene emission fluxes span **−0.05 to 15.30 mg m^{-2} h^{-1}**. The methodological result is the one worth carrying: **drone-measured horizontal variability is 10–30% within 30 m of the tower**, and **tower- and drone-derived flux estimates differ by about 30%**. That is a direct, published estimate of the representativeness error a mobile aerial platform introduces relative to a fixed mast — the same quantity a mobile PM platform needs and rarely reports. Limits: BVOCs are SOA precursors, not particles; intermittent campaigns over multiple years confound season with instrument configuration.

doi: 10.5194/acp-26-12521-2026

Ji et al. 2026 Atmos Res

Dust aerosol aging tracked by combined Mie, polarization and fluorescence lidar

Metadata only. Crossref carries the title, author list (8) and 5 September registration date but no abstract, and there is no Europe PMC record; the publisher's full text was not retrievable on this run. From the title the study links optical evolution during dust aging to chemical transformation using a three-channel lidar — a combination relevant to anyone attempting composition inference from optical instruments rather than mass. **No quantities, site, campaign length or retrieval performance are asserted here, because none could be verified.** Added to the abstract retry list.

doi: 10.1016/j.atmosres.2026.109312

Kaur et al. 2026 Aerosol Sci Technol

Plantower PMSX003N against FEM through a Utah winter

Three PMSX003N, three PMS5003T and three OPC-N3 co-located with FEM monitors at a Utah DEQ station for five months, plus laboratory size-binning against monodisperse DOS aerosol. The laboratory result drives everything: the PMSX003N **preferentially assigns particles >1 μm to the 2.5–5 μm bin**. In dust events this recovers coarse mass its predecessor loses — **$PM_{2.5}$ slope 1.85–1.91,** $R^2 > 0.7$ vs **0.15,** $R^2 < 0.37$; **PM_{10} slope 0.91–1.07** vs **∼0.02** — and under freezing fog, mist, snow and rain it substantially overestimates both fractions. Three units per model over one site-season is a small basis for a between-unit claim, and a slope near 1.9 is a large residual bias even where the correlation is good. Humidity-dependent correction is not optional for this part.

doi: 10.1080/02786826.2026.2716649

Kim C-h et al. 2026 Atmosphere

Block-missing monitoring data, five imputation models, and an honest sensitivity test

Eleven months of 20-minute measurements of 22 designated odour compounds at a livestock farm, a wastewater plant and an organic-waste plant in Eumseong, South Korea, with **multi-week to multi-month block gaps** — the failure mode that actually afflicts long-running low-cost networks, as opposed to the scattered single-point dropout most imputation papers assume. Compound-specific performance was assessed by **block holdout** against BiLSTM, BiGRU, TCN, Transformer and HistGradientBoosting. Odour activity values were computed *primarily from observed concentrations*, with the fully imputed series used only as a sensitivity comparison to quantify interpolation-induced bias, which is the correct discipline and is rarely followed. Ammonia was excluded outright for a low valid-observation rate. Odour contribution was dominated by volatile fatty acids and trimethylamine rather than hydrogen sulfide. Transferable to PM networks in method, not in analyte.

doi: 10.3390/atmos17090868

Liu T et al. 2026 Environ Sci Technol

Aerosol solute strength drives TMI-catalysed sulfate aerosol formation

Aerosol flow-reactor experiments on atmospherically relevant aqueous aerosols, decoupling solute strength, interfacial chemistry and Fe(III)–Mn(II) interaction. Sulfate formation from Mn(II) chemistry runs **>2 orders of magnitude faster** than models assuming dilute cloudwater; substantial insoluble Fe(III) suppresses the rate below an ionic-strength threshold of **16 mol kg^{-1}**. Implemented in an atmospheric model, this raises sulfate abundance during polluted haze episodes by **>50%**. The flow reactor cannot reproduce real aerosol organic coatings or mixing state, so the transfer to ambient particles is an extrapolation — but this is a rare case of a laboratory rate measurement closing a known model–measurement gap rather than opening a new one.

doi: 10.1021/acs.est.6c09521

Mohd Nadzir et al. 2026 Atmosphere

Minute-scale trace gas and PM variability in the Antarctic Peninsula from a low-cost node

A single low-cost multi-sensor package, May–June 2023, **58,652 valid observations** of NO_2, O_3, CO, SO_2, $PM_{2.5}$ and PM_{10} (TVOC in May only). Medians: $PM_{2.5}$ **6.29** and PM_{10} **7.29 μg m^{-3}**, O_3 38.17 ppb, NO_2 11.05 ppb, SO_2 0.01 ppb. The **$PM_{2.5}$/PM_{10} ratio of 0.96** indicates an almost entirely fine-mode aerosol, consistent with combustion rather than sea salt or crustal dust. Percentile-based event detection found **142 single- and 30 multi-pollutant enhancements**. The TVOC channel was 98% zeros and was rightly demoted. What is missing is any co-location or drift assessment — at these concentrations a few μg m^{-3} of baseline offset would change every conclusion, and an uncalibrated nephelometric PM channel in high humidity is exactly where that offset lives.

doi: 10.3390/atmos17090855

Raso et al. 2026 Atmosphere

Temporal resolution and the characterisation of episodic pollution events

Two consecutive New Year's Eve firework campaigns with certified reference instrumentation co-deployed with calibrated low-cost sensors, comparing daily, hourly and minute aggregation of PM_{10}, NO_2, CO and O_3. Regulatory averaging detects the episode and misses its structure: minute PM_{10} reached **215 μg m^{-3}**, a **peak amplification factor of 2.90–2.93** over the hourly value. The direct implication for distributed sensing is that the case for a dense network is not redundancy against reference-grade accuracy but access to a time scale the reference framework discards — and that acute-exposure metrics built on hourly data are systematically low for episodic sources. Two events at one site; the PAF is a description, not a transferable constant.

doi: 10.3390/atmos17090860

Rosas et al. 2026 Environ Sci Pollut Res Int

The $PM_{2.5}$/PM_{10} ratio separates two aerosol regimes that mass alone cannot

Continuous $PM_{2.5}$, PM_{10} and particle-bound PAH at a near-coastal urban site in Merida, Mexico, joined to FIRMS fire detections and HYSPLIT back-trajectories, contrasting the March–May regional biomass-burning season against the June–August African dust season. Concentrations are the wrong discriminator: **$PM_{2.5}$ and PM_{10} rise only modestly during dust**, but the **$PM_{2.5}$/PM_{10} ratio falls**, marking the coarse mineral contribution, while biomass burning loads the fine mode and the combustion tracers. A network reporting $PM_{2.5}$ mass alone would score the dust season as the milder of the two and miss the toxicologically distinct composition entirely. Part I of a two-part study, so the health framing in the title is anticipatory rather than delivered here. Single site, single urban domain.

doi: 10.1007/s11356-026-38193-x

Saad et al. 2026 Environ Monit Assess

Short-term air pollutant monitoring at the Alexandria port–urban exposure zone

Two monitoring campaigns, May–October 2022, across 11 in-port and 13 surrounding urban sites, using a portable multi-gas monitor and particulate sensors, with hourly meteorology from Ras El-Tin (campaign-mean wind 3.0–7.2 $m\,s^{-1}$, northerly to north-westerly). Fuel and bulk-fertiliser berths recorded significantly higher CO and NH_3 than urban sites; SO_2 and traffic pollutants outside the port dominate the urban background. **Particulate levels were low across the whole study area** — the authors say so, and the paper is correspondingly framed as screening-level spatial evidence, not exposure assessment. No sensor model, no reference co-location and no uncertainty budget are reported, which caps what the PM numbers can support. Carried mainly because African monitoring records are rare in this corpus.

doi: 10.1007/s10661-026-15828-7

Senarathna et al. 2026 [backfill] Environ Monit Assess

Seasonal and spatial non-transferability of LCS calibration across two tropical climatic zones

TSI BlueSky sensors in a harmonised network across Colombo (wet-zone) and Kandy, compared under linear regression, multiple linear regression and an echo-state network. Harmonised sensors agreed within **10%** of one another. Performance improves monotonically with averaging window: at 24 h, wet season $R^2 = 0.60$, MAE 0.93 $\mu g\,m^{-3}$, MAPE 7.81%; dry season $R^2 = 0.85$, MAE 1.98, MAPE 9.17%. The two transfer results are the point — applying a wet-season model to dry-season data yields **26.57% MAPE**, and a Colombo-fitted model degrades in Kandy, while a Kandy-specific model with temperature and RH reaches $R^2 = 0.84$–0.92. Limitation: no independent reference co-location is reported for the second zone at the same duration, so the spatial-transfer claim rests partly on inter-sensor agreement rather than reference truth.

doi: 10.1007/s10661-026-15623-4

Sengupta et al. 2026 [backfill] arXiv (preprint)

LSTM calibration of low-cost PM and NO_2 sensors against a regulatory equivalence test

Sequence model trained on co-located OxAria reference data in Oxford for $PM_{2.5}$, PM_{10} and NO_2, benchmarked against a Random Forest baseline. The framing argument is that RF treats each hourly observation independently and therefore cannot represent sensor lag or delayed environmental response; the LSTM with time-lagged features, harmonic time encodings and interaction terms beats it on R^2 across train, validation and test. The result that carries weight is the external check: validation against Equivalence Spreadsheet Tool 3.1 gives expanded uncertainties of **9.1% ($PM_{2.5}$), 12.42% (PM_{10}) and 22.11% (NO_2)** — i.e. PM meets, and NO_2 misses, the indicative-monitoring bar. Unrefereed, single network, and no drift analysis over a multi-year horizon, which is where sequence models are most likely to fail.

doi: 10.48550/arXiv.2604.21527

Seong & Hoque 2026 Front Pediatr

Attachment fraction ranks metal > wood > glass, and carpet is two microenvironments

Batch attachment and centrifugal detachment of *Corynebacterium* sp. on carpet, wood, metal and glass, with near-surface velocity fields resolved by particle image velocimetry and surfaces characterised for roughness, contact angle and porosity. Attachment fraction rose with exposure time on all materials, reaching ∼**0.59 on metal,** ∼**0.48 on wood,** ∼**0.36 on glass**. The useful observation is morphological: **carpet generates two distinct microenvironments** — a higher-velocity outer-fibre region and a sheltered low-velocity inter-fibre region — so a single resuspension coefficient for "carpet" is a category error. Wood produced the most spatially heterogeneous velocity field, glass the most uniform. Relevance to PM is indirect but real: resuspension source terms in indoor exposure models are usually fitted per material with no flow-field justification. One organism, no PM measurement, replicate count not stated.

doi: 10.3389/fped.2026.1895305

Shafi & Scafetta 2026 Environ Sci Pollut Res Int

Reconstructing a decade of missing Delhi monitoring records

Delhi went from **4 stations in 2014 to 45 in 2024**, leaving a fragmented record that defeats trend analysis. For each missing daily value across six pollutants, the workflow selects the four most-correlated stations as predictors, evaluates **35 regression models** in MATLAB's Regression Learner, takes the best, and iterates until the matrix is full. Tree ensembles and Gaussian/rational-quadratic kernels consistently beat multilinear regression; artificial-gap experiments validate across pollutants, stations and years, best for $PM_{2.5}$ and other regionally-driven species. The structural problem is unaddressed and important: stations are missing non-randomly, and imputing from correlated neighbours smooths exactly the local excursions that a network is deployed to detect. Reconstructed series should not be used for exceedance statistics.

doi: 10.1007/s11356-026-38181-1

Shi S et al. 2026 npj Clim Atmos Sci

Vessel speed reduction trades CO_2 against plume composition — and against the tracer that identifies the plume

UAV interception of 14 fresh ship plumes with single-particle mass spectrometry and population-balance inversion to reconstruct ultrafine and coarse modes. Two marker gaps emerge. **Only 35.8% of the reconstructed particle-number population carried detectable vanadium** (29.5% at 4.9 knots vs. 39.7% at 13.4 knots in a matched-vessel pair; 0–70.8% across a speed-blind manifold), so the canonical ship tracer is operation-dependent rather than conserved. Separately, ∼**20% of analysed particles met biomass-burning criteria**, a false-positive channel in ship-influenced samples. Sulfate-related ion signals ran 2.3–10.7-fold higher in V/VO-bearing particles. The authors are careful that laser desorption/ionization matrix effects may contribute to that contrast, which is the right caveat and limits how far the V–S co-location can be pushed mechanistically. Single campaign, 14 plumes, one matched pair — the speed comparison rests on very few vessels.

doi: 10.1038/s41612-026-01532-3

Veselovskii et al. 2026 Atmos Meas Tech

Five-channel fluorescence lidar retrieves smoke mass inside the boundary layer

Smoke and urban aerosol have distinct fluorescence spectra, and a multi-channel fluorescence lidar exploits that to separate them and retrieve smoke mass concentration with an estimated detection threshold of ∼**0.1 $\mu g/m^3$**. Over Moscow, smoke was detected in **59 of 67 sessions between April and October 2024**, mostly from long-range Atlantic transport; only 12 episodes traced to southern Russian fires. Within the PBL, transported North American wildfire smoke contributed ∼**1 $\mu g/m^3$** against **up to 50 $\mu g/m^3$** for regional fires. This is the source-resolved column-to-surface capability that mass-only networks lack, and 0.1 $\mu g/m^3$ is a genuinely low threshold. Single site, single instrument, and mass retrieval requires an assumed fluorescence-to-mass conversion whose uncertainty is the main open term.

doi: 10.5194/amt-19-5587-2026

Wang S et al. 2026 Fire Saf J

Smoldering chaparral emits 10–70× the particulate of flaming, and fuel moisture sets the multiplier

FTIR quantification of gas and particulate emissions from controlled flaming and smoldering combustion of four dominant California chaparral species at varying fuel moisture. **Smoldering produced 10–70-fold higher PM emission factors than flaming across all species**, with the amplification ratio itself varying by fuel type and moisture; higher moisture enhanced incomplete-combustion products, principally CO and hydrocarbons, with live eucalyptus highest at **37.18 ± 0.65 g/kg** hydrocarbons under smoldering. Cross-ecosystem comparison found chaparral broadly similar to western shrubland and grassland fuels except in nitrogen-containing species. The limitation that matters is scale: these are bench-scale burns, and the authors note combustion scale and mixing conditions shift emission profiles, so transfer to landscape fire inventories is an assumption rather than a result. Directly useful for anyone weighting wildfire smoke exposure by combustion phase rather than by total burned area.

doi: 10.1016/j.firesaf.2026.104687

Xing et al. 2026 Environ Sci Technol

DeepAQM — observation-constrained ML surrogate for the NOAA forecast system

A ConvLSTM–ResNet fusion reconstructs spatially continuous surface $PM_{2.5}$ and O_3 from VIIRS AOD, TEMPO NO_2 and EPA AirNow, and supplies them as initial conditions for 72 h UFS-AQM forecasts over the continental US. It reproduces UFS-AQM spatial patterns ($R^2 > 0.9$ for O_3, to 0.80 for $PM_{2.5}$, normalised mean bias within ±10%). The result that matters: observation-updated initial conditions lift site-level R^2 in the first 10 forecast hours from ∼**0.10–0.15 to 0.40–0.50 for $PM_{2.5}$**, and observation-based fine-tuning carries overall R^2 to **0.41 for $PM_{2.5}$** across the full 72 h. Note what that number is: a fourfold gain at short lead, decaying to a forecast that still explains under half the variance. TEMPO makes this a current-generation result rather than a reanalysis exercise.

doi: 10.1021/acs.est.6c04600

Yohanna & Lim 2026 Chemosphere

Regional and seasonal $PM_{2.5}$ and ozone variation across Malaysia, XGBoost + SHAP

Daily DOE network observations 2018–2023 for five Malaysian regions, augmented with ERA5 boundary-layer height, solar radiation and cloud cover; XGBoost against a multiple linear regression baseline, interpreted with SHAP. **$PM_{2.5}$ R^2=0.576, RMSE 4.87 μg m^{-3}**; O_3 R^2=0.543, RMSE 3.4 ppb — rising to R^2=0.85 only when persistence is included, which is the number to be sceptical of, since a persistence term will carry most of the skill in any daily-mean series. Kruskal–Wallis confirms regional (χ^2=3239.5) and seasonal (χ^2=477.8) structure. SHAP attributes PM variability to combustion tracers and meteorology, ozone chiefly to humidity and temperature. Useful as a tropical-monsoon reference for anyone tuning a regional model; not a calibration result.

doi: 10.1016/j.chemosphere.2026.145074

Zeng et al. 2026d J Hazard Mater

Oxidative potential and light extinction decouple, and aerosol water is the switch

Paired oxidative-potential and optical measurements at two urban sites in southern Taiwan, arranged into a quadrant framework on *volume-normalised* OP against light extinction. Under coupling, mass-normalised OP and mass extinction efficiency share their drivers, principally **water-soluble organic carbon**. Under decoupling, **high aerosol liquid water content** pushes the aerosol into an optics-dominant state through hygroscopic growth, while **low ALWC** yields the highest OP-to-extinction ratios — the "invisible killer" quadrant. Secondary nitrate loads extinction specifically; aerosol pH is a weaker secondary term. The direct consequence for distributed sensing: every nephelometric low-cost sensor estimates mass from scattering, so its error against the redox-active fraction is not random but conditioned on humidity, in the same direction as its already-known RH bias. Weakness: OP is one assay on one aerosol population at two sites, sampling duration is not stated in the abstract, and no causal chain from OP to a clinical endpoint is tested here.

doi: 10.1016/j.jhazmat.2026.143459

Zhang Z et al. 2026 Anal Chem

Microfluidic multispectral diffraction for black carbon quantification

A microfluidic chip does inertial separation, low-velocity sheath-flow regulation and high-voltage electrostatic capture to enrich submicron BC in the imaging region (**92% enrichment efficiency**); a lensless diffraction system then acquires hyperspectral fingerprints over 400–800 nm, which a graph model maps to concentration. Over **0.01–0.20 mg/m^3**, R^2=0.98. The discriminating claim is more interesting than the calibration: the spectral fingerprint separates BC from similarly-sized microplastics and pathogenic spores, which absorption-based instruments cannot do. Bench work only — no ambient co-location, no interference testing against real coated or aged BC, and 0.01 mg/m^3 (10 μg/m^3) is a high floor for ambient BC, which typically runs 0.5–5 μg/m^3.

doi: 10.1021/acs.analchem.6c03051

Exposure assessment & modelling 47 records

Carried from the daily issues — full summaries (47 of 47 records in this subtopic)

Arpornwichanop & Chuersuwan 2026 Environ Geochem Health

Life-stage-resolved inhalation exposure to $PM_{2.5}$-bound elements, Greater Bangkok

Secondary-data screening at one fixed site (selected 24 h filter days, Dec 2020–Sep 2021), combining seasonally weighted outdoor elemental concentrations with indoor penetration, time-activity, respiratory-deposition and simulated-lung-fluid solubility assumptions across pediatric life stages. Concentration and deposited-intake profiles rank **Al, Fe, B, Zn** highest; the solubility- and toxicity-adjusted comparative priority quotient ranks **As, Cd, Sb**. Modelled lifetime cancer risk spans **1.11×10^{-6} to 2.78×10^{-5}** across 0–100% Cr(VI) assumptions — an order of magnitude driven entirely by a speciation parameter nobody measured. The authors state explicitly that this is within-model prioritisation, not measured child exposure. That honesty is unusual and worth noting.

doi: 10.1007/s10653-026-03455-7

Arunachalam et al. 2026 Research Square (preprint)

Building-sector combustion mortality is three times prior estimates, and it is mostly NO_2

CMAQ Decoupled Direct Method sensitivities of NO_2, O_3 and $PM_{2.5}$ to commercial and residential building-sector precursor emissions for the contiguous US (2021), propagated through BenMAPR. Net burden ≈**14,000 premature deaths per year** (**95% CI 7,900–19,000**): **12,000 from NO_2**, **3,100 from $PM_{2.5}$**, and **1,000 deaths avoided** through O_3 reductions. Residential gas alone accounts for ≈**7,900**. The headline "three times higher than previously reported" is carried almost entirely by the NO_2 term, so the result is a statement about which NO_2 concentration–response function you accept, not about building emissions being larger than thought. Preprint, not peer reviewed. The negative O_3 term is the sign of a titration effect and is correctly reported rather than suppressed, which is a point in the paper's favour.

doi: 10.21203/rs.3.rs-10758198/v1

Boogaard et al. 2026 Curr Environ Health Rep

Advancing long-term outdoor air pollution exposure assessment for health studies

Overview of the completed HEI multinational initiative on novel long-term exposure assessment across North America and Europe. Exposure predictions from models that account for daily mobility patterns and outdoor–indoor infiltration correlated $>$**0.7** with conventional residential estimates, and the health effect estimates they produced were similar — differing in magnitude but *not direction*. The paper is honest that this is a comparison among models, not a validation against measurement: representative validation data for long-term exposure simply does not exist, which is the initiative's own headline gap. Read as the strongest current argument for spending resources on validation campaigns rather than additional model structure, and as a caution against over-reading small differences between LUR variants in cohort re-analyses.

doi: 10.1007/s40572-026-00560-8

Boyle et al. 2026 Cancer Epidemiol Biomarkers Prev

Cancer-specific air-pollution exposome metrics for the continental US

Combines CACES land-use-regression criteria-pollutant surfaces with the RSEI industrial-release model across US census tracts, 2005–2020, weighting each pollutant by meta-analytic breast- and lung-cancer risk estimates to produce two composite metrics (APBC, APLC). Both are strongly right-skewed — some tracts an order of magnitude above the mean — peak in large cities and along transport corridors, differ from RSEI notably in Los Angeles and New York, vary by race, ethnicity and housing, and cluster spatially. No cancer outcome is analysed here: this is metric construction, and the meta-analytic weights import the heterogeneity of their source studies into a tract-level surface that then reads as precise. Useful as a template for building a risk-weighted composite from a network, with that caveat attached.

doi: 10.1158/1055-9965.EPI-26-0493

Bravo et al. 2026 J Expo Sci Environ Epidemiol

Chicago's 2023 wildfire smoke, by tract

Hourly $PM_{2.5}$ on a **500 m** grid aggregated to daily 24-hour means for all **792 Chicago census tracts** in nine city regions, 1 May–31 July 2023, examined against demographic, socio-economic and health profiles. The title carries the finding: during the long-range wildfire-smoke episodes, the usual within-city exposure gradient compressed — smoke reached every region — while the pre-existing disparity in baseline $PM_{2.5}$ and in underlying health persisted underneath it. That combination is the argument for source-resolved rather than total-mass exposure metrics in environmental justice work, since a total-mass summer average would show convergence and hide the differential burden. Modelled rather than measured exposure at tract level, one summer, one city, and no health outcome is linked here.

doi: 10.1038/s41370-026-00956-6

Chi et al. 2026 Atmos Chem Phys

Heterogeneous nitrosation of amines by N_2O_4: a missing source of particulate nitrosamines

Born–Oppenheimer molecular dynamics plus metadynamics on methylamine and dimethylamine reacting with N_2O_4 at the air–water interface. Two competing routes: a *barrierless* direct N-nitrosation yielding nitrosamine cations and NO_3^-, and an amine-mediated N_2O_4 hydrolysis generating interfacial HONO in **2–16 ps**, which then nitrosates over a 7.65 $\mathrm{kcal\,mol^{-1}}$ barrier at 300 K. Purely computational, two amines, idealised interface — there is no measurement here and no yield that a chemical transport model can use directly. Its value is as a mechanism for particulate carcinogen formation that current CTMs omit entirely.

doi: 10.5194/acp-26-12261-2026

Engeroff et al. 2026 Research Square (preprint)

PM exposure while running: park, residential and roadside routes by time of day

Mobile personal PM_{10}, $PM_{2.5}$ and PM_1 over **60 instrumented 2 km runs** (20 each on a major road, a residential street and an urban park; 30 morning, 30 midday; pace 8:02–4:31 min/km) with local urban background as reference and linear mixed-effects models adjusting for meteorology. Lowest concentrations of every fraction: **midday in the park**. Highest PM_{10} anywhere: **morning on that same park route**. Exposure across all routes was lower at midday, tracking urban background, and humidity, pressure, wind speed and temperature all carried signal. The conclusion — park routes and rush-hour avoidance are sufficient — is reasonable but rests on 20 runs per route in one city, and the fact that the park produced both the minimum and the maximum should temper any general route advice. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10016764/v1

Gao J et al. 2026 Environ Sci Technol

Are inverse ozone–health associations an artefact of measuring the wrong oxidant?

A perspective, not a study. Population studies not infrequently find *lower* O_3 associated with *higher* disease risk. NO titration near traffic and differential measurement error are the standard explanations and do not fully account for it. The alternative advanced here: low measured O_3 can mark locations and times of active oxidative chemistry — O_3 being consumed, or parallel oxidant pathways running — which generate toxic secondary products that no monitoring network measures. If so, O_3 is functioning as a negatively-signed proxy for an unmeasured oxidant field. The same logic transfers directly to PM mass as a proxy for oxidative potential, which is why this is carried in a PM watch. No data and no test proposed beyond a call for expanded measurement, so it stands or falls on plausibility.

doi: 10.1021/acs.est.5c17322

Guo & Sudo 2026 Atmosphere

Global online PAH simulation in the CHASER chemistry-climate model

First global online simulation of naphthalene, phenanthrene, pyrene and benzo[a]pyrene in CHASER V4.0, coupling soot adsorption $K_{SA}(T)$ with organic-matter absorption $K_{OA}(T)$ under an extended Dachs–Eisenreich dual-mode scheme. Evaluated at urban, background and Arctic stations: pooled spatial R **0.78** on a $\log_{10}$ scale, **FAC2 80%** for BaP excluding sub-grid hotspots, seasonal R 0.62–0.93. The K_{SA} term fixes the long-standing under-prediction of particle-bound fraction at cold high latitudes, raising polar ϕ_{BaP} by up to $\sim$0.6, and elevated elemental carbon extends BaP lifetime to **1.72 d** against 0.57–0.72 d for lighter congeners, sustaining transoceanic outflow. The dependence on modelled EC is also the weak point: the partitioning result is only as good as the soot field underneath it.

doi: 10.3390/atmos17090861

Guo et al. 2026b Atmos Chem Phys

Kinetic isotope fractionation framework for sulfate source apportionment

Seasonal $PM_{2.5}$ observations in Nanjing combined with Bayesian isotope mixing and process-specific Rayleigh fractionation. Conventional isotope models assume complete SO_2 oxidation and therefore discard the kinetic isotope effects generated during incomplete processing; correcting for them shows transition-metal-ion-catalysed and NO_2-mediated pathways dominate secondary sulfate, and moves apportionment by $-$**10% coal combustion,** $+$**8% traffic** during summer photochemical episodes. Single-city, single-year, and the Rayleigh treatment assumes a closed reservoir, which is a strong assumption for an urban boundary layer. The direction of the correction nevertheless agrees with Liu T et al. in the same issue, from wholly independent evidence.

doi: 10.5194/acp-26-12231-2026

Gürler Akyüz & Akyüz 2026 J Environ Sci Health A

Size-resolved elemental composition in a coastal urban-industrial region

Seasonal $PM_{2.5}$ and $PM_{2.5-10}$ elemental composition at Zonguldak, Türkiye, where coal mining, coal combustion, iron-steel with coke production, maritime fuel oil, traffic and marine/crustal sources overlap. Heating-period $PM_{2.5}$ carries elevated **As, Sb, Cd, Pb, Se, V, Ni, Cr**; non-heating $PM_{2.5-10}$ is crustal, marine and resuspended dust with a persistent industrial residue. Enrichment factors against Ti show strong anthropogenic enrichment for combustion and metallurgy and near-natural levels for crustal and marine elements; PCA separates coal combustion, iron-steel, fuel-oil/shipping, non-exhaust traffic, crustal dust, marine aerosol and coal fly ash, and discriminant analysis confirms the season/size separation. Conventional in method, valuable as a composition baseline for a source mixture that is rarely characterised together.

doi: 10.1080/10934529.2026.2724190

Hernández Pardo et al. 2026 Atmos Chem Phys

Downward transport of tropical upper-tropospheric aerosols

Multi-year global EMAC simulations releasing 19 idealised tracers in the tropical upper troposphere under resolved advection, parameterised convection, turbulent mixing and deposition. Age of air at 500 hPa is $\approx$45 d for tropics-wide sources but >250 d for regional continental ones — transport diagnostics are dominated by source extent, not by transport physics. Advection dominates; injection height matters more than particle size in the 20–100 nm range. Tracers are idealised, carry no mass or composition, and there is no surface validation, so this constrains *where* upper-tropospheric particles arrive, not *how much* they contribute to surface PM. Carried at tier C for that reason.

doi: 10.5194/acp-26-12275-2026

Hulme et al. 2026 Environ Res Lett

Home and school exposure, jointly, for 919 Accra schoolchildren

Land-use regression annual means for $PM_{2.5}$, NO_2 and BC plus L_{day}, L_{night} and L_{den} noise, linked to geocoded home and school points for **919 of 1,034** children aged 8–12 across 90 Accra schools, then combined into a time-location-weighted total. Median home $PM_{2.5}$ **33.8 μg/m^3**, NO_2 55.5, BC 5.2×10^{-5} m^{-1}; school values slightly lower (29.9 / 53.5 / 4.9). Time-weighted totals exceeded WHO guidance for most children, and noise exceeded WHO road-traffic thresholds at home (L_{den} 66.3) and Ghana's educational-area standard at school. The joint air-plus-noise, home-plus-school design is the contribution — both are under-measured in sub-Saharan Africa. Annual LUR means cannot resolve the school day, so the time-weighting is coarser than it looks.

doi: 10.1088/1748-9326/ae8d40

Klebach et al. 2026 Atmos Chem Phys

MSA and sulfuric acid in tropical Indo-Pacific aerosol from aircraft

CAFE-Pacific, HALO aircraft, January–February 2024, nitrate chemical-ionisation mass spectrometry of methanesulfonic and sulfuric acid. Marine boundary layer concentrations reach **4×10^7 cm^{-3} MSA and 6×10^7 cm^{-3} SA**; the MSA/SA ratio increases with altitude in the lower free troposphere, consistent with temperature-dependent DMS oxidation. Two flights show marine deep convection lofting DMS to the upper troposphere with particle formation and growth after **10–20 h of OH exposure**, matching the kinetically modelled DMS lifetime. MSA frequently exceeds SA, which the current generation of new-particle-formation schemes does not anticipate. Aircraft inlets heat and evaporate acidic particles, so the gas/particle split at altitude is instrumental as much as atmospheric — the authors are explicit about this.

doi: 10.5194/acp-26-12355-2026

Kong J et al. 2026 J Environ Sci (China)

Triplet-state brown carbon as an endogenous oxidant

Photochemistry of 1,4-naphthoquinone as a brown-carbon proxy in deoxygenated aqueous systems under 355 nm irradiation, resolving two long-lived $^3(n,\pi^*)$ triplets (^{3}NQ* and ^{3}OH-NQ*). ^{3}NQ* abstracts hydrogen from organic substrates **without any external oxidant**, with a rate hierarchy set by bond dissociation energy and one-electron oxidation potential: alkanes $\sim10^4$, oxygenated alkanes $\sim10^5$, phenols $\sim10^6$ M^{-1} s^{-1}. Self-photoconversion regenerates a second triplet, amplifying the oxidative cascade. The claim is that ^{3}BrC* is the dominant phenol sink in BrC-rich haze under acidic, O_2-limited conditions — which is exactly the condition inside a haze particle and exactly the condition current models do not represent. Proxy compound, deoxygenated system; the rate constants are the transferable output.

doi: 10.1016/j.jes.2025.10.037

Lai et al. 2026 Environ Sci Technol

Two-thirds of ambient organosulfates have no assigned source

Organosulfates contribute up to **30% of aerosol mass**. Of **366** ambient OS molecular formulas compiled from field studies, only about **36%** map to a known precursor or formation mechanism, leaving **64% unclassified**. The proposal here is that some unclassified species are not freshly formed at all but higher-generation products of heterogeneous and aqueous multiphase transformation of existing OS: identified transformation products already account for $\sim$**17%** of reported formulas and explain an additional **5%** of the unclassified set. Modest arithmetic, but it reframes a source-apportionment problem as an aging problem. Compilation-based, dependent on the sampling and HRMS coverage of the underlying field studies, and 5% is a small demonstration of a mechanism claimed to be general.

doi: 10.1021/acs.est.6c08394

Lei et al. 2026 Atmos Environ

Source transition in $PM_{2.5}$-bound metals in a coastal megacity

Ships on metadata only — no abstract in the Crossref deposit and no other index carries the record on this run. The title indicates a multi-year, high-time-resolution observational record of $PM_{2.5}$-bound metals with a source-transition analysis and a health-implication component. Counted and listed; no numbers claimed. High retry priority: multi-year hourly metals is exactly the dataset class against which a low-cost network's composition proxies would have to be validated, and there are very few of them.

doi: 10.1016/j.atmosenv.2026.122336

Leung et al. 2026 Indoor Air

Abandoned wells, indoor VOCs, and a pilot honest enough to report its own limits

Passive indoor sampling for 13 VOCs in **17 households** (**October 2024**) and **15 households** (**February–March 2025**) in southwestern Ontario, with abandoned-well exposure estimated by inverse-distance-squared weighting over wells within 3 km and self-reported acute symptoms over the previous six months and during sampling. This is gas-phase, not particulate — carried here because the exposure-assignment design is the one a distributed indoor PM study would use and because the MAPSH project measures both. At $n<20$ per round with Spearman correlations and no personal monitoring, the study is powered to describe indoor concentrations against Canadian guidelines and not to detect a well-proximity association; the six-month symptom recall window compounds that. Read as a feasibility and concentration-range report.

doi: 10.1155/ina/8041546

Li Q et al. 2026 Environ Sci Technol

Health risk of fine PM across China with a pollutant-specific index, 2013–2023

A 1 km daily $PM_{2.5}$ surface for 2013–2023 built by machine learning (cross-validated R^2 **0.88 spatially, 0.82 temporally**), fused with meta-analysed short-term effect estimates into an $AQHI_{PM2.5}$. National mean fell **55.06 → 26.50 μg m^{-3}** (2013→2022) then rebounded to 27.46 in 2023. Per 10 μg m^{-3} over 3 d: **+0.42% all-cause, +0.61% cardiovascular, +0.84% respiratory mortality**. Northern and southwestern regions retain >80 medium/high-risk days a year; eliminating them would avert 6,985 (6,419–7,588) deaths, 7.42% of the attributable burden. The weak point is the meta-analysed exposure — response step, which imports between-study heterogeneity into a spatially resolved index and then reports it at 1 km as if the precision were the model's. For sensing work: the AQHI construction is directly portable to a dense network, but only if the network can supply the exposure surface it assumes.

doi: 10.1021/acs.est.6c00144

Liu S et al. 2026 J Environ Sci (China)

Terrain blocking and monsoon coupling in North China Plain $PM_{2.5}$ and O_3

Quantifies mountain-blocking effects on pollutant dispersion in the semi-closed North China Plain now that anthropogenic emissions have fallen far enough for terrain and meteorology to dominate the residual pattern. Within 120 km buffers of the major ranges, concentrations rise with altitude and then fall sharply above a threshold; the **effective blocking altitude is 22–929 m for winter $PM_{2.5}$ and 19–1060 m for summer O_3**. Concentrations rank plains > platforms > hills > mountains, and winter $PM_{2.5}$ and summer O_3 high-value zones aggregate in *opposite* north–south directions. Descriptive rather than causal — emissions and terrain co-vary strongly in the NCP — but the altitude thresholds are a concrete siting constraint for any regional network.

doi: 10.1016/j.jes.2026.01.013

Liu S et al. 2026 Environ Pollut

Wind-shear regimes and the vertical $PM_{2.5}$ profile in a basin

Six instrument types — surface network, wind profiler radar, microwave radiometer, aerosol lidar, radiosonde and UAV — plus WRF, over Taiyuan Basin. Unsupervised clustering gives three vertical-wind-shear regimes: strong directional near-surface shear with near-surface accumulation and the shallowest boundary layer; speed shear above 500 m with an *elevated* $PM_{2.5}$ maximum near that height; and deep shear below 2000 m with stronger winds and low concentrations. UAV profiles over 0–500 m confirm the three shapes. The 17 January 2024 episode makes the operational point: strong shear did not enhance vertical exchange while the shear layer sat above a shallow stable layer, and only coupled once daytime heating deepened the boundary layer into it. Directly relevant to siting — a surface monitor cannot see regime 2 at all. One basin, one winter.

doi: 10.1016/j.envpol.2026.129103

Mutic et al. 2026 medRxiv (preprint)

Paired silicone wristbands across home and early-childhood-education settings

A *protocol*, not results. Prospective repeated-measures pilot targeting **44 preschool children and 8 staff** across two sociodemographically distinct Atlanta early-childhood centres, with home- and centre-designated silicone wristbands exchanged between settings over three consecutive days and nights, one spot urine per child, and continuous indoor air-quality monitoring in two classrooms per site. Eight pre-specified feasibility indicators. The design premise is sound and underexploited — children spend >90% of time indoors and early-childhood centres are high-occupant-density microenvironments almost absent from the exposure literature — but wristbands integrate semi-volatile organics, not particle mass, so the PM component rests entirely on the fixed classroom monitors. Listed for the design; nothing to conclude from it yet.

doi: 10.64898/2026.08.27.26361470

Odu-Onikosi et al. 2026 Atmos Pollut Res

$PM_{2.5}$-bound toxic elements in Lagos

Ships on metadata only, same Elsevier deposit gap as above: no abstract, not in PubMed or Europe PMC, ScienceDirect blocked. Sampling duration, element list, apportionment method and risk estimates are all unretrievable at digest time. Counted and listed; no numbers claimed, and flagged for the abstract retry list. Noted because the author list includes P. A. Solomon and P. K. Hopke, and because speciated $PM_{2.5}$ from Lagos — a megacity of roughly 15 million with almost no published speciation record — is exactly the geographic gap the two African records above document.

doi: 10.1016/j.apr.2026.103193

Pachés et al. 2026 Atmos Pollut Res

Regional background $PM_{2.5}$ in the Valencian Community by clustering

Ships on metadata only. The Elsevier deposit carries no abstract, the record is not yet in PubMed, Europe PMC or Semantic Scholar, and ScienceDirect refused retrieval from this sandbox, so the network size, the clustering method and the resulting background estimates are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list. Background/local-increment separation is directly relevant to siting a low-cost network — the increment is what a dense network is deployed to resolve — so this is worth revisiting once the abstract deposits.

doi: 10.1016/j.apr.2026.103192

Papetta et al. 2026 Atmos Chem Phys

Source-dependent optical and mineral signatures of Mediterranean dust outbreaks

Four major 2021–2022 dust outbreaks from the Eastern, Western and Central Sahara and the Middle East, tracked across **24 AERONET stations** with IASI and MODIS MIDAS dust optical depth and HYSPLIT back-trajectories. Saharan events were coarse and scattering; the Middle East event carried **finer, more absorbing particles**, likely anthropogenically influenced. DOD-to-AOD ratio separates them — only one East-Central Saharan event stayed above 0.8 (near-pure dust), while the Middle East event had the lowest ratios, indicating mixing with anthropogenic or marine aerosol. A Cyprus in-situ case study with elemental and absorption measurements anchors the remote-sensing inference. For network work the transferable idea is the DOD/AOD purity screen: it is a cheap way to decide whether a coarse-mode excursion is genuinely dust before attributing it.

doi: 10.5194/acp-26-12395-2026

Petersen et al. 2026 Environ Sci Technol

Sulfate ions in aerosol water are their own source of radical oxidants

Direct spectroscopic evidence that the sulfate anion absorbs surface-level sunlight and ejects an electron, forming **$SO_4^{\bullet-}$ on a sub-microsecond timescale** in a concentration-dependent manner; long-timescale production rates quantified by radical trapping with LC-HRMS. Compounds emitted from biomass fires photosensitise the pathway, raising radical formation **up to five-fold** over direct photolysis across pH 1.6–5.5, more strongly at low pH. Estimated $SO_4^{\bullet-}$ production is **comparable to measured hydroxyl-radical production in aerosols** — i.e. a second oxidant budget nobody was counting. The downstream claim is the striking one: secondary brown carbon from this chemistry **turns the direct radiative forcing of sulfate-containing aerosol from negative to positive**, and the organosulfates formed are cloud-active. Bulk solutions at controlled ionic strength; extrapolation to real aerosol water is the open step.

doi: 10.1021/acs.est.6c05704

Rahmani et al. 2026 Rev Environ Health

Airborne microplastics indoors and outdoors: pooled concentrations and the instrument problem

PRISMA systematic review to October 2023: 27 eligible studies across 16 countries, 21 pooled under a random-effects model. Indoor concentrations exceed outdoor — moderately polluted indoor environments 1,380.28 vs. 102.45 MP m^{-3} outdoors — with indoor healthcare settings and outdoor urban areas the most contaminated sub-environments. The authors present the pooled figures as a descriptive summary rather than an estimate, correctly: heterogeneity is $\mathbf{I^2 > 99}$% and the dominant moderator is **detection instrument**, with FTIR reporting consistently higher counts than microscopy or Raman spectroscopy; sampling duration also matters. Sensitivity analysis excludes single-outlier dominance. For instrumentation work this is a clean external demonstration that method-driven variance can exceed environmental variance, which is exactly the risk in heterogeneous LCS fleets.

doi: 10.1515/reveh-2025-0059

Ren et al. 2026 iScience

Vertical contrast in $PM_{2.5}$ composition and secondary formation, Shiyan, central China

One year across three urban sites and one mountain site. Annual mountain $PM_{2.5}$ is **70% of urban** levels, but sulfate–nitrate–ammonium is a *larger* fraction there (49.3%). SNA is vertically near-homogeneous in spring, summer and autumn (coefficient of divergence <0.2) and stratifies only in winter, when most components are urban-enriched. Higher nitrogen oxidation ratio (0.27) and SOC/OC (0.65) at altitude indicate a more aged aerosol. Limitation: a single mountain site cannot separate altitude from the site's own upwind exposure, and no vertical profiling instrument is used — the "vertical distribution" claim rests on a two-point contrast. Relevant to siting logic for distributed networks: composition, not mass, is what varies with height here, and a mass-only node would report the mountain site as simply cleaner.

doi: 10.1016/j.isci.2026.117090

Sawan et al. 2026 Environ Sci Technol

Aviation drives ultrafine number, freight drives black carbon, in the same neighbourhood

Hybrid mobile and fixed-site campaign in a marginalized northwest Toronto neighbourhood downwind of Toronto Pearson and a freight corridor. UFP correlated with flight activity ($r = 0.58$, $p < 0.05$ **at 3.5 km**) and was enriched **1.7–3.3-fold** downwind of the airport, measured as the ratio of interquartile ranges under downwind versus other wind conditions, reaching **3.2-fold at residential sites ~8 km away**. BC instead tracked roadway proximity and freight and correlated weakly with flight counts. BC–UFP correlation was strong under northerly winds (mean $r = 0.62$) but collapsed to $r = 0.1$ downwind of the airport. Random Forest models ($R^2 = 0.79$ UFP, 0.87 BC) identified industrial land use and a highway as shared drivers, airport proximity as UFP-specific and freight trucking as BC-specific. The inference is observational and wind-sector stratified, so airport attribution rests on a co-location argument rather than on any tracer unique to jet exhaust. The lesson for network design is that mass-based siting would have missed the aviation signal entirely.

doi: 10.1021/acs.est.6c06666

Scalabrin et al. 2026 Atmos Pollut Res

Chemical tracers for tire wear particles in atmospheric aerosol: analytical approaches and challenges

Metadata only. Crossref registers the title, 11 authors and a 5 September creation date with no abstract, and the record has no Europe PMC counterpart. The title places it as a review of tracer chemistry for apportioning non-exhaust tire wear — a live problem, since tire wear is a growing share of traffic PM precisely as exhaust controls tighten, and the standard markers each fail differently. **No findings, recommended tracer or study count are asserted here.** A targeted Scholar Gateway full-text query returned only adjacent June 2026 reviews, not this paper. Added to the abstract retry list.

doi: 10.1016/j.apr.2026.103194

Schaefer et al. 2026 Atmos Chem Phys

Deep convection sorts ice-nucleating particles by activation temperature

Immersion-mode INP filter samples from the HALO aircraft up to **14.5 km** during CIRRUS-HL (summer 2021), taken in the inflow and outflow of deep convective clouds and from cloud-particle residuals. Inflow spectra contain INPs active at both high ($T > -15°C$) and low ($T < -20°C$) temperatures; **outflow spectra contain only the low-temperature population**. The reading is precipitation scavenging in the updraft removing the warm-active INPs while the cold-active ones are carried efficiently into the free troposphere. Cloud microphysics rather than exposure science, carried for the particle-transport result: convection is a size- and composition-selective vertical filter, which matters for any attempt to relate column-integrated aerosol to surface composition. Offline analysis, filter sampling, one campaign.

doi: 10.5194/acp-26-12505-2026

Solaiman et al. 2026 Environ Sci Atmos

Nighttime NO_3 oxidation of myrtenal: SOA yields and HOMs

Chamber study of secondary organic aerosol from the gas-phase oxidation of myrtenal, a biogenic aldehyde, by NO_3 radicals, reporting SOA yields, chemical composition and highly oxygenated molecules. The Crossref deposit truncates the abstract mid-sentence and no full record was retrievable from Semantic Scholar on this run, so the yields and HOM distributions cannot be quoted. Counted and listed; the nighttime NO_3 channel for oxygenated biogenics is under-represented in this watch's corpus and the record is worth revisiting once the full abstract deposits.

doi: 10.1039/d6ea00092d

Stergiou et al. 2026 Atmos Pollut Res

Spatiotemporal graph neural network bias correction for CMAQ over CONUS

Ships on metadata only. Crossref carries no abstract for this DOI and the record is not in PubMed, Europe PMC or Semantic Scholar as of this run, so nothing beyond title, authors and venue can be attributed to it. Counted and listed; no numbers claimed. Flagged for the abstract retry list — graph-structured bias correction of a regulatory CTM against monitor networks is directly on this watch's topic and should be summarised once the abstract deposits.

doi: 10.1016/j.apr.2026.103191

Ullah et al. 2026 Environ Pollut

Transboundary smoke moves almost nobody across China's standard and almost everybody across WHO's

Daily $PM_{2.5}$ from **51 cities** in Yunnan, Guangxi and Guangdong for March–April 2015–2019, gridded onto WorldPop, with the Indo-China Peninsula fire contribution isolated by paired GEOS-Chem runs with and without the fires. The population-weighted share of days above the **WHO 2021 24 h guideline (15 μg/m^3) reached 78–96% every year**; above **China's Grade II standard (75 μg/m^3) it was 1–6%**, an average daily exposed population of about **6 million**, under 3% of the corridor's residents. The two benchmarks describe different worlds from identical data, which is the paper's real contribution to monitoring policy. Principal weakness: the attribution runs at **2° × 2.5°**, roughly 200 km, so the fire contribution is a regional average imposed on city-scale observations and cannot support any within-province ranking.

doi: 10.1016/j.envpol.2026.129081

Wang J et al. 2026 J Environ Sci (China)

Water-soluble aerosol matter photosensitises guaiacol into aqueous SOA

Mechanism and kinetics of aqSOA formation from guaiacol mediated by photosensitive water-soluble inorganic and organic matter extracted from real aerosol. Photolysis of the extract generates both reactive oxygen species and triplet WSOM; **^{3}WSOM* is the predominant initiator** of guaiacol transformation, with steady-state concentration **5.35 ± 0.12 $\times 10^{-13}$ mol/L**, generation rate 3.43×10^{-7} mol L^{-1} s^{-1}, quantum yield **4.86%** and a rate constant with guaiacol above 10^9 L mol^{-1} s^{-1}. ROS then convert soluble intermediates into the insoluble ethers and low-volatility phenols that constitute the aqSOA. Bulk aqueous extracts are not cloud or particle water — ionic strength differs by orders of magnitude — which is the standing limitation of this whole method.

doi: 10.1016/j.jes.2025.11.050

Wang T et al. 2026 J Environ Sci (China)

IAQMS-Industry: 100 m $PM_{2.5}$ inside industrial parks

Couples the regional NAQPMS to a city-scale chemical transport model with explicit point-source locations and Gaussian plume dispersion, resolving $PM_{2.5}$ at **100 m** against a 3 km regional baseline, evaluated at Beijing Yizhuang and Tangshan. Normalised mean bias improves from −16.9%– −7.7% to **3.1–6.2%**, and industrial point sources account for **22.9–26.4%** of park $PM_{2.5}$ versus 1.6–13.7% in the regional model. The mechanism is straightforward — coarse grids smooth the concentration gradient and over-disperse — but the size of the attribution error is the useful number. This is the exact error a dense in-park sensor network would be deployed to measure, and the paper supplies the modelled expectation to test against.

doi: 10.1016/j.jes.2025.12.020

Wang X et al. 2026 Environ Monit Assess

Coupling coordination between bus-stop dust and soil metals

52 paired surface-dust and soil samples (104 total) at bus stops in Weiyang District, Xi'an, assayed for Cr, Mn, Cu, Zn, As and Pb and scored with Igeo and the Nemerow index. All but As and Mn exceeded Shaanxi soil background. Spatial patterns differ by medium — dust higher east, soil higher south — and sources resolve to traffic, natural and industrial. Hand-to-mouth ingestion dominated the exposure route and non-carcinogenic risk was higher for children than adults. The novel element is applying a coupling-coordination degree between the two media, which came out *moderate* and is read as partial interactive migration with shared sources. That index is a descriptive statistic borrowed from regional economics, not a transport model, so "interactive migration" is an interpretation rather than a demonstrated flux.

doi: 10.1007/s10661-026-15820-1

Wu Q et al. 2026 Environ Sci Technol

Nighttime new particle formation, and it is not always sulfuric acid

Roughly **1.5 years** of molecular-level measurement in coastal Dalian. Spring and winter nocturnal NPF is driven by H_2SO_4, whose dimer formation proceeds near the collision limit. In summer and warm autumn, H_2SO_4 and its dimers are scarce while oxygenated organic molecules are abundant, and **OOM dimers correlate with nascent particles at mean r=0.55**. Organic-driven nucleation has been reported for biogenic environments; showing it under anthropogenic influence at night is the novel claim. Implication for number-concentration monitoring: an ultrafine source term exists at night that photochemically-anchored parameterisations will miss. Correlation of 0.55 is suggestive rather than mechanistic, and one coastal site cannot establish how general the seasonal switch is.

doi: 10.1021/acs.est.6c04493

Xiao S et al. 2026 Environ Geochem Health

PAHs and oxygenated PAHs in Xi'an $PM_{2.5}$ under the clean air action

Winter $PM_{2.5}$-bound parent PAHs and OPAHs in Xi'an, a Fenwei Plain basin city, with PMF source apportionment and toxic-equivalent concentrations. Wintertime PAHs are lower than historical levels under the clean-air actions; biomass-burning-related mixed combustion is the largest PMF factor. During haze episodes both classes rise, but the increase is **statistically significant for OPAHs and not for parent PAHs**, and **OPAHs — BcdPQ in particular — dominate the toxic-equivalent burden**. Regional pollution is in-basin accumulation plus short-range transport from northern Shaanxi industry, modulated by synoptic pressure pattern. The finding that matters: a control programme that lowers parent PAHs need not lower the oxygenated derivatives that carry the toxicity, and routine monitoring measures the former.

doi: 10.1007/s10653-026-03449-5

Xu L et al. 2026 J Environ Sci (China)

Marine aging compacts soot and thickens its coating

TEM single-particle analysis of soot collected over the Bohai Sea and the Northern and Southern Yellow Sea after outflow from East Asia. **Over 98%** of soot-containing particles are coated, mean mixing-state index **0.83**. Fractal dimension rises along the transport path — 1.84 ± 0.05 (N. Yellow Sea), 1.90 ± 0.08 (Bohai), 1.96 ± 0.07 (S. Yellow Sea) — indicating progressive structural compaction, while D_p/D_{core} of 3.9–5.3 exceeds the 3.54 typical of continental regional transport. Compact, uniformly coated, thickly shelled soot absorbs substantially more than the lacy fresh particle that optical models assume. Any absorption-based black-carbon instrument calibrated on fresh soot will misreport aged marine air, and this paper quantifies by how much the morphology has moved.

doi: 10.1016/j.jes.2026.01.047

Xu X et al. 2026 J Environ Sci (China)

Ammonia enhances isoprene SOA through gas-phase cluster formation

Smog-chamber series on isoprene photooxidation by OH with and without NH_3, with nitrate-CIMS for the gas phase and HR-TOF-AMS for the particle phase. NH_3 increased SOA mass and accelerated isoprene oxidation. The mechanism is not the expected particle-phase acid–base reaction: CIMS shows NH_3 reacting homogeneously in the *gas* phase with organic acids and low-volatility oxygenated organic molecules to form extremely and ultra-low-volatility **NH_3–OOM clusters**, which then nucleate and condense. Quantum chemistry confirms spontaneous hydrogen-bonded association with RCOOH, R-OOH and R-OH groups. **Cluster formation accounts for 78% of the SOA enhancement.** Chamber conditions with elevated NH_3; the relevance is to agricultural and livestock regions where NH_3 and biogenic VOC co-occur.

doi: 10.1016/j.jes.2026.01.041

Yang & Li 2026 Environ Sci Technol

Modelling the whole indoor-radon distribution, not the area mean

Over **6 million** short-term radon measurements across the contiguous US, 2000–2021, used to predict *percentiles* at ZIP Code Tabulation Area level and then to model the within-ZCTA distribution rather than summarising each area by a mean or median. The consequence is quantitative: roughly **half of US ZCTAs have at least one in five households** above policy-relevant thresholds — a figure an area-average surface cannot produce, because the exceeding fraction is a property of the spread. Same argument applies to any PM exposure surface used for threshold-crossing statistics: modelling the mean and modelling the tail are different tasks. Radon test data are self-selected (homeowners who chose to test, often at point of sale), which biases both the level and the spread in unknown directions.

doi: 10.1021/acs.est.6c01333

Yu H et al. 2026 Environ Sci Technol

Sea–land breeze recirculation of black carbon in Qingdao

A full year of online BC (2021) split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis. Annual mean **1.78±1.29 μg/m^3**; fossil fuel **88.2%** of mass, biomass burning rising to **19% in winter**. The two fractions respond differently — BC_{ff} to wind-driven ventilation and transport, BC_{bb} to low temperature — and BC enhancement under sea–land breeze peaks at **2.0–3.5 m/s**, with diurnal persistence and directional shifts consistent with return transport and reaccumulation. Coastal siting guidance follows directly: a single coastal monitor under SLB conditions is sampling a recirculated air mass, not a fresh one. Aethalometer-model source splitting depends on assumed absorption Ångström exponents, which is the dominant uncertainty here.

doi: 10.1021/acs.est.6c10060

Zhang N et al. 2026 J Environ Sci (China)

Aging, composition and light absorption of nitrogen-containing organics in PM_1

59 Shanghai PM_1 samples analysed by UHPLC-Orbitrap MS. Nitrogen-containing organics are **31% of detected species in ESI− and 64% in ESI+**. Aging reduces molecular diversity: CHON− compounds fall in both number and mass fraction with aging degree, and carboxylic-rich alicyclic molecules drop from **37.7% to 21.2%** of the CHON− contribution. Mass absorption efficiency at 365 nm declines with aging alongside the Bep/(Bep+Bap) ratio and relative humidity, pointing to aqueous-phase oxidation as the dominant mechanism, with CHON− and CHN+ identified as the principal chromophores. Semi-quantitative by design — ESI response factors are compound-specific and uncalibrated — so the fractions are ranks, not concentrations.

doi: 10.1016/j.jes.2025.12.003

Zhao C et al. 2026 Environ Pollut

Mass fell a third; toxicity per microgram did not move

Sixteen priority $PM_{2.5}$-bound PAHs in Kunming, compared between a higher- and a lower-$PM_{2.5}$ stage. Raw declines: **$PM_{2.5}$ −32.7%, ΣPAHs −27.4%, BaP_{eq} −31.4%**; covariate-adjusted, **−28.8%** and **−33.5%**. But the particle-normalised intensity **$BaP_{eq}/PM_{2.5}$ changed by −7.6%** (**p=0.274**) and BaP_{eq}/ΣPAH by only −6.5% (p=0.026). Four-to-six-ring species held above **81%** of ΣPAH and BaP plus dibenzo[a,h]anthracene supplied over **82%** of BaP_{eq} in both stages; **40.3%** of simulated adult incremental lifetime cancer risks still exceeded 10^{-6}. PMF patterns are read conservatively as source-*related* rather than as resolved sources, which is the right caution. The inference is threatened by the two-stage design: "stage" bundles season, meteorology and policy, and no counterfactual separates them.

doi: 10.1016/j.envpol.2026.129094

Zhao H et al. 2026 Environ Monit Assess

Lead isotopes resolve dustfall sources across urban functional areas

Ten trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) in dustfall from **16 sites** across residential, busy-road, park and industrial areas of Baoding, with enrichment factors, geo-accumulation index and Pb isotope fingerprinting. Cu, Pb, Zn and Cd reach considerable pollution levels; the spatial ranking is **residential > busy road > industrial > park**, which is not the ordering most source-apportionment intuition predicts and the paper does not fully explain. Total non-carcinogenic risk stayed below threshold but **total carcinogenic risk exceeded the acceptable threshold** in several areas, worst on busy roads and in residential zones. Pb isotopes attribute **over 50%** to a combined coal-combustion plus traffic-emission group in most areas — combined, i.e. the isotope system does not separate the two here, which is the honest limit of the method in a city that burns coal and drives cars.

doi: 10.1007/s10661-026-15859-0

Zhu et al. 2026 Atmos Chem Phys

Stabilized Criegee intermediates and the sensitivity of sulfuric acid formation

Criegee reactions in MCM v3.3.1 updated to current IUPAC recommendations, driven through a large AtChem box-model ensemble, with interpretable XGBoost surrogates used to attribute the sCI-mediated SO_2 loss rate and its fractional share to controlling factors. The sCI loss rate is governed chiefly by **relative humidity and isoprene**; the drivers of the sCI *fraction* split diurnally — alkene speciation, O_3 and alkene share of VOCs by day, alkene speciation, RH and NO_x by night. Under high-sCI-fraction regimes, the positive response of total SO_2 oxidation to O_3, VOCs and alkene fraction is *amplified*, which a summer episode at Wuhai supports. Policy consequence: where alkene ozonolysis is active, **cutting VOCs controls sulfuric acid formation more effectively than cutting SO_2 alone** — and sulfuric acid is the new-particle-formation precursor. Box model with a machine-learned surrogate; the surrogate inherits every mechanistic assumption underneath it.

doi: 10.5194/acp-26-12479-2026

Cardiovascular & metabolic 7 records

Carried from the daily issues — full summaries (7 of 7 records in this subtopic)

Chen N et al. 2026 Eur J Prev Cardiol

Elemental carbon leads the $PM_{2.5}$ components on venous thromboembolism, barely

483,610 UK Biobank participants (mean age 56.5, 54.3% female), median **11.83** years, **9,988** incident VTE from hospital and death records. Per-IQR hazard ratios: elemental carbon **1.052 (1.005–1.101)**, sulfate **1.064 (1.001–1.130)**, ammonium **1.090 (1.010–1.176)**; organic matter and nitrate not significant. Residential greenness **0.944 (0.916–0.972)**, partly mediated by elemental carbon and ammonium. Heatwave hazard ratios span **1.149 (1.137–1.161)** to **1.407 (1.396–1.417)** across definitions. Two cautions. The three positive component intervals all have lower bounds within 0.01 of unity at 3×3 km resolution, so component *ordering* is not established. And the heatwave intervals are roughly three times narrower on the same events because that exposure is daily rather than annual — a real difference in information, not a real difference in certainty about air pollution.

doi: 10.1093/eurjpc/zwag465

De Potter et al. 2026 Eur Heart J Acute Cardiovasc Care

Socioeconomic and environmental determinants of STEMI incidence and in-hospital mortality

Nine years of Belgian national hospital discharge data, **51,544 STEMI admissions** (51.4 per 100,000 per year), with residential $PM_{2.5}$, NO_2, O_3, temperature, noise and greenspace from high-resolution models. A time-stratified case-crossover handled the acute question and multivariable logistic regression the mortality question. Per IQR, same-day: **$PM_{2.5}$ OR 1.022 (1.008–1.035)**, **NO_2 OR 1.044 (1.025–1.063)**; O_3 peaked over lag 0–2 at **1.063 (1.034–1.090)**. In-hospital mortality was 11.8% and tracked *socioeconomic* variables — lower individual and area income, retirement status — rather than any pollutant. Two readings matter: NO_2 outperforming $PM_{2.5}$ in single-pollutant models is the standard traffic-proximity confounding, and the incidence/mortality split says exposure and outcome severity are governed by different mechanisms.

doi: 10.1093/ehjacc/zuag118

Fang et al. 2026 Eur J Prev Cardiol

Lifestyle matters more where the air is worse, and most for early-onset heart failure

233,341 UK Biobank participants, 2006–2010 baseline to 2023, with an air pollution score over NO_2, NO_x, PM_{10}, $PM_{2.5}$ and $PM_{2.5-10}$ crossed against a six-component lifestyle score. Additive interaction is present and larger for early onset: RERI **0.44 (0.14–0.74)** for all heart failure, **1.33 (0.03–2.62)** early-onset, **0.39 (0.08–0.70)** late-onset, with the early-versus-late contrast at **p=0.08** — i.e. the headline age difference is not itself statistically resolved, and the early-onset interval very nearly touches zero. Read the other direction instead, where the contrast is firmer: lifestyle differences map to **3.35-fold (1.76–9.79)** larger early-onset incidence differences under high pollution against **1.52-fold (1.33–1.74)** under low. A composite pollution score cannot attribute anything to particles specifically.

doi: 10.1093/eurjpc/zwag456

Jiang Z et al. 2026 Atmos Environ

Multipollutant exposure and cardio-renal-metabolic multimorbidity trajectories — metadata only

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time. Design, cohort, exposure model, sample size and setting are all unknown and are recorded as such rather than inferred from the title. It is the **32nd** *Atmospheric Environment* record to enter this corpus abstract-free since 1 August, which makes this a chronic publisher deposit gap rather than an incident; the DOI is queued for the weekly re-check, and the record occupies a *Cardiovascular & metabolic* slot on the title's face only.

doi: 10.1016/j.atmosenv.2026.122340

Pan SG et al. 2026 Front Public Health

Ozone, not PM, survives two-pollutant adjustment against cardiac arrest — in one city-year

7,996 out-of-hospital cardiac arrests from the Tianjin prehospital emergency system, July 2023–June 2024 (mean 21.9/day), with a quasi-Poisson DLNM over lag 0–14 for temperature and lag0–1 single- and two-pollutant models for $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO and O_3. **Ozone was the only pollutant consistently associated**: per-IQR lag0–1, **+15.3%** overall, men **RR 1.22 (1.08–1.39)**, women **1.22 (1.04–1.43)**. The temperature result is reported with unusual honesty — a monotonic cumulative risk curve, minimum-mortality temperature $-1.9\,°C$, apparent heat peak at lag 1 (RR ≈1.57 at the 95th percentile) whose **confidence intervals include unity**, and the authors say so in the abstract. The binding limit is exposure length: **one city, twelve months**, so season and pollutant are almost fully confounded and the null PM result is uninformative rather than negative.

doi: 10.3389/fpubh.2026.1871041

Sun D et al. 2026 Lancet Planet Health

China's Clean Air Act and individual-level cardiovascular risk

China Kadoorie Biobank, n=34,862 CVD-free at baseline, mean age 51.3, split into intervention (n=25,497) and control (n=9,365) by the local government's PM reduction target under the 2013 Clean Air Act, with 10-year predicted CVD risk as the outcome and a difference-in-differences estimator. The intervention group's predicted risk rose **3.95% (3.18–4.72) less** than control, with larger differences under stricter enforcement. Per $10\,\mu g\,m^{-3}$, 2013–2021: $PM_{2.5}$ **1.80 (1.34–2.27)** percentage points, PM_{10} 1.24 (0.84–1.63), O_3 0.58 (0.33–0.83); overall pollutant change corresponded to a 6.6 percentage-point reduction in predicted risk. Two real limits: the outcome is a validated *prediction*, not observed incidence, and treatment assignment is a municipal policy target, so residual regional confounding is structural rather than adjustable. O_3 rose in both arms.

doi: 10.1016/j.lanplh.2026.101513

Wang L et al. 2026 Lipids

Joint exposure to small dense LDL-C and long-term $PM_{2.5}$ and incident stroke

CHARLS 2011–2020, **6,457** stroke-free adults, **710** incident strokes over a median 9 y; $PM_{2.5}$ from the ChinaHigh$PM_{2.5}$ product; Cox models with k-means trajectory clustering and restricted cubic splines. Per 1 SD: **sd-LDL-C HR 1.116 (1.018–1.224)**, **$PM_{2.5}$ HR 1.321 (1.222–1.428)**, with a significant multiplicative interaction (p=0.037) and **HR 1.713 (1.335–2.198)** for the dual-high stratum against dual-low. A sustained high sd-LDL-C trajectory added 36% (1.361, 1.086–1.706). sd-LDL-C outperformed conventional LDL-C for discrimination (AUC 0.578 vs. 0.534, p <0.001) — though both are poor in absolute terms, which the abstract does not say. sd-LDL-C is *estimated*, not measured, and $PM_{2.5}$ is a modelled annual mean at residence, so the interaction is between two proxies and is the result most likely to attenuate under better measurement.

doi: 10.1002/lipd.70072

Neuro / mental health — 4 records

Carried from the daily issues — full summaries (4 of 4 records in this subtopic)

Barbalat et al. 2026 Environ Sci Technol

Heat, cold and ambient pollution in the first 1000 days and autism risk indicators

ELFE national birth cohort, n=12,139 two-year-olds, M-CHAT-R screening score against weekly temperature and $PM_{2.5}$/PM_{10}/NO_2 from 1 km daily predictions (200 m for NO_2 and temperature in large urban areas), modelled with confounder-adjusted distributed lag non-linear models. **Prenatal air pollution was not associated with the screening score**; night-time heat in the early third trimester was, and survived adjustment for $PM_{2.5}$ and PM_{10}. Postnatally the pattern is sex-specific: higher $PM_{2.5}$ and PM_{10} at months 7–11.5 raised scores in males relative to females. The outcome is a screening instrument, not a diagnosis, and the sex-stratified postnatal windows are the kind of finding a DLNM will produce by construction unless pre-registered. Useful as a negative result on the prenatal PM window.

doi: 10.1021/acs.est.5c18062

Kuznetsov et al. 2026 Environ Res

Particulate matter and life satisfaction, across and within twin pairs

TwinLife, n=4,504 young-adult twins (2,249 MZ / 2,255 DZ), mean age 23.0, up to six waves 2014–2022, decomposing the PM–wellbeing association into between- and within-pair components. The effect runs **between families** (b=−0.05, SE 0.01, p <.001) and strengthens with age and calendar year for PM_{10}. Within pairs, after removing shared family variance, the association is null in DZ twins and only nominally significant in MZ twins for PM_{10} (b=−0.09, SE 0.04, p=0.04) and not at all for $PM_{2.5}$ (p=.15). That is the honest reading: most of the apparent PM effect on subjective wellbeing is confounded by family SES and neighbourhood, and the design is what makes that visible. A rare case of a study whose null is the contribution.

doi: 10.1016/j.envres.2026.125565

Paulino et al. 2026 Int J Neurosci

Exposome burden and cognitive recovery after mild TBI

44 adults with mild TBI (GCS 13–15, 50% female) prospectively enrolled at an academic clinic; 15 a priori exposome variables reduced by forward selection to 13 and summarised by PCA into a cumulative burden score, against residualised 6-month MoCA dichotomised into worse- and not-worse-than-expected recovery. The latent axis explains **16.0%** of variance and is loaded by education, area deprivation, insurance coverage, **$PM_{2.5}$** and adverse childhood events; burden was higher in the worse-than-expected group (median PC1 1.34 vs −0.76, p=0.010, holding under permutation). Carried for the $PM_{2.5}$ loading only. At n=44 with 13 variables and an outcome-informed selection step, this is hypothesis-generating; the loading structure is mostly socioeconomic and $PM_{2.5}$ may be riding on it.

doi: 10.1080/00207454.2026.2728613

Ren X et al. 2026 Front Public Health

Black carbon dominates a $PM_{2.5}$ mixture on adolescent self-injury — in cross-section

18,398 adolescents in grades 7–12 across 15 schools in four Chinese provinces, with non-suicidal self-injury by ANSAQ and multidimensional biological rhythm by SBRDA. $PM_{2.5}$ and its constituents were positively associated with NSSI risk, black carbon most strongly: **OR 1.76 (1.60–1.94)**, and dominant in the mixture models (**WQS weight 0.810**, Qgcomp effect 0.671). This is the largest effect on today's forest panel and rests on the weakest design there. A single cross-sectional survey round cannot order exposure and outcome in time; exposure is assigned by school or residence, so the black-carbon signal is also a traffic-proximity and urbanicity signal; and rhythm disorder is plausibly a mediator, a confounder or a consequence, which stratification cannot distinguish. The mixture weight is the informative part — it says the components are not interchangeable — and the odds ratio is not the part to quote.

doi: 10.3389/fpubh.2026.1887542

Respiratory & allergic — 5 records

Carried from the daily issues — full summaries (5 of 5 records in this subtopic)

Cui L et al. 2026 Clin Rev Allergy Immunol

Epidemic thunderstorm asthma in China as an Artemisia phenotype

54 primary studies (43 international, 11 Chinese — Yulin, Hohhot, Chongqing, Lanzhou, Shenzhen) from 436 records, risk of bias by Newcastle–Ottawa and Murad, certainty by GRADE. Three patterns recur: Chinese events are associated with **mugwort (*Artemisia*)** rather than ryegrass sensitisation; **40–60% of Chinese cases and up to 78% of Australian and UK cases had no prior asthma diagnosis**; and individual Chinese localities report recurrence. No meta-analysis, and GRADE certainty for direct pollen rupture, pollution-mediated severity and pre-emptive treatment is **low to very low**. The measurement gap is explicit and is the actionable part: Chinese reports identify sensitisation only to genus level and **do not speciate the airborne pollen**, so no aerobiological trigger threshold can be set. The authors' first research priority is field aerosol sampling.

doi: 10.1007/s12016-026-09194-y

Gómez-Olarte et al. 2026 Environ Pollut

Early-life air pollution, respiratory outcomes and stimulated cytokine response

German birth cohort, n=606 one-year-olds for bronchitis and wheezing, cytokine subcohort n=508 (LPS-stimulated TNF-α, MCP-1, IL-8, IL-6, IL-10), residential $PM_{2.5}/PM_{10}/NO_2$ per IQR of 2.0/2.8/9.1 $\mu g\,m^{-3}$, with BKMR for the mixture and stratification by sex and temperature. In boys, $PM_{2.5}$ gave **aOR 1.50 (1.02–2.23)** for bronchitis and **+23.8% (11.2–37.9)** IL-8; in girls, **+31.9% (18.6–46.7)** IL-8, +12.8% (3.9–22.6) TNF-α and +35.7% (14.4–61.1) IL-10. The mixture association with IL-8 and TNF-α was non-linear and appeared only above the 60th percentile of all pollutants. Small cohort, and the sex split doubles the tests; the consistent direction across five cytokines is what carries it. $PM_{2.5}$, not NO_2, is the primary predictor.

doi: 10.1016/j.envpol.2026.129079

Li H et al. 2026 ERJ Open Res

Multi-omics Mendelian randomisation of $PM_{2.5}$ and childhood asthma

Two-sample MR on large-scale GWAS summary data gives **OR 1.897 (1.063–3.388), p=0.030** for genetically instrumented $PM_{2.5}$ on childhood asthma. A TWAS identified **70 genes** co-expressed with both, enriched in lysosome- and leukocyte-mediated immunity, with **MEAF6** validated as protective and **RNF40** as a risk gene. Two-step MR put **FUT10 as a positive mediator carrying 19.3%** of the effect, with CD200 and MANBA negative mediators; a phenome-wide MR flagged associations of these targets with other diseases, which the authors correctly present as a caution against treating them as drug targets. The limits are the usual MR limits and they are severe here: the $PM_{2.5}$ instrument is a genetic proxy for a residentially correlated environmental exposure, so horizontal pleiotropy through socioeconomic position is not excluded, and the interval spans a factor of three.

doi: 10.1183/23120541.01444-2025

Wangupadcha et al. 2026 J Glob Health

A null result, reported as null, at 42 $\mu g/m^3$

Time-series of **924** respiratory visits (ICD-10 J00–J99, mean 6.37/day) among residents aged ≥**60** living within 5 km of a newly established industrial area in central Thailand, November 2023–March 2024, against daily $PM_{2.5}$ from the nearest government monitor, with negative binomial single-lag and moving-average models adjusted for temperature, humidity, wind speed, day of week and trend. Mean $PM_{2.5}$ was **42.43 $\mu g/m^3$ (SD 15.12)**, far above WHO guidance. The largest estimate was at lag 4: **IRR 1.055 (0.995–1.119)** per 10 $\mu g/m^3$ — positive, not significant, and described that way by the authors rather than spun. Subgroup estimates were numerically larger in women and at 60–69 with non-significant interaction tests. The study is underpowered by construction: **five months, 924 events, one monitor** for a 5 km radius. Worth carrying as a well-behaved null and as a monitoring-density argument.

doi: 10.7189/jogh.16.04269

Wright et al. 2026 Curr Environ Health Rep

Air pollution and respiratory outcomes in Africa — a review that mostly documents absence

PRISMA search of PubMed, ScienceDirect and Elicit for African case-control studies 2015–2025 returned **13 studies from 10 countries**. Reported concentrations are the finding: indoor $PM_{2.5}$ **96–177** $\mu\mathbf{g/m^3}$ in biomass-using homes and ambient $PM_{2.5}$ to **259** $\mu\mathbf{g/m^3}$, an order of magnitude above WHO guidance. Nitrogen oxides tracked reduced lung function in children; household air pollution tracked under-five mortality and low birthweight; associations with acute respiratory infection were inconsistent across settings. **Only three studies measured temperature and none examined heat–pollution interaction**, which is the stated motivation for the review and which it therefore cannot deliver. Restricting to case-control designs also excludes the cohort and panel literature, so this is a floor on the African evidence base, not a census.

doi: 10.1007/s40572-026-00561-7

Reproductive & developmental 7 records

Carried from the daily issues — full summaries (7 of 7 records in this subtopic)

Chen Y et al. 2026 Sci Rep

$PM_{2.5}$ components in early pregnancy and congenital heart disease

2,601,300 live births and stillbirths, Guangdong 2014–2017, with satellite-derived component exposures resolved to the week of gestation and quantile g-computation for mixture weights. Effects peak in **pregnancy week 1**: sulfate **OR 2.23 (2.19–2.27)**, black carbon **1.95 (1.91–1.98)**, joint weights 86.8% and 13.2%. Read the intervals before the point estimates. A near-doubling of CHD odds with a 95% CI of width 0.08 implies a precision no exposure model at this spatial scale supports, and a week-1 peak in a satellite product with month-scale effective resolution is more consistent with seasonal and spatial confounding than with a window-specific teratogenic mechanism. The scale of the register is real; the causal contrast is not established.

doi: 10.1038/s41598-026-59940-7

Gao et al. 2026 Environ Sci Technol

Urinary biomarkers of airborne organics in pregnancy and small-for-gestational-age risk

TIMFEM prospective cohort, n=1,425 mother–infant pairs, 20 urinary biomarkers of airborne organic pollutants by LC-MS/MS (p-phenylenediamines, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine), 18 detected in >40% of participants, with leave-one-out in vitro screening in human trophoblast cells at exposure-guided doses. Each ln-unit increase in **6PPD was associated with 27% higher SGA risk and 6PPD-Q with 23%**; mixture models added TPG, cotinine and 1-aminopyrene. The mechanism the cell work supports is metabolic rather than cytotoxic — reduced pyruvate entry into the TCA cycle and impaired isocitrate-to-α-ketoglutarate conversion at doses that barely touched proliferation. No confidence intervals are given for the epidemiological estimates in the abstract, so they are not plotted here. Relevance: these are tyre-and-road and traffic-associated organics that no PM mass channel resolves.

doi: 10.1021/acs.est.6c04451

Mbishi et al. 2026 Environ Pollut

Perinatal PM and infant head circumference: unambiguous, and small

119,110 Flemish children with BMI-, length- and head-circumference-for-age z-scores from routine well-baby clinic records, perinatal PM_{10} and $PM_{2.5}$ from the RIO detrended-kriging model, and mixed-effects models carrying a 5-df natural cubic spline in child age with exposure-by-age interaction and child-level random intercepts and slopes. At 12 months, per IQR: head circumference **−0.0499 (95% CI −0.0606, −0.0393)** z for PM_{10} and **−0.0645 (−0.0771, −0.0518)** for $PM_{2.5}$, with inverse associations sustained between 6 and 24 months. BMI- and length-for-age associations *change sign* across age, and green-space associations are inconsistent across measures. The authors state that estimates are small and that most nominally significant associations are of limited magnitude — roughly a fifteenth of a standard deviation. Exposure is municipality-scale interpolation, which at this n makes precision a poor guide to accuracy.

doi: 10.1016/j.envpol.2026.129064

Tanner et al. 2026 Environ Epigenet

Prenatal exposure mixture and aromatase brain-promoter methylation in cord blood

Barwon Infant Study birth cohort, **n = 906**, testing 25 prenatal exposures across five classes — manufactured chemicals, air pollution, pharmacological, nutritional and sunlight-related — against methylation of the brain-specific *CYP19A1* P1.f promoter using weighted quantile sum regression. The mixture index was associated with higher P1.f methylation, **adjusted mean difference 0.71 (95% CI 0.11–1.32)**, $p = 0.021$, interpreted as reduced brain aromatase activity. Bisphenols, reduced sunlight, household mould, phthalates, low folate and air pollution carried the largest weights. The inference-threatening limitation is structural: WQS assigns weights within a single index, so air pollution's individual contribution is neither estimated nor separable, and cord blood is a proxy for a brain-specific promoter. Tier C for this watch — PM is one component of a 25-exposure mixture.

doi: 10.1093/eep/dvag027

Wei Q et al. 2026 Int J Epidemiol

Weekly component windows: black carbon at 12–15 weeks, nitrate at 5–12 and 22–24

2,680 mother–newborn pairs and **9,339** growth ultrasounds (abdominal circumference, head circumference, femur length, estimated fetal weight), Guangzhou 2017–2020, with *weekly* black carbon, organic matter, nitrate, sulfate and ammonium from the TAP model entered into distributed lag models with GEE, plus machine-learning mixture analysis. Femur-length reduction is most strongly associated with **black carbon at gestational weeks 12–15**; head-circumference deficits with **nitrate at weeks 5–12 and 22–24**; estimated fetal weight with **nitrate at weeks 6–8**. Black carbon and nitrate dominate the mixture. This is the exposure resolution the Flemish cohort above lacks, at 45× fewer children. The threat to inference is that five components drawn from one gridded product co-vary strongly, so "which component" is partly a question about the exposure model rather than about biology; the abstract reports no effect sizes for the identified windows.

doi: 10.1093/ije/dyag185

Xu ZL et al. 2026 Biomed Environ Sci

Component-specific $PM_{2.5}$ windows and oocyte yield in assisted reproduction

51,122 ART cycles with individual $PM_{2.5}$ and its sulfate, nitrate, ammonium, organic-matter and black-carbon fractions assigned over recent (0–3 months), distal (4–12 months) and cumulative windows; negative binomial regression on total, mature and normally fertilised oocyte counts, DLNM for window identification, mixture models for joint effect. Higher exposure tracked poorer oocyte outcomes throughout, with **stronger effects for the distal window** and two sensitive periods — 1 month and 6–11 months before retrieval. Sulfate and ammonium dominated the mixture. The 6–11 month window is biologically coherent with primordial-to-antral follicle transit, which is the paper's strongest argument. Its weakness is that ART patients are a selected population and the exposure model is residential, so a distal window is also the window over which residential misclassification is largest.

doi: 10.3967/bes2026.072

Zanini et al. 2026 Environ Pollut

Maternal air-pollution exposure and fetoplacental Doppler markers

Ships on metadata only. The Crossref deposit carries no abstract, the record is not in PubMed or Europe PMC, and Semantic Scholar returns 404 for the DOI, so cohort size, exposure model and Doppler effect sizes are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list — fetoplacental haemodynamics is a mechanistically informative intermediate between maternal PM exposure and birth outcome and is thinly represented in this watch's corpus.

doi: 10.1016/j.envpol.2026.129046

Mechanistic toxicology 11 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Arora et al. 2026 J Environ Sci (China)

Oxidative potential of fresh versus ozone-aged $PM_{2.5}$ across Chinese sources

A commentary, not primary data — it summarises Ma et al. (10.1016/j.jes.2024.04.023), which measured DTT-assay oxidative potential on fresh and O_3-aged $PM_{2.5}$ from multiple source types. **Biomass-burning particles carried up to 35× the OP of suburban $PM_{2.5}$**, driven by water-soluble organics and transition metals, and O_3 aging generally *lowered* OP while producing complex chemical transformation. Counted and listed here because the source-dependence claim is the one this watch tracks, but no number in it is new and the aging direction runs opposite to the usual assumption that atmospheric processing increases toxicity. Treat as a pointer to the 2024 primary paper.

doi: 10.1016/j.jes.2026.05.042

Cao et al. 2026 Environ Sci Atmos

Black carbon interfacial chemistry and multiscale aging

A review that puts BC *particle state* — surface reactivity, coating thickness, morphology, aging timescale — rather than BC mass at the centre, and follows it through to optics, cloud activation, removal, lifetime and toxicity. The framing matters for instrumentation because every one of those downstream properties is a function of a variable that filter-based absorption and single-particle soot photometry constrain only partially, and that no low-cost optical sensor constrains at all. No primary data and no quantitative synthesis; the Crossref deposit carries only a graphical-abstract blurb, so the parameterisation recommendations were not verifiable at digest time. Listed as a reference framework rather than as evidence.

doi: 10.1039/d6ea00093b

Garcia Dominguez et al. 2026 Environ Sci Technol

Ultrafine particle deposition and clearance in an avian lung

Zebra finches exposed to 50 or 100 nm model particles at 5 or 25 °C, all within environmentally relevant ranges, with deposition quantified in lung and — in a subset — heart, liver and red blood cells. Lung deposition was highest for the smallest particles, consistent with diffusional deposition. Deposited *fraction* was unchanged by temperature but deposited *dose rate* rose in the cold, driven entirely by elevated minute ventilation: a clean demonstration that thermoregulatory demand, not respiratory physiology, sets avian dose. **44% of the initially deposited 100 nm burden remained at two weeks**, i.e. clearance is slow. Model particles, one species, small subset for extrapulmonary organs — but this is the only record in today's corpus that measures ultrafine dose in a living respiratory system, and the avian lung's cross-current architecture makes it a genuinely different test case from rodent work.

doi: 10.1021/acs.est.6c05957

Han G et al. 2026 (preprint) Research Square (preprint)

Picropodophyllotoxin attenuates $PM_{2.5}$ injury in pulmonary endothelial cells via mTOR-dependent autophagy

Human pulmonary artery endothelial cells exposed to $PM_{2.5}$ at 25–100 μg/mL for 24–48 h with picropodophyllotoxin (2–50 μM) or dexamethasone (50 μM). $PM_{2.5}$ reduced viability dose- and time-dependently with increased LDH leakage and ROS and suppressed SOD/CAT and SGK1; PPT restored viability, SGK1 and mTOR phosphorylation and attenuated TLR4, MyD88, LC3II and Beclin 1 upregulation. **Not peer reviewed, single cell line, no animal or human data, and no dose–response statistics reported in the abstract.** The exposure concentrations are far above realistic alveolar deposition, which is the standard weakness of this design and limits the finding to pathway plausibility rather than therapeutic inference. Included for completeness of the mechanistic map, not as evidence of effect.

doi: 10.21203/rs.3.rs-10446885/v1

Heo et al. 2026 Tuberc Respir Dis (Seoul)

Three weeks of PM_{10} at the air–liquid interface: barrier down, differentiation lost, Notch up

Differentiated human small airway epithelial ALI cultures exposed to PM_{10} for 7, 14 and 21 days. Transepithelial electrical resistance **fell progressively**; club-cell and ciliated-cell markers fell while basal-cell markers were preserved, i.e. a differentiation block rather than uniform cytotoxicity. Surfactant **SP-A and SP-B mRNA fell**, yet **secreted SP-A rose while SP-B secretion fell** — a dissociation the paper reports without over-explaining. IL-1β, IL-6, IL-8 and TNF-α rose; cleaved Notch1 and Notch3, phospho-NF-κB, IκBα, HES1 and HEY1 all increased. The chronic, repeated-dose ALI design is a genuine advance on single-dose submerged culture. Limits are the usual ones and are not minor: **PM_{10} dose, source and donor number are not stated in the abstract**, and no pathway inhibition experiment establishes that Notch is causal rather than concurrent.

doi: 10.4046/trd.2026.0072

Jerome et al. 2026 J Immunotoxicol

World Trade Center dust: neuroinflammation and myelin response at 3 and 12 months

Twelve-week-old male spontaneously hypertensive rats given intratracheal WTC dust at **2.43 ± 0.78 mg/day on two consecutive days**, then euthanised at 3 and 12 months. At 3 months, increased Black-Gold II myelin staining without change in myelin basic protein or CNPase; at 12 months, microglial activation (Iba1) and elevated brain **IL-1β**. The authors read the discrepant myelin markers as a compensatory response preceding degeneration. Three limitations bound the inference: a two-day bolus instillation is not a chronic inhalation exposure and bypasses the upper airway; the SHR strain is hypertensive by design, so the neuroinflammatory phenotype is confounded with a vascular one; and males only. Useful as a mechanistic anchor for the First-Responder cognitive literature, not as a dose–response.

doi: 10.1080/1547691X.2026.2722654

Luo et al. 2026b Clin Rev Allergy Immunol

Environmental pollutants and innate immunity via group 2 innate lymphoid cells

Narrative review covering O_3, PM, diesel exhaust particles and cigarette smoke alongside nanomaterials and microplastics, organised around ILC2 biology. The proposed common route is epithelial barrier compromise followed by alarmin release — IL-33, IL-25, TSLP — with ILC2s as the sensor coupling environmental detection to type-2 inflammation, plus non-canonical regulation of ILC2 plasticity and functional adaptation. No new data, no quantitative synthesis, and PM is not separated from the other exposures, so nothing here constrains a PM-specific dose or particle-property relationship. Carried as a citation anchor for the epithelial-alarmin axis in respiratory PM work; tier C.

doi: 10.1007/s12016-026-09193-z

Meldrum & Leonard 2026 Front Immunol

Diesel particle priming reprograms the airway response to a subsequent allergen

Murine model with deliberate temporal separation between diesel exhaust particle exposure and house dust mite challenge — so neither co-exposure nor prior sensitisation — read out with compartment-resolved bulk transcriptomics (lung tissue vs airway lumen) plus targeted single-cell profiling. DEP priming amplified leukocyte recruitment on subsequent HDM challenge and did so **neutrophilically, with no eosinophil response**. DEP alone drove NF-κB programmes; HDM drove NF-κB and interferon; the primed response was qualitatively different from either, with elevated ISG expression and expanded neutrophil subpopulations concentrated in the luminal compartment. The design is the contribution: it isolates a memory-like effect of a particle exposure that has already cleared. Murine, single particle type, no dose–response.

doi: 10.3389/fimmu.2026.1869644

Na HW et al. 2026 J Toxicol

Airborne metals reconstructed from worker sebum, then dosed onto skin

ICP-MS on sebum from **62 indoor and 40 outdoor workers** in Shanghai across 24 metals, with large outdoor–indoor contrasts (cobalt 925×, arsenic 135×, barium 29×, manganese 17×). Sebum-mimicking mixtures were then formulated at the measured profiles and applied to keratinocytes, sebocytes and ex vivo scalp skin: the outdoor mixture reduced viability, raised lipid peroxidation, induced γH2AX foci, upregulated *IL1A*, *IL1B*, *TNFA*, *IL6* and *PTGS2*, and activated NF-κB and MAPK. The design is the interesting part — a biomonitoring matrix used to specify the in vitro dose rather than an arbitrary slurry concentration. The weakness is that sebum metal load is a cumulative deposition proxy of unknown relation to inhaled or dermally absorbed dose, and indoor/outdoor is a crude exposure contrast with occupation confounded into it.

doi: 10.1155/jt/7925949

Ren Y et al. 2026 Research (Wash D C)

Lung-on-chip for PM, appraised model by model

Critical review of *in vitro* platforms for PM–lung interaction, from 2D monolayers through 3D bioprinted constructs and organoids to lung-on-chip devices, with each class assessed for what it can and cannot reproduce. The central argument is that breath-mimicking cyclic mechanical strain is not an incidental refinement but a determinant of how particles interact with the epithelial interface — deposition, clearance and mechanotransduction all change under motion, so static models systematically misrepresent the dose actually delivered. Useful as an orientation to which platform answers which question. It is a narrative review with no systematic search, no pooled comparison and a closing pitch for the authors' own "highly integrated and intelligent" platform, so the appraisal is not disinterested.

doi: 10.34133/research.1414

Wang Y et al. 2026 Environ Sci Technol Lett

α-Dicarbonyl aqueous products modulate cellular ROS in opposite directions

Glyoxal or methylglyoxal reacted with ammonium sulfate or glycine in the aqueous phase; HRMS shows ammonium routes yield imidazole-type nitrogen heterocycles and glycine routes yield more pyrrole-related species. Alone, neither product set induces much ROS in A549 cells. Co-exposed with 1,4-naphthoquinone, Cu^{2+} or both, **imidazole-dominated mixtures enhance ROS and pyrrole-rich mixtures suppress it**, with acellular assays attributing the split to promotion versus suppression of quinone redox cycling and metal-associated pathways. This is a direct challenge to any oxidative-potential assay read as a scalar: the same secondary organic aerosol aging pathway can raise or lower measured ROS depending on the nitrogen source. Cell-line work at unstated atmospheric relevance of concentration, so treat as mechanism, not as an exposure-response.

doi: 10.1021/acs.estlett.6c00722

Occupational & indoor 19 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (18 OF 19 RECORDS IN THIS SUBTOPIC)

Chen Y et al. 2026 Atmos Environ

Solid-fuel PAH exposure, phenanthrene metabolites, DNA damage and inflammation

Ships on metadata only — no abstract in the Crossref deposit and no PubMed, Europe PMC or Semantic Scholar record on this run. Title and authorship indicate urinary phenanthrene metabolites as the exposure index in a rural northern Chinese solid-fuel population, with DNA-damage and systemic-inflammation biomarkers as outcomes. Counted and listed; no numbers claimed. High priority for the retry list: household solid-fuel combustion is the single largest PM exposure contrast available in observational work and this watch carries very little of it.

doi: 10.1016/j.atmosenv.2026.122339

Derwein et al. 2026 Build Environ

Masks, filtration and ventilation on exhaled-air transmission — metadata only

No abstract was deposited to Crossref, Europe PMC or Semantic Scholar at index time, and ScienceDirect refused retrieval from this environment, so nothing beyond title, authorship and venue can be asserted. Recorded so the DOI enters `seen.json` and the record is not re-screened; queued for the weekly abstract re-check. Listed under *Occupational & indoor* on the title's face, which is a provisional assignment.

doi: 10.1016/j.buildenv.2026.115206

Feng R et al. 2026 Environ Sci Technol Lett

Clean heating renovation cuts nitro-PAHs but not nitrated phenols

Size-resolved *personal* exposure to particulate nitrated phenols (NPs) and nitro-PAHs among rural users of clean coal, raw coal chunk and biomass, paired with A549 cytotoxicity and NF-κB reporter assays. Biomass users carried up to **11.5×** the $PM_{2.5}$-bound n-PAH exposure of clean-coal users, while NPs varied **≤1.3-fold** across fuels (clean coal 64.0±12.1 vs raw coal 57.4±25.8 ng/m^3, p=0.545). Toxicologically the two classes diverge as well: n-PAHs were more cytotoxic by IC_{50}, NPs activated NF-κB more strongly via p-IκB-α and COX-2. The policy reading is sharp — a fuel switch evaluated on PM mass or on PAHs alone will score a benefit that the NP fraction does not deliver. Group sizes are not stated in the abstract and the in vitro work uses representative compounds, not the collected particles.

doi: 10.1021/acs.estlett.6c00629

Gu et al. 2026b Environ Int

Nitrogen-containing organics in household solid-fuel combustion PM vs. urinary inflammatory and amino-acid profiles

Listed on Crossref metadata only. The deposit carries no abstract, and the record is absent from Europe PMC, PubMed and Semantic Scholar as of this run, so **no finding, effect size, cohort size or location is asserted here**. The title indicates an exploratory personal exposure design pairing speciated N-containing organic compounds in solid-fuel combustion PM with targeted urinary inflammatory and amino-acid metabolites — a personal-monitoring plus targeted-metabolomics pairing that is directly on-axis for this watch. Queued for the abstract retry list; it will be summarised, not re-listed, once text is available.

doi: 10.1016/j.envint.2026.110491

Ibrahim 2026b Qeios (preprint)

The case for routine indoor PM monitoring, argued without data

A commentary arguing that Malaysia's ICOP IAQ 2010 framework, though it contains preventive elements, is operated as a complaint-triggered investigation regime, and that **particulate matter and outdoor-to-indoor pollutant transport** should join CO_2 as routinely monitored parameters in building management. The specific climatic argument — that hot, humid conditions couple ventilation, condensation and biological growth in ways a temperate-derived code handles badly — is sound and underexploited in the LCS literature. Everything else is assertion: single author, non-peer-reviewed platform, no primary data, and the Malaysian office, hospital and school studies it cites are described rather than synthesised. Included as an argument about monitoring design, not as evidence.

doi: 10.32388/b506md

Illa et al. 2026 Environ Sci Technol

Exposure pathways when psychoactive drugs are smoked indoors

Mechanistic indoor chemical-fate and exposure model across seven compounds spanning orders of magnitude in volatility and hydrophobicity, with scenarios varying smoker retention rate and smoking frequency, resolving first-, second- and third-hand pathways and inhalation, mouthing-ingestion and dermal routes. Dominant pathway is strongly chemical-specific: volatile compounds (nicotine) go by inhalation; intermediate hydrophobicity by dermal absorption; involatile hydrophobic compounds (Δ^9-THC) receive meaningful contributions from all three. For non-users the SHS/THS balance is set jointly by partitioning and by the ratio of active-smoking to post-smoking time. Entirely modelled — no measurements, no validation — so the useful output is the ranking of pathways, not the doses.

doi: 10.1021/acs.est.6c02953

Jaiswal et al. 2026 Research Square (preprint)

Nature-based indoor filtration in an EDM gear-manufacturing shop

Single-site five-month before/after intervention in Gurugram, India. Baseline was severe: **$PM_{2.5}$ 330 μg m^{-3} and CO 88 ppm**. Plant-based filter units plus a redesigned vent/exhaust layout, sited using airflow visualisation, cut **CO by 90% and $PM_{2.5}$ by 42% on average** — leaving $PM_{2.5}$ around 190 μg m^{-3}, which the authors say plainly is still not a healthy range. No control period, no controlled comparison against a conventional HEPA unit, and the CO and PM reductions differ by a factor the paper does not explain mechanistically. Preprint, not peer reviewed. Listed for the occupational baseline number, which is the part that is hard to obtain.

doi: 10.21203/rs.3.rs-10511730/v1

Jamali et al. 2026 Int J Biol Macromol

All-cellulose HEPA nanofilter in a wooden frame

Fog-sprayed, freeze-dried bacterial-cellulose nanofibres on a wood-derived cellulose-microfibre tissue substrate, in a modular wooden frame with a starch-adhesive gasket. Removal efficiency rises with BCNF grammage to **2 g/m^2** and falls above it — consistent with pore closure raising velocity through the remaining paths. Optimum gives **99.998% $PM_{0.3}$ at 150 Pa**, inside the EN 1822 H13–H14 band, and $PM_{0.3}$ capture *peaks at high RH* under cyclic humidity, the opposite of the usual electret degradation. The claim being made is biodegradability at H13 performance, and the tests reported are single-pass efficiency and pressure drop; there is no loading lifetime, no microbial growth assessment on a cellulose substrate held at high RH, and no mechanical durability data. Promising material, incomplete filter.

doi: 10.1016/j.ijbiomac.2026.154178

Jin et al. 2026b Build Environ

Air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling

Metadata only. The Crossref deposit carries no abstract and the record is not yet in PubMed, Europe PMC or Semantic Scholar. Listed and counted for completeness; no finding, effect size or method detail is attributed to it, and it is on the abstract retry list.

doi: 10.1016/j.buildenv.2026.115193

Kusi et al. 2026 Research Square (preprint)

Waste-tyre combustion toxicology: a systematic search that could not pool

PRISMA-style search of PubMed, Scopus, ScienceDirect and Google Scholar for 2016–2025 human and experimental studies of waste-tyre combustion emissions — particulate matter, PAHs, VOCs and heavy metals — reporting biochemical, molecular, biomonitoring or toxicological outcomes, appraised with SYRCLE, JBI and NIH instruments as appropriate to design. **Heterogeneity in exposure conditions, designs and reported outcomes precluded quantitative synthesis**, which the authors state plainly. The value is in mapping organ-specific mechanistic pathways for an exposure that is ubiquitous in informal waste economies and almost absent from regulated monitoring networks. Preprint, not peer reviewed; a narrative synthesis of a heterogeneous literature inherits every selection effect in that literature.

doi: 10.21203/rs.3.rs-10530000/v1

Li et al. 2026f Aerosol Air Qual Res

Non-uniform oil-mist distribution and local ventilation in a machining workshop

Listed on Crossref metadata only — no abstract deposited, and no record in Europe PMC, PubMed or Semantic Scholar. **No quantitative claim is made here.** The title indicates spatial characterisation of oil-mist aerosol in a machining workshop with a local ventilation intervention, which is the occupational analogue of the indoor-siting problem: where a workshop's aerosol field is strongly non-uniform, a single fixed monitor is a poor estimator of personal exposure regardless of its accuracy. Queued for the abstract retry list.

doi: 10.1007/s44408-026-00161-y

Lv H et al. 2026 ACS Appl Mater Interfaces

Polyimide microsphere/nanofibre gradient aerogels for high-temperature filtration

A single-component polyimide gradient aerogel built to beat the dust-holding limit of uniform nanofibre mats, which cake at the surface and choke. Dual-layer structure: ≈730 nm coarse fibres under ≈300 nm medium fibres, with a composite functional layer of ≈3.34 μm microspheres and ≈60 nm ultrafine fibres on top. Performance: **99.996% $PM_{0.3}$ efficiency at 45.08 Pa** initial pressure drop, **127 g m^{-2} dust-holding capacity**, and **99.91% retained at 300 °C**. Single-component construction is claimed for recyclability and uniform thermal expansion. What is not reported is loaded pressure-drop evolution or efficiency after thermal cycling, which is what actually determines service life in a stack.

doi: 10.1021/acsami.6c10042

Ma L et al. 2026 Can J Urol

Surgical smoke from laser laparoscopic partial nephrectomy with an aspirating handle

Ex vivo porcine kidney comparison of electrocoagulation, ultrasonic, 980 nm diode and 532 nm KTP laser resection, with and without a novel aspirator-integrated handle. The KTP green laser produced the most smoke; coagulation-layer thickness ran diode (**4.16 ± 0.21 mm**) > green (1.55 ± 0.21) > electrocoagulation (0.35 ± 0.08) > ultrasonic (0.52 ± 0.08). With the aspirator, diode-laser formaldehyde fell **4.95 → 1.69 mg m^{-3}**, TVOC **8.34 → 1.11 mg m^{-3}** and CO **124 → 24 ppm**. Gases only — the study reports no particle-size distribution or number concentration, which is the fraction that reaches the alveoli and the reason surgical smoke is an occupational aerosol problem rather than a gas one. In vitro, no personal sampling.

doi: 10.32604/cju.2026.070096

Mbazima et al. 2026 Int J Environ Health Res

Chalk dust puts teachers over a deposition threshold while staying under the occupational limit

Sixty personal breathing-zone samples from 12 Johannesburg secondary-school teachers (7 female, 5 male), collected Monday–Friday 08:00–14:35 with GilAir-3 pumps, a 4 μm cut-point cyclone and 25 mm PVC filters at 1.85 L/min, then run through a multiple-path particle dosimetry model at low, moderate and high exertion for nasal and oronasal breathing. **PM_4 ranged 0.019–0.761 mg/m^3, mean 0.126 ± 0.134 mg/m^3**; chalk use ($\beta = 0.468$) and sampling duration ($\beta = 0.305$, both $p < 0.001$) predicted concentration. Breathing route dominated deposition, and alveolar clearance was slower than tracheobronchial in both sexes. The authors' conclusion — that risk persists below the occupational exposure limit — rests on modelled rather than measured dose, and 12 teachers in one region is a small and non-random base. Note the standard deviation exceeds the mean, so the distribution is heavily right-skewed and the mean is a poor summary.

doi: 10.1080/09603123.2026.2726420

Micallef & Scerri 2026 Build Environ

A flow-conditioned infiltration metric for traffic-related pollution ingress

Ships on metadata only — Crossref carries no abstract and the record is not otherwise indexed on this run. The title indicates an outdoor-to-indoor infiltration metric for traffic-related air pollution in naturally ventilated buildings, conditioned on flow regime rather than assumed constant. Counted and listed; no numbers claimed. Worth retrying: a flow-dependent infiltration factor is exactly the term that makes outdoor-monitor-based exposure assignment wrong for naturally ventilated stock, and it is the term a paired indoor/outdoor sensor pair can actually estimate.

doi: 10.1016/j.buildenv.2026.115195

Wei J et al. 2026 Front Immunol

Welding fume immunology, with the exposure-assessment gap named explicitly

Mini review of metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils, monocytes and circulating immune cells. Consistent signals across human exposure studies, controlled inhalation, cell models and workplace investigations: acute-phase proteins, cytokines, chemokines and neutrophilic responses, with some evidence of altered T-helper activity and blunted responsiveness to bacterial stimuli. TRPA1/TRPV1 sensory signalling is proposed as a rapid route to cough and airway reflex, and the review is careful to say that direct welding-fume-specific human evidence for that pathway is limited. The stated obstacle to synthesis is the one that matters here: **fume composition and particle metric vary so much between studies that biomarker results cannot be pooled**. No quantitative estimate, no pooling, no systematic search protocol reported.

doi: 10.3389/fimmu.2026.1882845

Wood et al. 2026b Indoor Environments

Influence of ventilation rate on the potential for indoor particulate matter exposure

Metadata only. Same situation — no abstract in the Crossref deposit and no other index carries it yet. Flagged here because the title is squarely in scope for indoor PM sensing; it is on the abstract retry list and will be summarised if the abstract appears.

doi: 10.1016/j.indenv.2026.100191

Yang et al. 2026 Atmosphere

Ventilation optimisation for a 300 m welding workshop with distributed dust sources

CFD of airflow and welding-fume transport in a 300 × 28 × 21 m hall, varying exhaust-to-supply ratio and side-exhaust-port height, and contrasting low-density aluminium-alloy fume with high-density carbon-steel fume. Two air-velocity peaks appear at 0–2 m and 8–12 m above floor; the **top exhaust port outperforms the side port** throughout, higher ESR accelerates upward migration and suppresses lateral dispersion, and a badly set side-port height produces either short-circuiting or excess lateral spread. Optimal side-port height **5 m for aluminium, 6 m for carbon steel**. Simulation only — no measured personal exposure, and fume is treated as a passive density-differentiated tracer rather than a coagulating aerosol.

doi: 10.3390/atmos17090843

Not carried by a daily issue — extractive precis or metadata only, not an assessed summary (1 records)

Srinivasan et al. 2026 Natl Med J India

Occupational wood dust exposure and lung function impairment: a systematic review and meta-analysis

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.25259/nmji_668_2024

Burden, policy & mitigation 7 records

Carried from the daily issues — full summaries (7 of 7 records in this subtopic)

Li Y et al. 2026 Environ Sci Technol

Multidimensional inequality of CO_2 and air-pollutant footprints in China

Consumption-based per-capita footprint accounting for CO_2 and air pollutants across 2020–2060 under the dual-carbon policy. The top 10% carry **28.5 t CO_2 and 50.6 kg NO_x** per capita against **1.2 t and 4.7 kg** for the bottom 50%. The structural finding is that **over 90% of footprint inequality is intraprovincial**, not between provinces, and the gap widens to 2060 — which means province-level policy instruments are aimed at the wrong axis. Composition differs by stratum: the top 1% footprint is investment-driven (96% of SO_2), the bottom 50% consumption-driven (88% of NH_3). Input–output footprint accounting carries well-known allocation sensitivity, and the 2060 horizon is a scenario, not a projection.

doi: 10.1021/acs.est.6c04314

Lim et al. 2026 Nat Commun

Public knowledge, attitudes and practices on air quality in the ten most populous countries

Quota sample of **10,618 respondents across 1,372 localities** in Bangladesh, Brazil, China, Indonesia, India, Mexico, Nigeria, Pakistan, Russia and the United States, linked to modelled $PM_{2.5}$. Exposure is *inversely* associated with knowledge about air quality but *positively* associated with protective practices, attitudes and concern — people in dirtier air act more and know less. Interactions with age, sex, education, residence, political ideology, marital, employment and smoking status are all significant. Quota sampling is not probability sampling, so national point estimates should not be read as prevalence; the cross-country contrasts are the durable part. A useful anchor for anyone arguing that a dense sensor network has a communication function as well as a measurement one.

doi: 10.1038/s41467-026-75908-7

Liu Z et al. 2026 Environ Sci Technol

Health co-benefits of China's 2035 NDC arrive before the climate benefits do

Integrated assessment over 2025–2060 puts the NDC's net benefit at **+1,500.5 billion USD**, decomposing into **3,924.3 billion** in long-run climate benefits and **1,659.5 billion** in health co-benefits. The finding is the timing and the distribution, not the total: in the near term the NDC *reduces* climate benefit, especially in underdeveloped regions, because coal generation balances renewable intermittency and rapid afforestation releases soil organic carbon; long-run benefits reach all regions but stay concentrated in high-income ones. Air-quality health co-benefits are the component that accrues locally and immediately, which is the argument for treating PM monitoring as mitigation-policy infrastructure. Monetised co-benefits inherit every assumption in the concentration–response and valuation chain and should be read as scenario contrast, not as a forecast.

doi: 10.1021/acs.est.6c06024

Shi et al. 2026 Environ Sci Technol

Carbon–air pollution cobenefits and emission leakage in China's steel industry

Unit-level emission database covering **2,866 enterprises and ∼4,030 units**, relating regional energy-industrial configuration to mitigation priorities. Steel CO_2 peaked at 1.99 Gt in 2019 while SO_2 and PM fell **more than 80%** over 2011–2022 — the decoupling this watch keeps seeing. Over 2023–2050 the priorities diverge: production restraint gives the largest cumulative CO_2 reduction (22.1%) while **obsolete-capacity retirement dominates air-pollutant reduction (22.3%)**, so a carbon-optimal pathway is not a PM-optimal one. Regionally differentiated pathways induce **spatial emission leakage**, stronger for CO_2 (17.39% in 2030, falling to 2.38% by 2050) than for pollutants, and national mean pollutant concentrations fall only 5.4–12.3% by 2030 relative to 2022. Sector-specific and inventory-based.

doi: 10.1021/acs.est.6c08461

Simion et al. 2026 Lancet Reg Health West Pac

Climate-related health attribution in Southeast Asia: what is and is not being studied

Scoping review, **5,294 records screened, 77 included**. Air pollution is the subject of **70% of studies**, linked to up to 31% of all-cause mortality and an estimated **296,000 annual deaths from wildfire smoke during El Niño events**; heat is second, associated with up to 8.61% of mortality in Bangkok. Floods, droughts, extreme rainfall, storms and sea-level rise are near-absent. The paper's own framing — that the research portfolio is misaligned with the regional hazard profile — is the transferable finding; the attributable fractions themselves are heterogeneously derived across 77 studies with no standardised attribution framework, which the authors name as the central methodological gap. For a monitoring programme the operative point is that the region's dominant PM source is episodic biomass burning, which mass-only fixed networks under-sample in both space and time.

doi: 10.1016/j.lanwpc.2026.101952

Song et al. 2026 Environ Sci Technol

Shifting $PM_{2.5}$ control from mass to toxicity for health equity worldwide

Perspective synthesising toxicological, epidemiological and modelling evidence against the uniform-toxicity assumption underlying mass-based standards. The empirical hook is that age-standardised attributable death rates diverged worldwide over the past decade despite falling mass. Incomplete-combustion sources — residential solid fuel and wildfire above all — carry the highest toxicity per unit mass from primary emission through secondary ageing. Toxicity-adjusted exposure is substantially elevated in South Asia, Southeast Asia, sub-Saharan Africa, Andean Latin America and China, and the **8.2% of the world that is low-income bears 24.0% of the cumulative toxicity-adjusted exposure burden**. It is a perspective, not a quantitative assessment: the toxicity weights are inherited from a heterogeneous literature and the equity arithmetic is only as good as they are.

doi: 10.1021/acs.est.6c05282

Zhao N et al. 2026 J Hazard Mater

Attainment pressure under China's revised GB 3095-2026 standard in Shandong

Daily observations of six criteria pollutants across 16 Shandong cities, 2015–2025, condensed into a relative attainment pressure index combining concentration gap and additional non-attainment days, with random forest, SHAP and monotonicity-constrained XGBoost on socioeconomic predictors. **$PM_{2.5}$, PM_{10}, SO_2, NO_2 and CO fell 41.0–79.6%; O_3 did not move, staying at 166–191 $\mu g\,m^{-3}$.** Under the transition-stage limits a western high-pressure belt emerges — Heze 98.7, Dezhou 91.3, Liaocheng 89.1. Civilian vehicle ownership and secondary-industry share are the leading predictors for particulates and NO_2; the socioeconomic set explains $PM_{2.5}$ far better than O_3. Even under the deep-transition pathway **eight of sixteen cities remain above 30 $\mu g\,m^{-3}$ in 2030**. One province, and the scenario arithmetic assumes the historical predictor–concentration relation holds forward.

doi: 10.1016/j.jhazmat.2026.143460

Other clinical endpoints 9 records

Carried from the daily issues — full summaries (9 of 9 records in this subtopic)

Batisse et al. 2026 Environ Epidemiol

Prenatal ultrafine particles and childhood cancer in 965,000 children

Cohort of over **965,000** children born in Montreal or Toronto 1999–2020, with high-resolution outdoor UFP number concentration *and mean UFP size* assigned to six-digit residential postal codes from annual tax records, followed to age 14 (mean 9.1 years) against the Canadian Cancer Registry. **1,625** incident cancers. Per 10,000 particles/cm^3 of prenatal exposure: acute lymphoblastic leukaemia **HR 1.12 (0.94–1.34)**, neuroblastoma **1.15 (0.85–1.57)**; both intervals include the null, both persisted across exposure windows and sensitivity analyses, and other sites were null. Adjusting for UFP size as well as number is the methodological advance — number and size are correlated and separately toxicologically relevant. The honest summary is a consistent, imprecise positive signal, not a demonstrated association.

doi: 10.1097/ee9.0000000000000528

Chen K et al. 2026 Sichuan Da Xue Xue Bao Yi Xue Ban

$PM_{2.5}$ components and incident frailty in CHARLS

CHARLS 2011–2020, n**=9,603** middle-aged and older adults, frailty index outcome, exposures from the Tracking Air Pollution product, time-dependent Cox models. Per IQR of the preceding year: $PM_{2.5}$ **HR 1.100 (1.015–1.193)**; by component, organic matter **1.142 (1.055–1.236)**, black carbon **1.131 (1.041–1.229)**, ammonium **1.098 (1.010–1.194)**, sulfate 1.088 (0.996–1.187) and nitrate 1.081 (0.996–1.172). Carbonaceous components rank above bulk mass and above the secondary inorganic ions, and the two that cross the null are exactly the two secondary ions — an ordering consistent with combustion-source toxicity rather than with mass. The components co-vary strongly in the same exposure surface, so the ranking is suggestive and the six estimates are not six independent tests.

doi: 10.12182/20260760103

Ding E et al. 2026 Environ Sci Technol

Compound climate extremes and influenza, with a submultiplicative interaction

630,120 laboratory-confirmed influenza cases from hospitals in **335 Chinese cities**, 2005–2019, analysed by case-crossover with distributed lag non-linear models, machine-learning variable ranking, and an LSTM forecasting proof of concept for Beijing. Temperature, humidity and particulate matter rank as the primary contributors. The result worth keeping: co-exposure to multiple extreme events carries higher joint risk than single exposure, but the interaction is **submultiplicative on the multiplicative scale with no evidence of additive interaction** — so the joint effect is less than the product and the naive stacking of extreme-event risks overstates it. Adding extreme-event terms improved daily forecasting in the Beijing model. Caveats: the abstract reports no effect sizes at all, hospital-based case ascertainment varies across 335 cities and 15 years, and one-city forecasting is a demonstration, not a validation.

doi: 10.1021/acs.est.6c04755

Donaldson et al. 2026 Sports Med Open

Race-day air pollution and marathon finishing time across 2.76 million finishes

2,756,553 net finishing times from six World Marathon Majors (Berlin, Boston, Chicago, London, New York, Tokyo), 75 marathon-year events, 2010–2024, with race-day NO_2 and $PM_{2.5}$ from CAMS global reanalysis and linear mixed models adjusting heat index, wind and temporal trend. Each 1 $\mu g\,m^{-3}$ NO_2 was associated with **+1.43 min (95% CI 0.66–2.21) in men and +1.56 min (0.63–2.48) in women**, with a steep gradient by ability — recreational 2.12 vs. elite 0.34 min per $\mu g\,m^{-3}$, persisting as a percentage of mean time (0.67% vs. 0.18%) — and by age (2.17 min in 18–29-year-olds vs. 1.32 in $\geq$60). **$PM_{2.5}$ was null** (-0.13 min, -0.67 to 0.40 in men). The exposure is reanalysis at city scale, not course-level or personal, and NO_2 at that resolution is a traffic-proximity proxy; the $NO_2/PM_{2.5}$ divergence is therefore at least partly a measurement-error contrast rather than a toxicological one. Still the largest exercise-performance exposure dataset assembled.

doi: 10.1186/s40798-026-01099-6

Feng et al. 2026 Joint Bone Spine

Air pollutant mixtures, plasma metabolic and proteomic signatures, and incident arthritis

UK Biobank, **401,676** arthritis-free at baseline, **78,188** incident cases over a median 13 y. Mixture exposure quantified by weighted quantile sum regression; metabolic and proteomic signatures derived by elastic net; Cox models plus generalised propensity scores; causal mediation for the signatures. The dominant mixture contributors are subtype-specific — NO_x/NO_2 for rheumatoid arthritis, PM for osteoarthritis, NO_x and PM for gout, PM for psoriatic arthritis — and the omics signatures mediated **4.92–35.05%** of the association. The abstract reports no hazard ratios or intervals for the WQS index itself, which is why nothing from this paper appears in the forest plot. WQS indices are also not comparable across studies, and the mediation proportions rest on a no-unmeasured-confounding assumption that plasma omics cannot support. The subtype-resolved contributor pattern is the transferable result.

doi: 10.1016/j.jbspin.2026.106129

Park J et al. 2026 Int J Epidemiol

Ozone, disability and emergency department use in Korea

Time-stratified case-crossover on **3,484,877** ED visits from the Korean National Health Insurance Service (2015–19, 2022–23), exposure the 3-day moving average of daily 8-hour maximum O_3, adjusted for daily mean temperature and holidays. Per IQR ($\sim$22 ppb): **+1.94% (1.64–2.25)** in persons with disabilities during the warm season and +1.93% (1.62–2.25) year-round, against **+1.36% (0.92–1.80)** in persons without. A roughly 40% larger relative effect in a population defined administratively rather than clinically, with an interaction term tested across temperature strata. No particulate metric is included, which is both the limitation and, read against Gao et al. in this issue, part of the point: a disability register can identify a susceptible group that pollutant-specific monitoring networks were never designed to serve.

doi: 10.1093/ije/dyag189

Wang C et al. 2026 Environ Sci Technol

Green space and cardiovascular health where the air-quality pathway is 75% of the model and the effects are fragile

Two-stage gradient-boosting plus structural equation modelling on a clinical cohort of **23,312** patients in Nanjing. **Air-quality-related pathways account for more than 75% of the relative statistical contribution to model variance**, exceeding psychological-restoration proxies — which, if it holds, says green-space cardiovascular benefit in a dense polluted city is mostly an air-quality mechanism. Distinct networks emerge for chronic ischaemic heart disease (vegetation vitality inverse to NO_2 and SO_2, linked to metabolic markers) and heart failure. The paper is unusually candid about its own weaknesses and they are disqualifying for the headline: the macroscale aerodynamic interpretation is called speculative for want of morphological or fluid-dynamics data, and the age-stratified canopy-density $\times$ $PM_{2.5}$ slopes are **nonsignificant with extremely small effect sizes**. Relative variance contribution is not effect size.

doi: 10.1021/acs.est.6c08296

Zhang & Yu 2026 (preprint) Res Sq (preprint)

Air pollution is detectable in ED demand and useless for forecasting it

Single-centre daily time-series at Tongji Hospital Guanggu Branch, Wuhan, 1 Jan 2024 to 26 Jun 2026: **906 dates, 391,344 visits**, mean 431.9/day (SD 59.2), falling to 319.3/day over Spring Festival. Quasi-Poisson models give, per IQR of lag 0–2 pollutant-specific AQI, RR **1.028 (1.015–1.042)** for $PM_{2.5}$, 1.030 (1.017–1.043) for PM_{10}, 1.049 (1.034–1.064) for NO_2 and an inverse 0.980 (0.965–0.996) for O_3; each 1 °C is +0.85% visits. The contribution is the temporally-ordered validation: a history-plus-calendar model achieves MAE 33.6 visits/day, meteorology reduces it to 33.2, and **adding air-pollution AQI does not improve it at all**. Single centre, city-level AQI rather than catchment exposure, unrefereed. The inverse O_3 term is almost certainly seasonal confounding. Read as a warning that a statistically robust exposure–response can carry no incremental predictive information.

doi: 10.21203/rs.3.rs-10496137/v1

Zheng M et al. 2026 Ann Med

Occupational particulate exposure ranks with alcohol in an ecological cancer model

GBD 2021 incidence for adults $\geq$**55** across **203 countries**, 1990–2021, classified into quartile-based co-burden categories for oesophageal and lip/oral-cavity cancer, with 58 candidate exposures screened by random forest and SHAP to 36, then to 25 after variance-inflation screening, entering negative binomial models with population offsets. Countries split into consistent (73), oesophageal-dominant (61) and LOC-dominant (69). **Occupational exposure to particulate matter, gases and fumes carries RR 1.29 (1.13–1.49)** for oesophageal cancer, comparable to high alcohol use (1.32, 1.13–1.53) and above low nut/seed intake (1.16). The authors state the correct caveat themselves: these are **country-level ecological associations** and establish no individual-level causality. GBD exposure estimates for occupational PM are modelled, not measured, so the estimate is a model-on-model contrast.

doi: 10.1080/07853890.2026.2721134

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Barbalat et al.	Association of heat, cold, and ambient pollution in the first 1000 days of life with autism spectrum disorder risk indicators in two-year-olds: findings from the ELFE national birth cohort	*Environ Sci Technol*	Prospective birth cohort + DLNM	Weekly mean $PM_{2.5}$, PM_{10}, NO2 from 1 km daily outdoor predictions (200 m for NO2/ temperature in large urban areas)	France
Kuznetsov et al.	Particulate matter air pollution and life satisfaction: links across and within twin pairs	*Environ Res*	Twin-pair longitudinal comparison	Residential $PM_{2.5}$ and PM_{10}, repeated across six waves	Germany
Paulino et al.	Neural exposome burden stratifies cognitive trajectories in a subacute-to-chronic mild TBI outpatient cohort: an exploratory dimension-reduction study	*Int J Neurosci*	Prospective cohort + PCA	Residential $PM_{2.5}$ as one of 13 exposome variables entering a principal-component burden score, alongside education, area deprivation and adverse childhood events	North America
Ren X et al.	Association between $PM_{2.5}$ constituents and non-suicidal self-injury among Chinese adolescents: modifying role of biological rhythm	*Front Public Health*	Cross-sectional survey	$PM_{2.5}$ total mass and constituents in logistic and weighted quantile sum regression, stratified by multidimensional biological-rhythm score	China
Cardiovascular & metabolic					
Chen N et al.	Associations of $PM_{2.5}$ components, residential greenness, and heatwaves with incident venous thromboembolism	*Eur J Prev Cardiol*	Prospective cohort (mixtures)	Elemental carbon, organic matter, sulfate, ammonium and nitrate at 3x3 km; NDVI in a 300 m buffer; heatwaves under several temperature thresholds	United Kingdom
De Potter et al.	Influences of socioeconomic and environmental factors on STEMI incidence and related mortality: insights from the Belgian Environmental Myocardial Infarction Risk (EMIR) study	*Eur Heart J Acute Cardiovasc Care*	Time-stratified case-crossover + DLNM	Acute and chronic residential $PM_{2.5}$, NO2, O3, temperature, noise and greenspace from high-resolution models; per IQR	Belgium
Fang et al.	Joint association of air pollutants and lifestyles on early-/late-onset heart failure: a population-based prospective cohort study	*Eur J Prev Cardiol*	Prospective cohort	Composite air-pollution score over NO2, NOx, PM_{10}, $PM_{2.5}$ and $PM_{2.5}$-10, crossed with a six-component lifestyle score; multiplicative and additive interaction	United Kingdom

First author	Focus	Journal	Design	Metric	Region
Jiang Z et al.	Long-term exposure to multiple air pollution and trajectory risk of cardio-renal-metabolic multimorbidity	*Atmos Environ*	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)
Pan SG et al.	Ozone and possible heat effects on out-of-hospital cardiac arrest: a time-series study	*Front Public Health*	Distributed-lag time-series	Daily mean temperature over lag 0-14 in a quasi-Poisson DLNM, plus lag0-1 $PM_{2.5}$, PM_{10}, NO2, SO2, CO and O3 in single- and two-pollutant models	China
Sun D et al.	Exploring China's Clean Air Act and associated cardiovascular disease risk: a prospective, quasi-experimental, and causal inference modelling study	*Lancet Planet Health*	Quasi-experimental (difference-in-differences)	Long-term $PM_{2.5}$, PM_{10} and O3; municipal PM-reduction target as the intervention assignment	China
Wang L et al.	Joint exposure to small dense LDL-C and long-term $PM_{2.5}$ and risk of incident stroke	*Lipids*	Prospective cohort + trajectory clustering	Annual mean $PM_{2.5}$ from the ChinaHigh$PM_{2.5}$ dataset, 2011-2020	China
Reproductive & developmental					
Chen Y et al.	The impact of $PM_{2.5}$ component exposure during early pregnancy on congenital heart disease: evidence from Guangdong, China	*Sci Rep*	Case-control (registry)	Satellite-derived $PM_{2.5}$ components (OM, sulfate, nitrate, ammonium, BC) by week of early pregnancy; quantile g-computation for mixture weights	China
Gao et al.	Exposure biomarkers of airborne organic pollutants in pregnant women and developmental risks: evidence from epidemiology to in vitro assessment	*Environ Sci Technol*	Prospective cohort + in vitro screening	20 urinary biomarkers of airborne organics (6PPD, 6PPD-Q, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine) by LC-MS/MS	China
Mbishi et al.	Early-life exposure to particulate matter and green spaces: Associations with infant growth	*Environ Pollut*	Prospective cohort	Perinatal PM_{10} and $PM_{2.5}$ from the RIO detrended-kriging interpolation model; municipal vegetation split into low, high and total green space	Belgium
Tanner et al.	Prenatal environmental determinants of aromatase brain-promoter methylation in cord blood: chemical, airborne, pharmacological, and nutritional factors	*Environ Epigenet*	Birth cohort + weighted quantile sum regression	Air pollution as 1 of 25 prenatal exposures in a WQS mixture index	Australia
Wei Q et al.	Prenatal exposure to $PM_{2.5}$ components and fetal growth: identifying critical gestational weeks and key toxic components in a Chinese birth cohort	*Int J Epidemiol*	Prospective birth cohort	Weekly black carbon, organic matter, nitrate, sulfate and ammonium from the TAP real-time exposure model, entered into distributed lag models with GEE plus ML mixture analysis	China

First author	Focus	Journal	Design	Metric	Region
Xu ZL et al.	Timing matters: component-specific $PM_{2.5}$ exposure windows and oocyte development in assisted reproduction	*Biomed Environ Sci*	Retrospective cohort + DLNM	Individual $PM_{2.5}$ and its sulfate, nitrate, ammonium, organic matter and black carbon fractions over recent (0-3 mo), distal (4-12 mo) and cumulative (0-12 mo) windows	China
Zanini et al.	Maternal exposure to air pollution and fetoplacental Doppler markers during pregnancy	*Environ Pollut*	Metadata only	Maternal residential air-pollution exposure during pregnancy	Europe
Respiratory & allergic					
Cui L et al.	Epidemic thunderstorm asthma in China: a systematic review of an Artemisia-associated phenotype	*Clin Rev Allergy Immunol*	Systematic review / meta-analysis	Airborne pollen as the causal particle; most Chinese reports identify sensitisation only to genus-level Artemisia, without species-level aerobiological confirmation	China
Gomez-Olarte et al.	Early-life exposure to ambient air pollution and respiratory outcomes: mixture and sex-stratified associations with inflammatory cytokine responses	*Environ Pollut*	Prospective birth cohort	Residential annual $PM_{2.5}$, PM_{10}, NO2; per IQR 2.0 / 2.8 / 9.1 ug m-3	Germany
Li H et al.	Multi-omics causal inference of childhood asthma triggered by ambient particulate matter	*ERJ Open Res*	Mendelian randomisation + TWAS	Genetically instrumented ambient $PM_{2.5}$ from GWAS summary statistics	Europe
Wangupadcha et al.	Short-term effects of ambient $PM_{2.5}$ on respiratory healthcare utilisation in older adults: a time-series study in an industrialising area of Central Thailand	*J Glob Health*	Time-series	Daily $PM_{2.5}$ from the nearest government monitor, single-lag and moving-average negative binomial models per 10 $\mu g/m^3$	Thailand
Wright et al.	Air pollution and heat impacts on respiratory morbidity and mortality outcomes in Africa: a systematic review towards a meta-analysis	*Curr Environ Health Rep*	Systematic review / meta-analysis	Reported indoor and ambient $PM_{2.5}$ and nitrogen oxides across included case-control studies; no pooled exposure-response	Sub-Saharan Africa
Mechanistic toxicology					
Arora et al.	Oxidative potential of fresh vs. O3-aged $PM_{2.5}$ across urban and rural sources in China	*J Environ Sci (China)*	Commentary / methods letter	$PM_{2.5}$ oxidative potential by DTT assay, fresh vs ozone-aged, source-resolved	China
Cao et al.	Interfacial chemistry and multiscale aging of atmospheric black carbon: from surface reactions to coating, morphology, and model parameterization	*Environ Sci Atmos*	Narrative / critical review	Black carbon particle state - surface reactivity, coating thickness, morphology and aging timescale - as the link between emission and optical, cloud and toxicological behaviour	Global / multi-region
Garcia Dominguez et al.	Ultrafine airborne particle deposition and clearance in an avian lung under ecologically relevant conditions	*Environ Sci Technol*	Animal inhalation exposure + dosimetry	50 and 100 nm model ultrafine particles at 5 or 25 degrees C	Sweden
Han G et al. 2026	Picropodophyllotoxin attenuates $PM_{2.5}$-induced pulmonary injury via modulation of oxidative stress and mTOR-mediated autophagy	*Research Square (preprint)*	Experimental / toxicology	$PM_{2.5}$ dosed in vitro at 25-100 ug/mL	East Asia (ex-China)

First author	Focus	Journal	Design	Metric	Region
Heo et al.	PM_{10} exposure impairs differentiation and induces inflammation in human small airway epithelial cells cultured at the air-liquid interface	*Tuberc Respir Dis (Seoul)*	In vitro exposure	Repeated PM_{10} dosing of differentiated air-liquid-interface cultures; TEER, lineage markers, surfactant proteins, cytokines, Notch and NF-kB signalling	South Korea
Jerome et al.	Prolonged brain cytokine elevation and microgliosis following exposure to World Trade Center dust particulate matter in male spontaneously hypertensive rats	*J Immunotoxicol*	Animal exposure study	WTC dust, intratracheal, 2.43 (+/-0.78) mg/day x 2 d	United States
Luo et al.	Impacts of Environmental Pollutants on Innate Immunity: The Role of Group 2 Innate Lymphoid Cells	*Clin Rev Allergy Immunol*	Narrative review	PM, diesel exhaust particles, O3, cigarette smoke, nanomaterials, microplastics	Multi-country
Meldrum & Leonard	Pollutant particle priming amplifies airway compartment specific neutrophil and macrophage inflammatory programs following house dust mite acute exposure	*Front Immunol*	Murine inhalation exposure	Diesel exhaust particles, temporally separated priming dose before allergen challenge	United Kingdom
Na HW et al.	Adverse effects of airborne heavy metals on human skin cells: insights from sebum analysis of indoor and outdoor workers	*J Toxicol*	Biomonitoring + in vitro	24 metals quantified by ICP-MS in sebum, then reconstituted as sebum-mimicking metal mixtures at the measured indoor and outdoor profiles	China
Ren Y et al.	Unveiling particulate matter-lung interactions from static models to dynamic lung-on-chips	*Research (Wash D C)*	Method review	Compares 2D culture, 3D bioprinted constructs, organoids and lung-on-chip platforms for reproducing PM deposition and breath-mimicking mechanical strain	Global / multi-region
Wang Y et al.	Contrasting cellular reactive oxygen species modulation by alpha-dicarbonyl aqueous reaction products via interactions with redox-active particulate matter	*Environ Sci Technol Lett*	Bench / device experiment	Aqueous reaction products of glyoxal or methylglyoxal with ammonium sulfate or glycine, characterised by HRMS, then co-exposed with redox-active PM surrogates	Laboratory
Occupational & indoor					
Chen Y et al.	Household solid fuel combustion-related PAH exposure in rural northern China: associations of phenanthrene metabolites with DNA damage and systemic inflammation	*Atmos Environ*	Metadata only	Household solid-fuel PAH exposure indexed by urinary phenanthrene metabolites	China
Derwein et al.	Combined effects of face mask blocking, filtration, and ventilation strategy on exhaled air transmission and infection risk	*Build Environ*	Metadata only (no abstract)	Not retrievable: the publisher deposited no abstract to Crossref, Europe PMC or Semantic Scholar at index time	Not stated (metadata only)

First author	Focus	Journal	Design	Metric	Region
Feng R et al.	Clean heating renovation differentially reduces nitro-PAH and nitrated phenol exposure: implications from real-world personal monitoring and NF-kB-based toxicity	*Environ Sci Technol Lett*	Personal exposure monitoring + in vitro	Size-resolved personal exposure to particulate nitrated phenols and nitro-PAHs among users of clean coal, raw coal chunk and biomass	China
Gu et al.	Nitrogen-containing organic compounds in household solid-fuel combustion PM and residential urinary inflammatory and targeted amino-acid profiles: an exploratory personal exposure study	*Environ Int*	Metadata only	Personal PM from household solid-fuel combustion; N-containing organic speciation	Not stated (metadata only)
Ibrahim	From complaint-driven response to preventive management: Strengthening indoor air quality management in Malaysian buildings	*Qeios* (*preprint*)	Commentary / methods letter	Argues for routine indoor monitoring of particulate matter and outdoor-to-indoor transport alongside CO2, rather than complaint-triggered investigation under ICOP IAQ 2010	Malaysia
Illa et al.	How are we exposed to chemicals when drugs are smoked indoors?	*Environ Sci Technol*	Mechanistic indoor fate and exposure model	Indoor smoke aerosol; 7 compounds spanning volatility and hydrophobicity; first-, second- and third-hand pathways	Global (computational)
Jaiswal et al.	Using nature-based solutions in mitigating indoor air quality: a case study of EDM emission from a gear manufacturing unit, India	*Research Square* (*preprint*)	Trial / intervention	Indoor $PM_{2.5}$ and CO before and after five months of NBS filtration plus ventilation redesign	India
Jamali et al.	Wooden-framed all-cellulose HEPA nanofilter for sustainable air purification	*Int J Biol Macromol*	Laboratory instrument development	PM0.3 removal efficiency and pressure drop of a fog-sprayed, freeze-dried bacterial-cellulose nanofibre layer on a cellulose-microfibre substrate	Laboratory
Jin et al.	Synergistic optimization of air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling	*Build Environ*	Metadata only	Bioaerosol transport and deposition during dental ultrasonic scaling	China
Kusi et al.	Mechanistic pathways of biochemical and toxicological responses to waste tyre combustion emissions: A systematic review	*Research Square* (*preprint*)	Systematic review (PRISMA)	Particulate matter, PAHs, VOCs and heavy metals from open waste-tyre burning, mapped to organ-specific biochemical and toxicological endpoints	Global / multi-region
Li et al.	Study on the Non-uniform Distribution Characteristics of Oil Mist and Local Ventilation Method in Machining Workshop	*Aerosol Air Qual Res*	Metadata only	Oil-mist aerosol, spatial distribution	China
Lv H et al.	Structurally engineered polyimide microsphere/nanofibrous gradient aerogels with enhanced dust-holding capacity for high-temperature air filtration	*ACS Appl Mater Interfaces*	Laboratory instrument development	PM0.3 filtration efficiency, pressure drop and dust-holding capacity of a single-component polyimide gradient aerogel	Laboratory

First author	Focus	Journal	Design	Metric	Region
Ma L et al.	Evaluating surgical smoke in laser-assisted laparoscopic partial nephrectomy with a novel handle: an in vitro study	*Can J Urol*	Bench / device experiment	Surgical-smoke formaldehyde, TVOC and CO with and without an integrated aspirator handle	Laboratory
Mbazima et al.	Exposure and respiratory tract deposition of particulate matter among teachers using chalkboards in selected public secondary schools in the Johannesburg region	*Int J Environ Health Res*	Measurement campaign	Respirable PM4, personal breathing-zone sampling + MPPD deposition model	Sub-Saharan Africa
Micallef & Scerri	A flow conditioned infiltration metric for traffic-related air pollution ingress and indoor air quality in naturally ventilated buildings	*Build Environ*	Metadata only	Outdoor-to-indoor infiltration of traffic-related air pollution, conditioned on flow regime	Europe
Srinivasan et al.	Occupational wood dust exposure and lung function impairment: a systematic review and meta-analysis	*Natl Med J India*	Systematic review / meta-analysis	Occupational wood dust exposure, categorical (exposed vs unexposed); no quantitative dust concentration pooled	Global / multi-region
Wei J et al.	Welding fume exposure and respiratory innate immune activation: metal-rich particles, inflammatory biomarkers, and immune-informed occupational prevention	*Front Immunol*	Mechanistic review	Metal-rich fine and ultrafine welding-fume particles as a stimulus to airway epithelium, alveolar macrophages, neutrophils and circulating immune cells	Global / multi-region
Wood et al.	Influence of ventilation rate on the potential for indoor particulate matter exposure	*Indoor Environments*	Metadata only	Indoor PM exposure potential as a function of ventilation rate	United Kingdom
Yang et al.	Optimization of ventilation systems in large welding workshops with multiple dust sources: a case study of a 300 m long welding workshop	*Atmosphere*	Chemical transport modelling	Welding fume transport by CFD; low-density aluminium vs high-density carbon-steel fume	China
Sensing, forecasting & instrumentation					
Alderotti et al.	Raman spectroscopy resolves nitrogen-driven metabolic acclimation to urban air pollution in Quercus ilex	*Sci Total Environ*	Bench / device experiment	Leaf Raman band intensities (chlorophyll, carotenoid, flavonoid) as a non-destructive biomonitoring index along an NO2 and PM_{10} gradient	Europe
Alraddadi et al.	Explainable one-hour-ahead air quality index forecasting across Saudi cities using pollutant and meteorological data	*Atmosphere*	Algorithm development	Hourly AQI (PM-dominated) with and without surface meteorological predictors	Saudi Arabia
Blaga et al.	City-scale calibration of a low-cost $PM_{2.5}$ network for regulatory-compliant air-quality assessment	*PLOS Clim*	Sensor co-location + seasonal MLR calibration	$PM_{2.5}$, Clarity Node-S vs NEPA reference stations	Romania
Cai J et al.	Scale-dependent drivers shape airborne microbiomes across the Shanghai subway network	*Environ Sci Technol*	Environmental surveillance campaign	Paired TSP (<=100 um) and $PM_{2.5}$ sampling with microbial taxonomy, transcriptional potential, $PM_{2.5}$ chemical composition, microclimate and network topology	China

First author	Focus	Journal	Design	Metric	Region
Chaparro et al.	Monitoring airborne fine particles with biological and technological sensors for preventive conservation of cultural heritage sites	*Sci Total Environ*	Metadata only	Airborne fine particles measured in parallel by biomonitors and instrumental sensors	Latin America
Chen C et al.	Characteristics and gas-particle partitioning of amine-containing particles from firework emissions: new insights from machine learning and single-particle mass spectrometry	*J Hazard Mater*	Real-time sensor campaign	Single-particle aerosol mass spectrometry of amine-containing particles; modified aerosol acidity Rra′	China
Chen et al.	Lake Breeze Effects on the Urban Environment in Chicago	*Environ Sci Technol*	Sensor-network event analysis (lake-breeze compositing)	$PM_{2.5}$ and O3 from the Microsoft Eclipse low-cost sensor network plus meteorological stations, 1 year	United States
Cui C et al.	Explainable machine learning prediction of long-term variations in cloud condensation nuclei at a high mountain site in the North China Plain	*Environ Sci Technol*	Data-fusion / ML surrogate model	CCN number concentration reconstructed from particle number size distribution and chemistry; bulk activation ratio and hygroscopicity parameter	China
Delkash	Receptor-scale imprint of barometric pressure variability on community air monitoring near an elevated-temperature landfill: a multi-method analysis across two mitigation interventions	*Environ Pollut*	Metadata only	Community-monitor receptor concentrations against barometric pressure variability, across two landfill mitigation interventions	United States
Galioto et al.	Physics-informed machine learning for urban nitrogen dioxide forecasting in Palermo, Italy	*Atmosphere*	Chemical transport modelling	Hourly NO2 at 11 m resolution: COPERT-Italy emission factors dispersed through the SIRANE street-network model, then fused with measurements in an ML stage	Europe
Giang et al.	Cumulative and real-time performance of consumer-grade electronic radon monitors in a Canadian single-family home over three months	*J Radiol Prot*	Sensor co-location / calibration	Radon activity concentration from 45 consumer monitors against an AlphaGuard PQ2000 and two alpha-track detectors; $PM_{2.5}$, CO2, T and RH co-recorded	North America
Guth et al.	Exploring the variability in emissions for broadcast and pile prescribed burns using low-cost sensors	*Atmosphere*	Field emission-factor campaign	Low-cost-sensor emission factors for CO and $PM_{2.5}$, with filter-based EC/OC and speciated organics, resolved by modified combustion efficiency	North America
He J et al.	Molecular packing engineering enables a high performance ppb-level room-temperature NO2 sensor	*ACS Sens*	Laboratory instrument development	Gas-phase NO2 (not particulate): chemiresistive response of cyano-substituted distyrylbenzene films with tuned alkyl-chain length and molecular packing	Laboratory

First author	Focus	Journal	Design	Metric	Region
Hong et al.	Predicting fungal and bacterial bioaerosol concentrations across diverse building types: a framework integrating generalized additive models and machine learning	*Indoor Air*	Cross-sectional + prediction model	Culturable bacterial and fungal bioaerosol concentrations vs indoor temperature and relative humidity	South Korea
Ichikawa et al.	Tower- and drone-based BVOC observations in a suburban Tokyo forest: methodological insights and MEGAN comparison	*Atmos Chem Phys*	Multi-instrument field evaluation	Isoprene and monoterpene mixing ratios and emission fluxes at multiple heights on a 30 m flux tower, cross-checked by drone sampling and against MEGAN	Japan
Ji et al.	Quantifying the aging of dust aerosols using Mie-polarization-fluorescence lidar: linking optical evolution to chemical transformations	*Atmos Res*	Metadata only (no abstract)	Dust optical properties by Mie-polarization-fluorescence lidar	China
Kaur et al.	Performance evaluation of Plantower's PMSX003N: laboratory and field analysis	*Aerosol Sci Technol*	Sensor co-location / calibration	$PM_{2.5}$ and PM_{10} from three PMSX003N, three PMS5003T and three Alphasense OPC-N3 against FEM reference; laboratory size-bin response to monodisperse DOS aerosol	North America
Kim C-h et al.	OAV-based source characterization of designated odor compounds in long-term continuous monitoring data with severe missingness	*Atmosphere*	Monitoring-record analysis	22 designated odour compounds at 20 min resolution; block-holdout imputation with BiLSTM, BiGRU, TCN, Transformer and HistGradientBoosting, then odour activity value and odour contribution	South Korea
Liu T et al.	Aerosol solute strength drives transition-metal-ion-catalyzed sulfate aerosol formation	*Environ Sci Technol*	Aerosol flow reactor + model implementation	Aqueous aerosol sulfate from SO2 + O2 with Fe(III)/Mn(II) catalysis; ionic strength threshold 16 mol/kg	Global / multi-region
Mohd Nadzir et al.	Short-term variability of trace gases and particulate matter in the Antarctic Peninsula atmosphere using a low-cost air sensor	*Atmosphere*	Real-time sensor campaign	Minute-scale NO2, O3, CO, SO2, $PM_{2.5}$, PM_{10} (plus TVOC in May) from a single low-cost multi-sensor node	Antarctica
Raso et al.	Influence of temporal resolution on the characterization of episodic air pollution events: experimental evidence from New Year's Eve fireworks	*Atmosphere*	Real-time sensor campaign	PM_{10}, NO2, CO, O3 at minute, hourly and daily resolution; certified reference instruments alongside calibrated low-cost sensors	Italy
Rosas et al.	Biomass burning and African dust influence on seasonal aerosol chemistry over the Western Caribbean (Part I): Atmospheric variability and human health	*Environ Sci Pollut Res Int*	Field campaign / remote sensing	Continuous $PM_{2.5}$, PM_{10} and particle-bound PAH with the $PM_{2.5}$/PM_{10} ratio as a regime discriminator, joined to FIRMS fire detections and HYSPLIT back-trajectories	Mexico

First author	Focus	Journal	Design	Metric	Region
Saad et al.	Short-term monitoring of air pollutants at the port-urban exposure zone, Alexandria Port	*Environ Monit Assess*	Portable sensor monitoring campaign	$PM_{2.5}$, PM_{10}, CO2, CO, NO2, SO2, NH3 and VOC by portable multi-gas monitor and particulate sensors	Egypt
Senarathna et al.	Enhancing low-cost $PM_{2.5}$ air quality monitoring sensors through sensor calibration using regression and echo-state network models	*Environ Monit Assess*	Sensor co-location + regression / echo-state network calibration	$PM_{2.5}$, TSI BlueSky vs reference, wet and dry season	Sri Lanka
Sengupta et al.	A temporal deep learning framework for calibration of low-cost air quality sensors	*arXiv (preprint)*	Sensor co-location + LSTM sequence calibration	$PM_{2.5}$, PM_{10} and NO2, LCS vs co-located reference (OxAria)	United Kingdom
Seong & Hoque	Influence of boundary layer conditions and surface properties on bacterial interaction with indoor built surfaces	*Front Pediatr*	Laboratory metrology	Attachment fraction and centrifugal detachment of Corynebacterium sp. on carpet, wood, metal and glass, with near-surface velocity fields by particle image velocimetry	Laboratory
Shafi & Scafetta	Management of missing air pollution data within urban environments using machine learning regressions: a case study for Delhi, India	*Environ Sci Pollut Res Int*	Data-fusion / ML surrogate model	Iterative reconstruction of missing daily $PM_{2.5}$, PM_{10}, O3, NO2, SO2 and CO from the four most-correlated neighbouring stations; 35 candidate regressors per gap	South Asia
Shi S et al.	Potential aerosol tradeoffs of vessel speed reduction for fresh ship plume composition and tracer coverage	*npj Clim Atmos Sci*	Measurement campaign	Ultrafine and coarse modes of fresh ship plumes; V/VO and sulfate markers	China
Veselovskii et al.	Evaluation of smoke mass concentration within the PBL based on observations of fluorescence lidar with several discrete channels	*Atmos Meas Tech*	Remote sensing / lidar retrieval	Smoke mass concentration retrieved from five-channel fluorescence lidar spectra, separating smoke from urban aerosol; detection threshold $0.1\ \mu g/m^3$	Europe
Wang S et al.	Impact of fuel moisture on chaparral emissions under flaming and smoldering conditions	*Fire Saf J*	Chamber / laboratory	PM and gas emission factors by FTIR, flaming vs smoldering	North America
Xing et al.	Multisource observation-constrained tuning for air quality forecasting in a machine learning-enhanced NOAA Unified Forecast System	*Environ Sci Technol*	Data-fusion / ML surrogate model	Surface $PM_{2.5}$ and O3 fields reconstructed from VIIRS AOD, TEMPO NO2 and EPA AirNow, used as initial conditions for 72 h UFS-AQM forecasts	North America
Yohanna & Lim	Regional and seasonal variation of $PM_{2.5}$ and ozone across Malaysia using XGBoost and SHAP	*Chemosphere*	Gradient-boosting model + SHAP interpretability	Daily $PM_{2.5}$ and O3 from the DOE monitoring network plus ERA5 reanalysis, 2018-2023	Malaysia

First author	Focus	Journal	Design	Metric	Region
Zeng et al.	Decoupling between $PM_{2.5}$ oxidative potential and light extinction: Roles of chemical composition and aerosol thermodynamic state	*J Hazard Mater*	Source apportionment	Volume-normalised oxidative potential and light extinction placed on a common quadrant framework, with water-soluble organic carbon, aerosol liquid water content, aerosol pH and PMF source factors as explanatory terms	Taiwan
Zhang Z et al.	A microfluidic multispectral diffraction platform for quantitative analysis of atmospheric black carbon aerosols	*Anal Chem*	Laboratory instrument development	Black carbon mass concentration from lensless multispectral (400-800 nm) diffraction fingerprints after microfluidic inertial separation and electrostatic capture	Laboratory
Exposure assessment & modelling					
Arpornwichanop & Chuersuwan	Life-stage-resolved screening of inhalation exposure to $PM_{2.5}$-bound elements in Nonthaburi, Greater Bangkok, Thailand	*Environ Geochem Health*	Exposure model (secondary data)	$PM_{2.5}$-bound elements; respiratory-deposited intake adjusted by simulated-lung-fluid solubility	Thailand
Arunachalam et al.	Residential gas combustion drives building-sector air pollution mortality in the U.S.	*Research Square (preprint)*	Health impact model	CMAQ Decoupled Direct Method sensitivities of NO2, O3 and $PM_{2.5}$ to building-sector precursor emissions, propagated through BenMAPR to premature deaths	United States
Boogaard et al.	Advancing long-term outdoor air pollution exposure assessment for health studies	*Curr Environ Health Rep*	Programme-level synthesis of exposure-model comparisons	Long-term outdoor $PM_{2.5}$, NO2 and BC; mobility- and infiltration-adjusted exposure models	Global / multi-region
Boyle et al.	Creating outdoor air pollution exposome metrics for assessing breast and lung cancer risks in the United States	*Cancer Epidemiol Biomarkers Prev*	Metric development + retrospective application	Census-tract criteria-pollutant concentrations (CACES land-use regression) plus RSEI industrial-release model, weighted by cancer-specific meta-analytic risk estimates	United States
Bravo et al.	No one is immune: neighborhood trends and disparities in air pollution exposure in the city of Chicago during summer 2023 wildfires	*J Expo Sci Environ Epidemiol*	Exposure model (secondary data)	Daily 24 h tract-level $PM_{2.5}$ from hourly 500 m gridded estimates, split by wildfire-smoke and non-smoke days	North America
Chi et al.	Heterogeneous nitrosation of amines driven by dinitrogen tetroxide: a missing source of particulate nitrosamines	*Atmos Chem Phys*	Born-Oppenheimer molecular dynamics + metadynamics	Particulate nitrosamines from methylamine / dimethylamine + N2O4 at the air-water interface	Global (computational)

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Engeroff et al.	Air pollution exposure during urban running: a randomized crossover comparison of particulate matter exposure on park, roadside, and residential routes with consideration of rush-hour timing	*Research Square (preprint)*	Mobile monitoring campaign	Mobile personal PM_{10}, $PM_{2.5}$ and PM1 during 2 km runs, referenced to local urban background $PM_{2.5}/PM_{10}$	Germany
Gao J et al.	Understanding unexpected ozone-health associations: are secondary oxidative pollutants key?	*Environ Sci Technol*	Narrative / critical review	Measured O3 as a proxy for an unmeasured oxidant field; argues low O3 can coincide with active oxidative chemistry producing toxic unmeasured secondary products	Global / multi-region
Guo & Sudo	Global simulation of polycyclic aromatic hydrocarbons (PAHs) with the CHASER chemistry-climate model	*Atmosphere*	Chemical transport modelling	Gas-particle partitioning of NAP, PHE, PYR and BaP under a coupled soot-adsorption / organic-matter-absorption scheme	Global
Guo et al.	Resolving systematic errors in sulfate source apportionment: a field-validated kinetic isotope fractionation framework	*Atmos Chem Phys*	Kinetic model + field validation	Seasonal $PM_{2.5}$ sulfate; stable-isotope Bayesian mixing plus Rayleigh fractionation	China
Gurler Akyuz & Akyuz	Seasonal and size-resolved elemental composition and source apportionment of atmospheric particulate matter in a coastal urban-industrial region of Zonguldak, Turkiye	*J Environ Sci Health A*	Field measurement	$PM_{2.5}$ and $PM_{2.5}$-10 elemental composition; enrichment factor, PCA and discriminant analysis	Turkiye
Hernandez Pardo et al.	Downward transport of tropical upper-tropospheric aerosols: multi-year idealized simulations	*Atmos Chem Phys*	Global chemistry-climate tracer simulation	19 idealised aerosol tracers, 20-100 nm, released in the tropical upper troposphere	Global / multi-region
Hulme et al.	Exposure of children to multiple long-term air pollutants ($PM_{2.5}$, BC, NO2) and noise at home and school in Accra, Ghana	*Environ Res Lett*	Exposure model (secondary data)	Annual mean $PM_{2.5}$, NO2 and BC from land-use regression at geocoded home and school points, combined into a time-location-weighted total; Lday/Lnight/Lden noise alongside	Sub-Saharan Africa
Klebach et al.	Aircraft observations suggest an important contribution of methanesulfonic and sulfuric acids to tropical Indo-Pacific aerosol	*Atmos Chem Phys*	Aircraft measurement campaign	Gas- and particle-phase MSA and H2SO4 by nitrate chemical-ionisation mass spectrometry	Indo-Pacific
Kong J et al.	Investigating the photosensitized chemistry of proxies for tropospheric brown carbon	*J Environ Sci (China)*	Bench / device experiment	Excited triplet-state brown carbon proxy (1,4-naphthoquinone); aqueous-phase oxidation kinetics	Laboratory

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Lai et al.	What are the sources of unclassified organosulfates in the atmosphere?	*Environ Sci Technol*	Narrative / critical review	Molecular formulas of ambient particulate organosulfates, which contribute up to 30% of aerosol mass, mapped to known precursors and formation routes	Global / multi-region
Lei et al.	Source transition and health implication of $PM_{2.5}$-bound metals: multi-year high-resolution observation in a coastal megacity	*Atmos Environ*	Metadata only	Hourly $PM_{2.5}$-bound metals, multi-year record	China
Leung et al.	Investigating the impacts of abandoned oil and gas wells on indoor air quality and health symptoms in Ontario, Canada: A pilot study	*Indoor Air*	Observational exposure study	13 passively sampled indoor VOCs (gas phase, not particulate) against inverse-distance-squared-weighted abandoned-well exposure within 3 km, with self-reported acute symptoms	Canada
Li Q et al.	Health risk of fine particulate matter across China with a pollutant-specific index, 2013-2023	*Environ Sci Technol*	Machine-learning exposure model + meta-analysis + risk index	Daily 1 km $PM_{2.5}$, 2013-2023, machine-learning estimate; $PM_{2.5}$-specific AQHI	China
Liu S et al.	Quantifying the combined effects of semi-closed terrain and monsoon on $PM_{2.5}$ and O3 pollution aggregation in the North China Plain	*J Environ Sci (China)*	Spatial analysis / GIS	Gridded $PM_{2.5}$ and O3 stratified by landform and altitude; Terrain-Wind Close Index	China
Liu S et al.	Wind shear structures and their association with the vertical distribution of $PM_{2.5}$ in Taiyuan Basin	*Environ Pollut*	Multi-platform observation synthesis	Vertical $PM_{2.5}$ profiles from aerosol lidar and UAV, against vertical wind shear regimes from wind profiler radar, microwave radiometer, radiosonde and WRF	China
Mutic et al.	Characterizing preschool children's multi-matrix air pollutant exposures across home and early childhood education settings: a paired silicone wristband study protocol	*medRxiv (preprint)*	Cross-sectional time-activity survey	Paired home- and centre-designated silicone wristbands plus continuous indoor air-quality monitoring in two classrooms per site	United States
Odu-Onikosi et al.	Characteristics, sources, and health risks of $PM_{2.5}$-bound potentially toxic elements (PTEs) in Lagos, Nigeria	*Atmos Pollut Res*	Metadata only	$PM_{2.5}$-bound potentially toxic elements with source apportionment and health-risk characterisation	Sub-Saharan Africa
Paches et al.	Estimation of background $PM_{2.5}$ concentrations in the Valencian Community applying clustering technique	*Atmos Pollut Res*	Metadata only	Regional background $PM_{2.5}$ separated from local increment by clustering of monitoring-network records	Europe
Papetta et al.	Source-dependent optical and mineral signatures of dust outbreaks over the Mediterranean	*Atmos Chem Phys*	Multi-platform observation synthesis	Dust optical depth, AOD, single-scattering albedo and asymmetry factor; AERONET + IASI + MODIS MIDAS + HYSPLIT	Mediterranean

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Petersen et al.	Direct photolysis and photosensitized reactions of sulfate ions in aerosol water form radical oxidants, secondary brown carbon, and organosulfates	*Environ Sci Technol*	Bench / device experiment	Sulfate-radical (SO4 anion radical) production in concentrated aerosol water; transient absorption plus radical trapping with LC-HRMS	Laboratory
Rahmani et al.	Airborne microplastics: a comprehensive analysis of indoor and outdoor pollution patterns	*Rev Environ Health*	Systematic review + random-effects meta-analysis	Airborne microplastic number concentration (MP/m3), indoor vs outdoor	Multi-country
Ren et al.	Contrasting chemical characteristics and secondary formation of $PM_{2.5}$ from mountain and urban sites in central China	*iScience*	Source apportionment	$PM_{2.5}$ mass and SNA / OC-EC speciation, annual	China
Sawan et al.	Uncoupling aviation and roadway contributions to ultrafine particle and black carbon exposure in a marginalized neighborhood	*Environ Sci Technol*	Source apportionment	Ultrafine particle number and black carbon, wind-sector stratified	North America
Scalabrin et al.	Chemical tracers for assessing the contribution of tire wear particles in atmospheric aerosol: analytical approaches, challenges and future perspectives	*Atmos Pollut Res*	Metadata only (no abstract)	Non-exhaust tire wear particles via chemical tracers	Europe
Schaefer et al.	Upward transport and segregation of ice-nucleating particles in deep convective clouds	*Atmos Chem Phys*	Aircraft measurement campaign	Immersion-mode INP concentration-temperature spectra from filter samples of convective inflow, outflow and cloud-particle residuals	Europe
Solaiman et al.	Secondary organic aerosol formation from the gas-phase oxidation of myrtenal by NO3 radicals: SOA yields, chemical composition, and highly oxygenated molecules	*Environ Sci Atmos*	Smog chamber experiment	SOA mass yields and HOM composition from a biogenic aldehyde oxidised by NO3 at night	Laboratory
Stergiou et al.	A spatiotemporal graph neural network framework for retrospective bias correction of CMAQ air quality simulations over the continental United States	*Atmos Pollut Res*	Algorithm development	CMAQ-simulated criteria pollutants including $PM_{2.5}$, monitor-anchored bias field	United States
Ullah et al.	Transboundary biomass-burning influence on springtime $PM_{2.5}$ population exposure across southern China during 2015-2019	*Environ Pollut*	CTM simulation + source attribution	Population-weighted share of days above the WHO 2021 24 h guideline (15 $\mu g/m^3$) and China GB 3095-2012 Grade II (75 $\mu g/m^3$), with the Indo-China Peninsula fire contribution isolated by a paired GEOS-Chem run	China
Wang J et al.	Water-soluble matter mediated phototransformation of guaiacol leading to formation of aqueous secondary organic aerosol: mechanism and kinetics	*J Environ Sci (China)*	Bench / device experiment	Aqueous SOA from guaiacol photosensitised by water-soluble aerosol extracts; ROS and triplet WSOM	Laboratory
Wang T et al.	An integrated multiscale air quality modelling framework for industrial park pollution: linking local emissions to regional transport	*J Environ Sci (China)*	Chemical transport modelling	$PM_{2.5}$ at 100 m resolution from NAQPMS coupled to a city-scale CTM with explicit Gaussian point-source plumes	China

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Wang X et al.	Interactive migration of heavy metals between dust and soil demonstrated by the coupling coordination analysis	*Environ Monit Assess*	Source-apportionment method	Cr, Mn, Cu, Zn, As and Pb in paired surface dust and soil, scored with Igeo and the Nemerow index and coupled across the two media	China
Wu Q et al.	Molecular characterization of night-time new particle formation in a coastal urban environment: season-dependent roles of sulfuric acid and oxygenated organics	*Environ Sci Technol*	Long-term field measurement campaign	Nocturnal H2SO4, its dimers and oxygenated organic molecules measured alongside nascent particle number, by nitrate- and Vocus-type mass spectrometry	China
Xiao S et al.	Pollution characteristics and health risks of PAHs and OPAHs in $PM_{2.5}$ of an inland basin city in China under the influence of the clean air action plan	*Environ Geochem Health*	Source-apportionment method	$PM_{2.5}$-bound parent PAHs and oxygenated PAHs, PMF source resolution, toxic-equivalent concentration	China
Xu L et al.	Marine air promotes structural compaction and coating growth of soot aerosols after long-range transport from East Asia	*J Environ Sci (China)*	Field measurement	Single-particle soot morphology and mixing state by TEM; fractal dimension and Dp/Dcore	China
Xu X et al.	Enhancement of secondary organic aerosol formation from isoprene photooxidation by ammonia	*J Environ Sci (China)*	Smog chamber experiment	Isoprene SOA mass and composition; nitrate-CIMS gas phase plus HR-TOF-AMS particle phase	Laboratory
Yang & Li	From percentiles to distributions: nationwide ZCTA-level modeling of indoor radon exposure across the contiguous United States, 2000-2021	*Environ Sci Technol*	Exposure model (secondary data)	Within-area distributions of indoor radon concentration modelled at ZCTA level, rather than area means or medians	North America
Yu H et al.	Machine learning attribution of atmospheric black carbon in a coastal city: source variability and the sea-land breeze recirculation mechanism	*Environ Sci Technol*	Source-apportionment method	Online BC split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis	China
Zhang N et al.	Effects of aerosol aging on composition and light-absorbance of nitrogen-containing organic compounds: evidences from ultra-high-resolution mass spectrometry analysis	*J Environ Sci (China)*	Field measurement	PM1 nitrogen-containing organics by UHPLC-Orbitrap MS; mass absorption efficiency at 365 nm	China
Zhao C et al.	Incomplete reduction of particle-bound PAH toxicity under lower $PM_{2.5}$ conditions: Evidence from Kunming, China	*Environ Pollut*	Source apportionment	16 priority $PM_{2.5}$-bound PAHs, BaP-equivalent burden, particle-normalised toxicity intensity ($BaPeq/PM_{2.5}$) and incremental lifetime cancer risk, split by lower- and higher-$PM_{2.5}$ stage	China

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Zhao H et al.	Lead isotope characteristics and source analysis of dust fall in different functional areas of a typical city in northern China	*Environ Monit Assess*	Source-apportionment method	Dustfall trace elements (V, Mn, Co, Cr, Ni, Cu, Zn, As, Cd, Pb) with Pb isotope source fingerprinting	China
Zhu et al.	Revisiting the critical role of stabilized Criegee intermediates in sulfuric acid formation: coupling mechanistic updates with interpretable machine learning	*Atmos Chem Phys*	Chemical transport modelling	Gas-phase SO2 oxidation rate via stabilized Criegee intermediates; sCI fraction of total SO2 loss	China
Burden, policy & mitigation					
Li Y et al.	Multidimensional inequality of CO2 and air pollutant footprints in China under dual-carbon policy	*Environ Sci Technol*	Modelling / inventory	Per-capita consumption-based footprints of CO2, SO2, NOx, NH3 and primary PM	China
Lim et al.	Public responses about air quality in the world's ten most populous countries	*Nat Commun*	Multi-country quota survey + exposure linkage	Modelled ambient $PM_{2.5}$ linked to respondent locality	Global / multi-region
Liu Z et al.	Significant near-term inequity under China's 2035 NDC due to a time lag in mitigation effects across regions	*Environ Sci Technol*	Integrated assessment modelling	Air-quality health co-benefits monetised alongside climate benefits under the 2035 NDC pathway, resolved by region and by decade	China
Shi et al.	Coupled energy-industrial transitions shape heterogeneous carbon-air pollution cobenefits and spatial emission leakage in China's steel industry	*Environ Sci Technol*	Modelling / inventory	Unit-level SO2 and primary PM emissions alongside CO2, with modelled ambient response	China
Simion et al.	Attributing climate-related health burdens in Southeast Asia: a scoping review	*Lancet Reg Health West Pac*	Scoping review	Air pollution, heat and extreme-weather attribution studies	Southeast Asia
Song et al.	Shifting $PM_{2.5}$ control from mass to toxicity for health equity worldwide	*Environ Sci Technol*	Perspective / evidence synthesis	Source-specific $PM_{2.5}$ toxicity weighting applied to global exposure surfaces	Global / multi-region
Zhao N et al.	Uneven attainment pressure under China's tightened ambient air quality standard: socioeconomic predictors and differentiated control pathways for $PM_{2.5}$ and O3 in Shandong	*J Hazard Mater*	Spatial regression (GWR)	Daily $PM_{2.5}$, PM_{10}, SO2, NO2, CO and O3, 2015-2025; relative attainment pressure index against GB 3095-2026	China
Other clinical endpoints					
Batisse et al.	Exposure to outdoor ultrafine particle number concentrations and incidence of childhood cancer: a population-based cohort study	*Environ Epidemiol*	Prospective cohort (registry linkage)	High-resolution outdoor UFP number concentration and mean UFP size at six-digit residential postal codes from annual tax records; prenatal and postnatal windows	North America
Chen K et al.	Effect of long-term exposure to $PM_{2.5}$ and its chemical components on frailty incidence in middle-aged and older adults in China: a prospective cohort study	*Sichuan Da Xue Xue Bao Yi Xue Ban*	Prospective cohort (registry linkage)	One-year mean $PM_{2.5}$ and its black carbon, organic matter, sulfate, nitrate and ammonium fractions from the Tracking Air Pollution dataset, per IQR	China

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Ding E et al.	Climate factors, air pollutants, and extreme events shape influenza risk in China: a hospital-based case-crossover and machine-learning study	*Environ Sci Technol*	Time-series / case-crossover	Exposure-lag-response for meteorology, air pollution and compound extreme events by DLNM, variable ranking by ML, and an LSTM forecasting proof of concept for Beijing	China
Donaldson et al.	Air Pollution Impairs Marathon Performance: A Cross-Sectional Analysis of 2.7 Million Finishers Across Six World Marathon Majors	*Sports Med Open*	Observational - cross-sectional	Race-day NO2 and $PM_{2.5}$ from CAMS global reanalysis, per 1 $\mu g/m^3$	Multi-country
Feng et al.	Plasma metabolic and proteomic signatures of mixed ambient air pollutant exposure and incident arthritis subtypes	*Joint Bone Spine*	Prospective cohort (mixtures) + omics mediation	Annual $PM_{2.5}$, PM_{10}, NOx, NO2 combined in a weighted quantile sum index	United Kingdom
Park J et al.	Association between ozone and emergency department visits in people with disabilities in South Korea: a nationwide case-crossover study	*Int J Epidemiol*	Time-stratified case-crossover	Daily 8 h maximum ozone, 3-day moving average (lag 0-2), per IQR (22 ppb), adjusted for daily mean temperature; no particulate metric	South Korea
Wang C et al.	Age-dependent heterogeneity trends in the associations between urban green space and cardiovascular health in a humid-subtropical megacity	*Environ Sci Technol*	Cross-sectional + prediction model	Two-stage gradient-boosting plus structural equation model; air-quality pathways (NO2, SO2, $PM_{2.5}$) carry >75% of the relative contribution to model variance	China
Zhang & Yu 2026	Environmental and calendar correlates of daily emergency department demand in Wuhan, China: a single-centre time-series study with temporal validation	*Res Sq (preprint)*	Time-series / case-crossover	Pollutant-specific AQI ($PM_{2.5}$, PM_{10}, NO2, O3), lag 0-2, per IQR	China
Zheng M et al.	Spatiotemporal co-occurrence patterns and related risk factors of esophageal and lip and oral cavity cancers among older adults across 203 countries and territories	*Ann Med*	GBD secondary analysis	Occupational exposure to particulate matter, gases and fumes as one of 25 country-level exposures in negative binomial models with population offsets	Global / multi-region

Provenance and method

Period **2026-08-30 to 2026-09-05**, seven daily issues pooled. **This rollup queried no API.** It aggregates `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over `PMRW_START–PMRW_END`; every record was screened, summarised and DOI-gated at daily-issue time and is reproduced verbatim. **150 records, 150 unique DOIs** — no duplicate publication this period, unlike the two repaired last week. Daily totals: 14, 16, 33, 15, 38, 27, 7. **Week-over-week comparisons** are against the 23–29 August rollup (109 records), with the 16–22 August (112), 9–15 August (134) and 2–8 August (146) rollups used where a longer run is cited; the 26 July–1 August week (295) is excluded from "record" claims because it contains the founding retrospective back-scan. **Effect estimates** are pooled as recorded, on heterogeneous contrasts, with no meta-analysis and no heterogeneity statistic. **Figures** are regenerated over the pooled range, not composited from the daily images. **Labels** — subtopic, design, particle-metric, geography — are single-label by dominant contribution, assigned at daily-issue time, and are a judgement rather than a controlled vocabulary.